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Weinstock

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(54) **SHARED OUTPUT CONNECTOR ASSEMBLY FOR TWO DRINK MAKER DISPENSER ASSEMBLIES**

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CPC A23G 9/281
See application file for complete search history.

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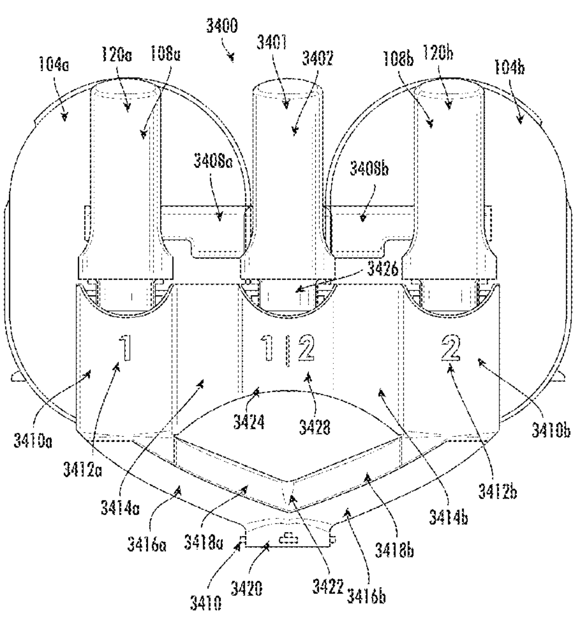
(57) **ABSTRACT**

Provided is a shared output connector assembly for connecting a first dispenser assembly and a second dispenser assembly. The shared output connector assembly includes a primary lever and at least one projection configured to mechanically connect the primary lever to a first lever of the first dispenser assembly and a second lever of the second dispenser assembly. When the shared output connector assembly is attached to the first dispenser assembly and the second dispenser assembly, the first lever and the second lever are independently actuatable. When the primary lever is actuated, the at least one projection causes the first lever and the second lever to actuate.

27 Claims, 51 Drawing Sheets



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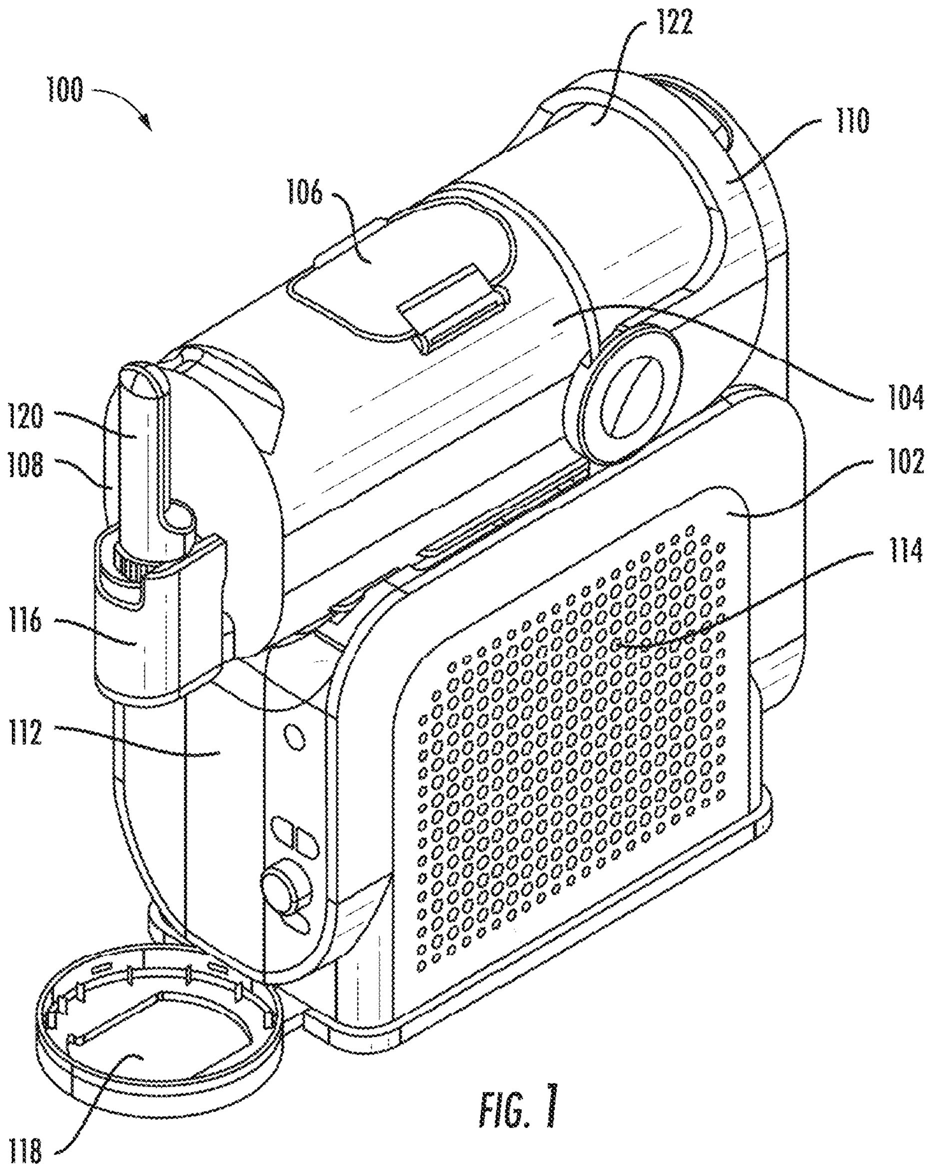
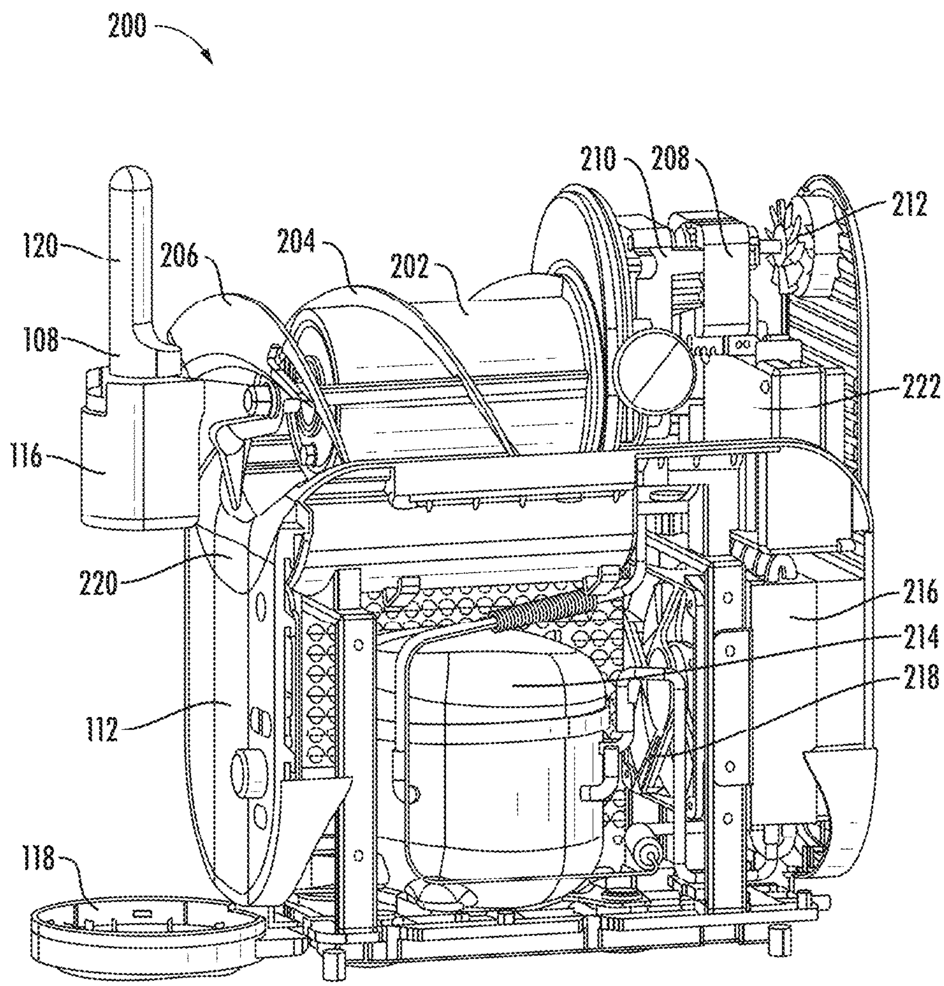


FIG. 1



200
210
208
212
120
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108
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116
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214
218
112
118

FIG. 2

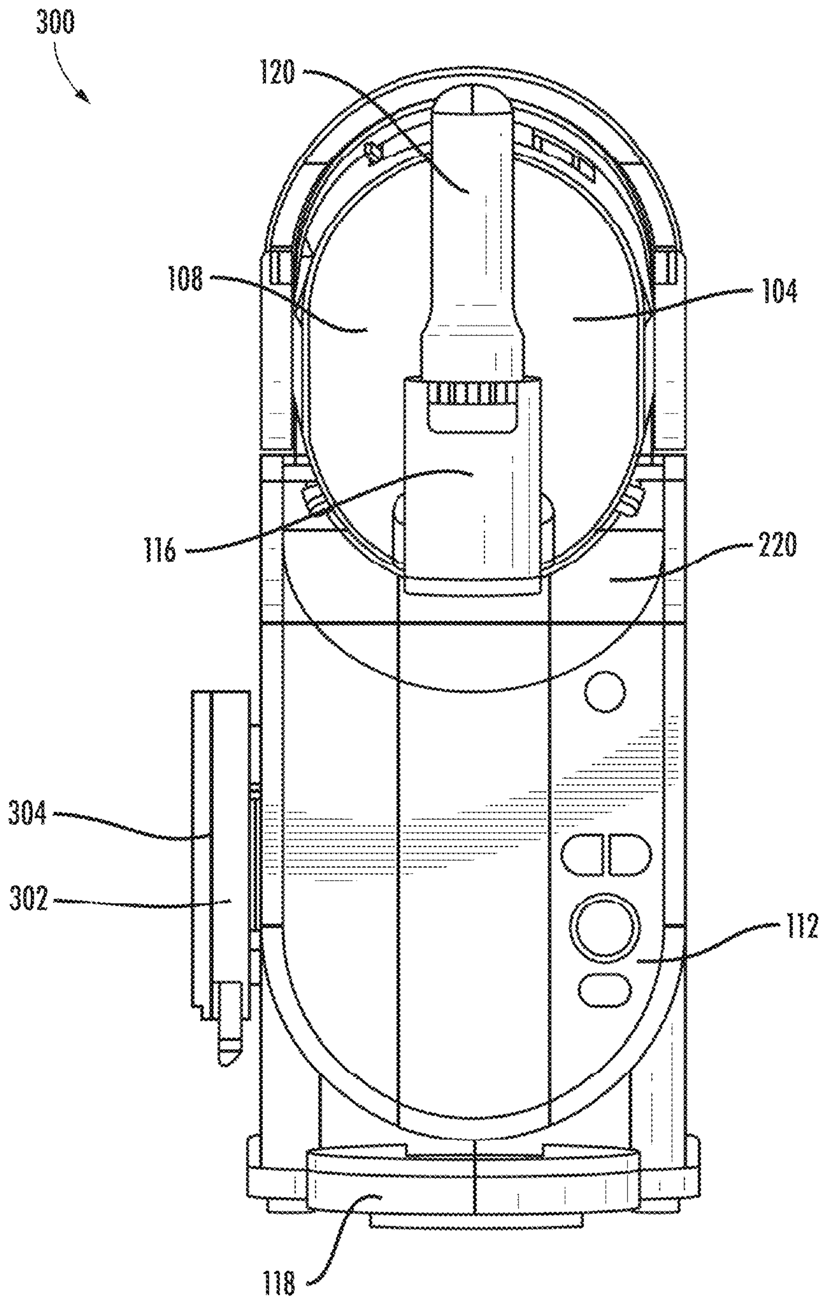


FIG. 3

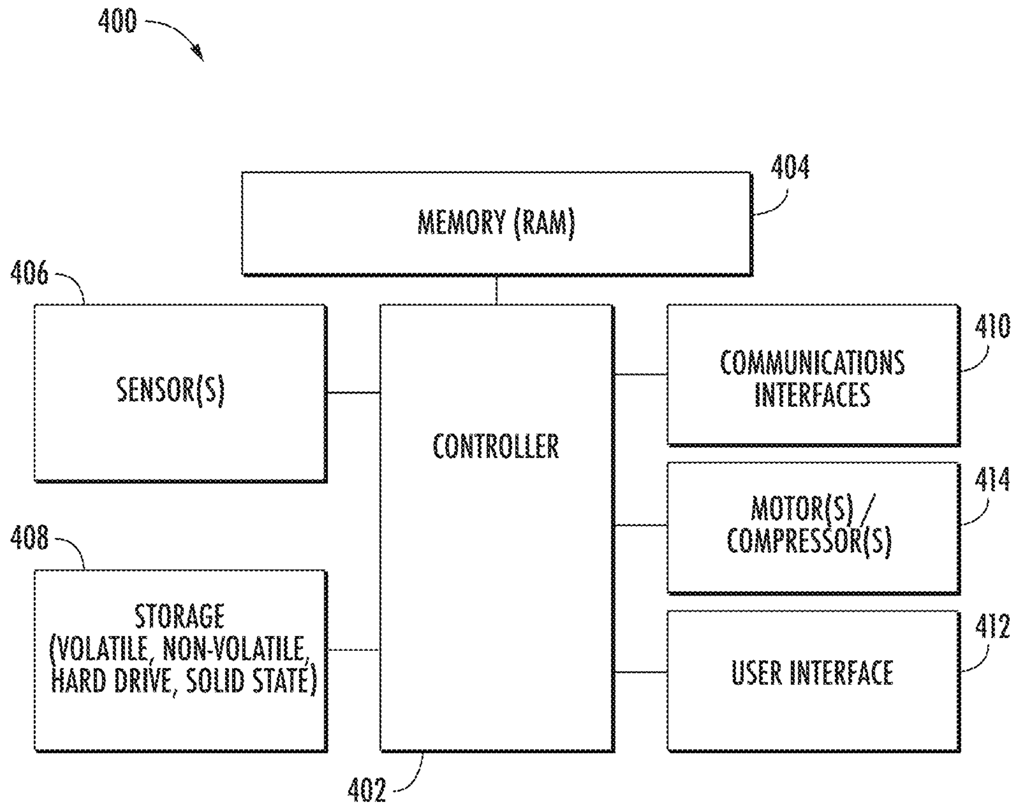

400
MEMORY (RAM)
404
406
SENSOR(S)
CONTROLLER
COMMUNICATIONS INTERFACES
410
MOTOR(S) / COMPRESSOR(S)
414
408
STORAGE (VOLATILE, NON-VOLATILE, HARD DRIVE, SOLID STATE)
USER INTERFACE
412
402

FIG. 4

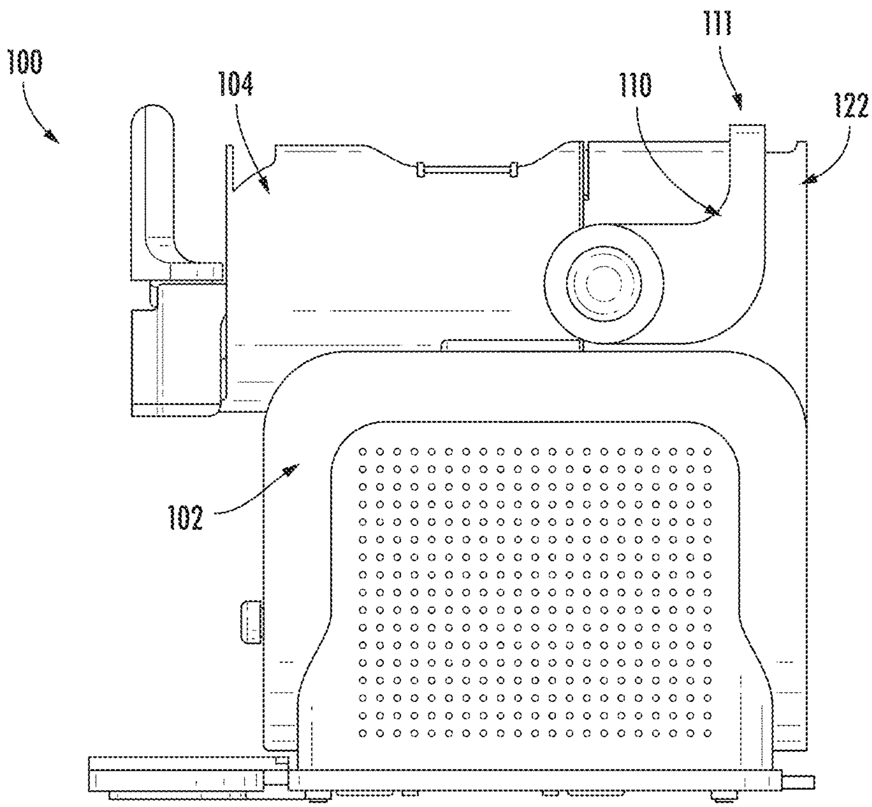


FIG. 5A

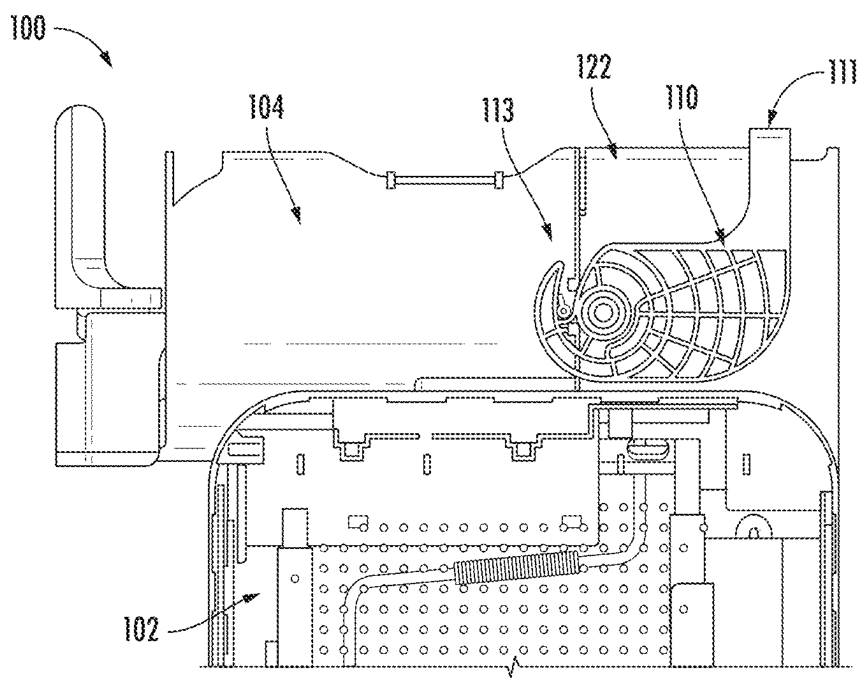


FIG. 5B

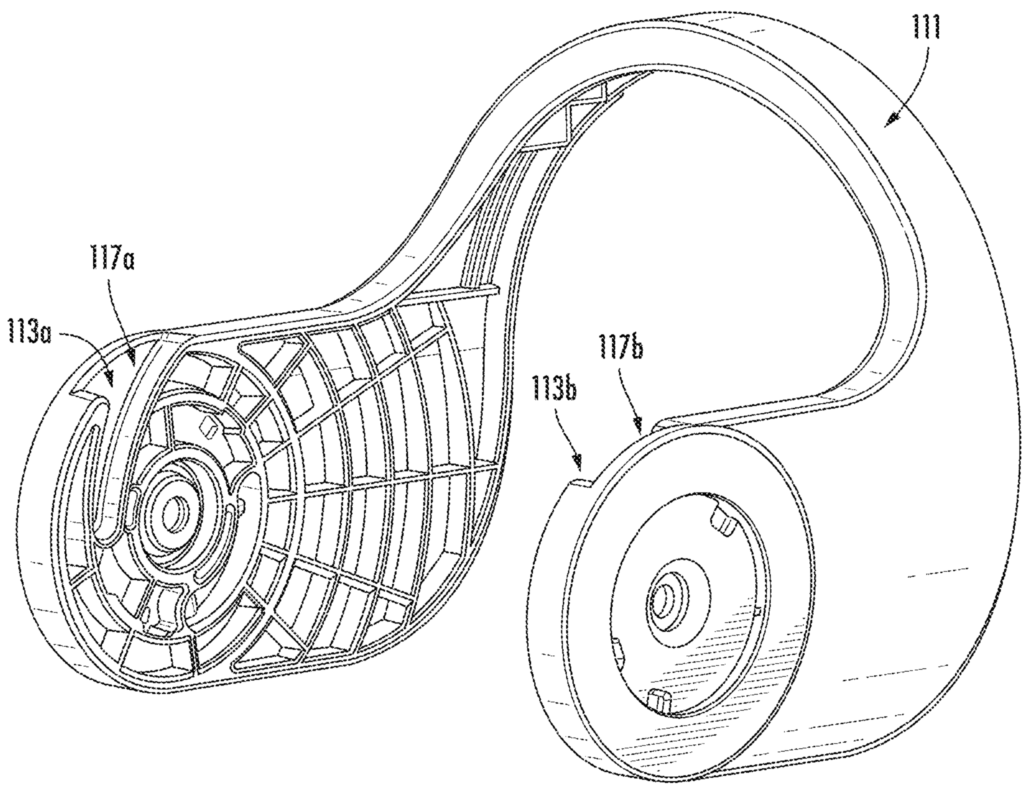


FIG. 6

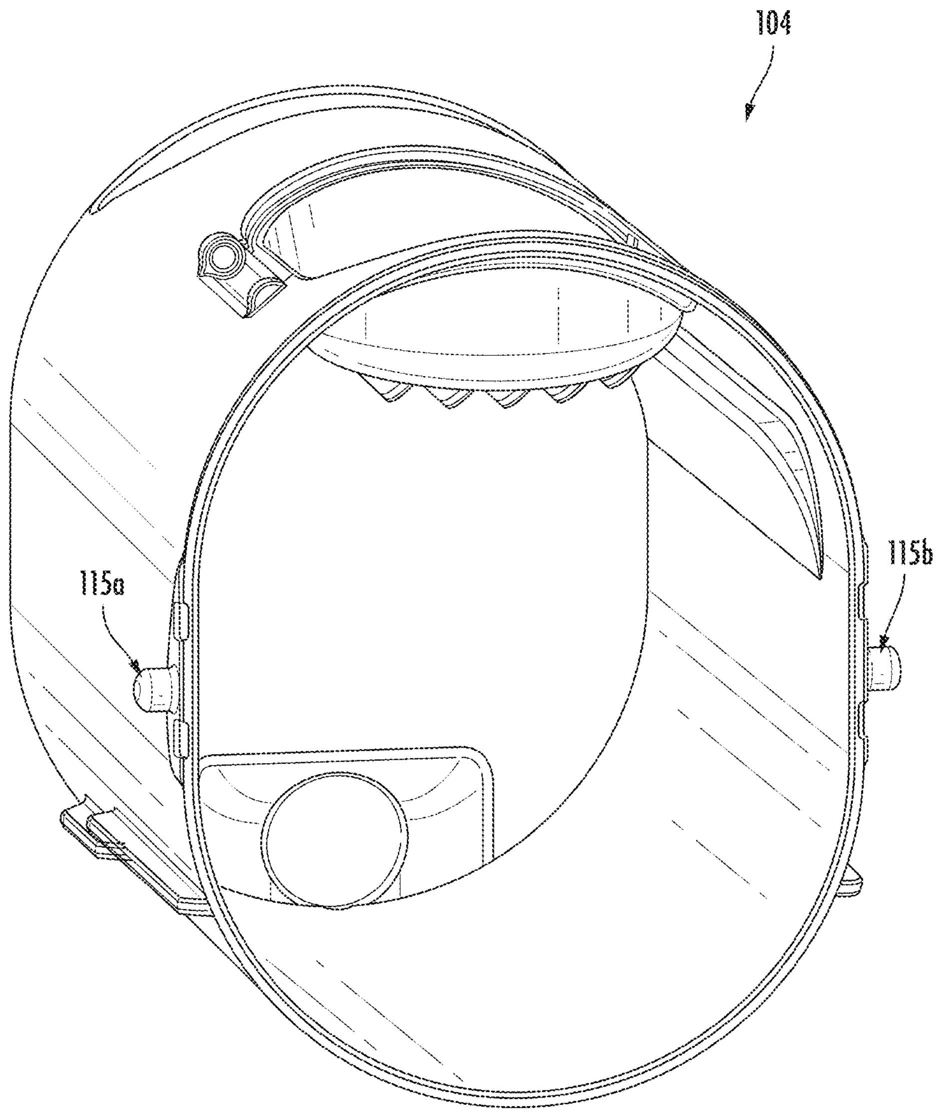


FIG. 7A

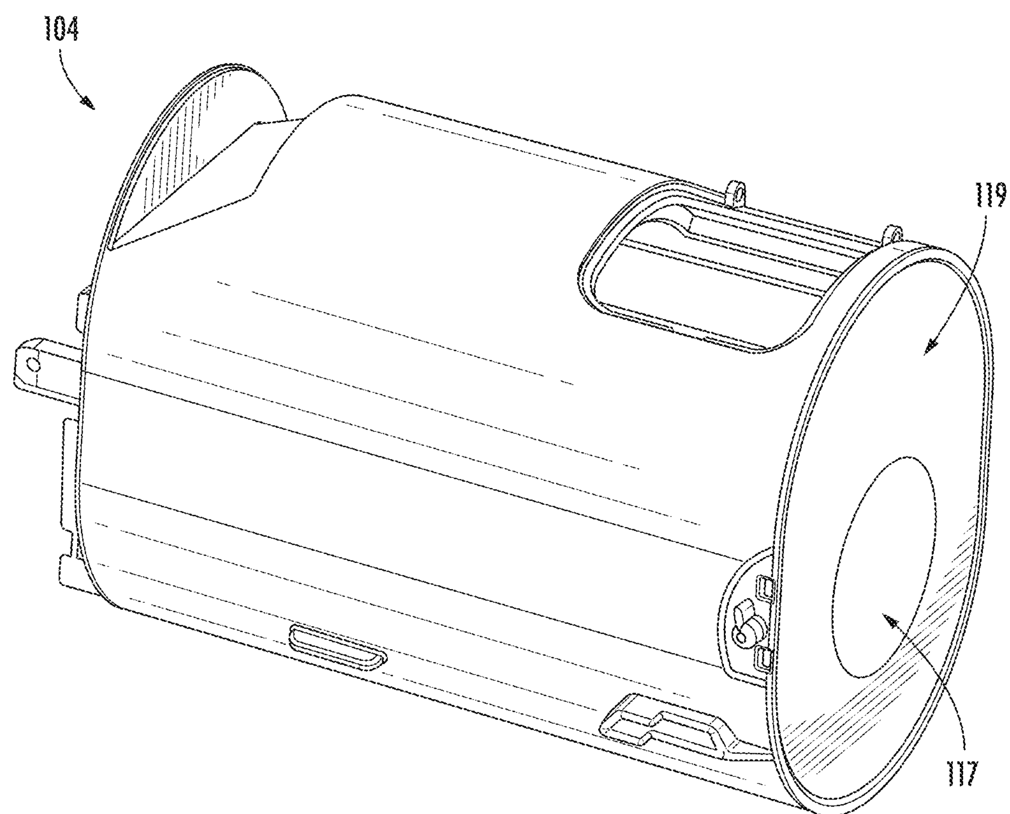


FIG. 7B

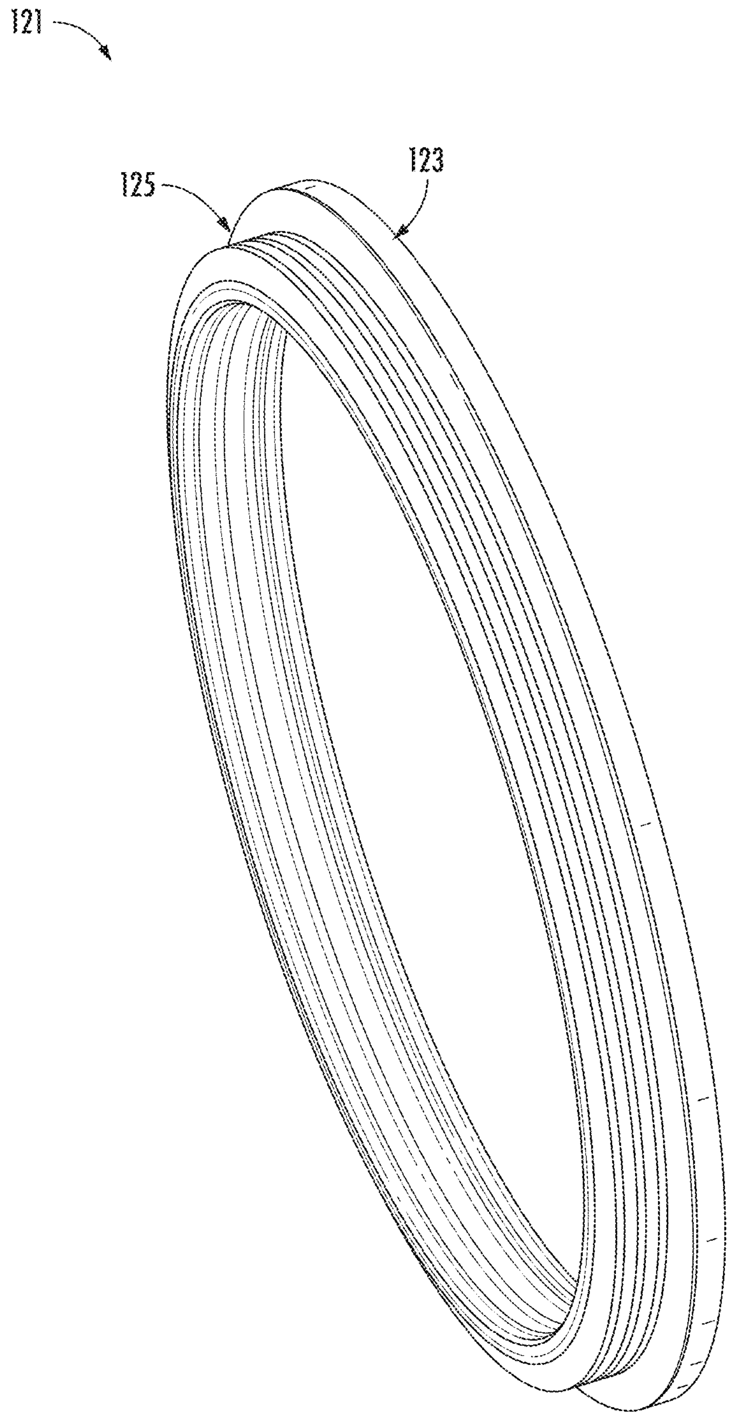


FIG. 8

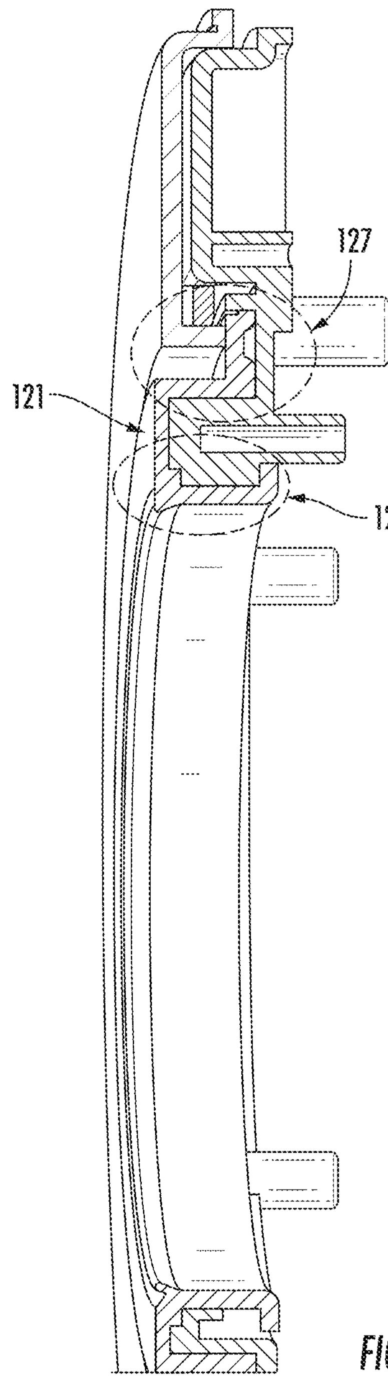


FIG. 9

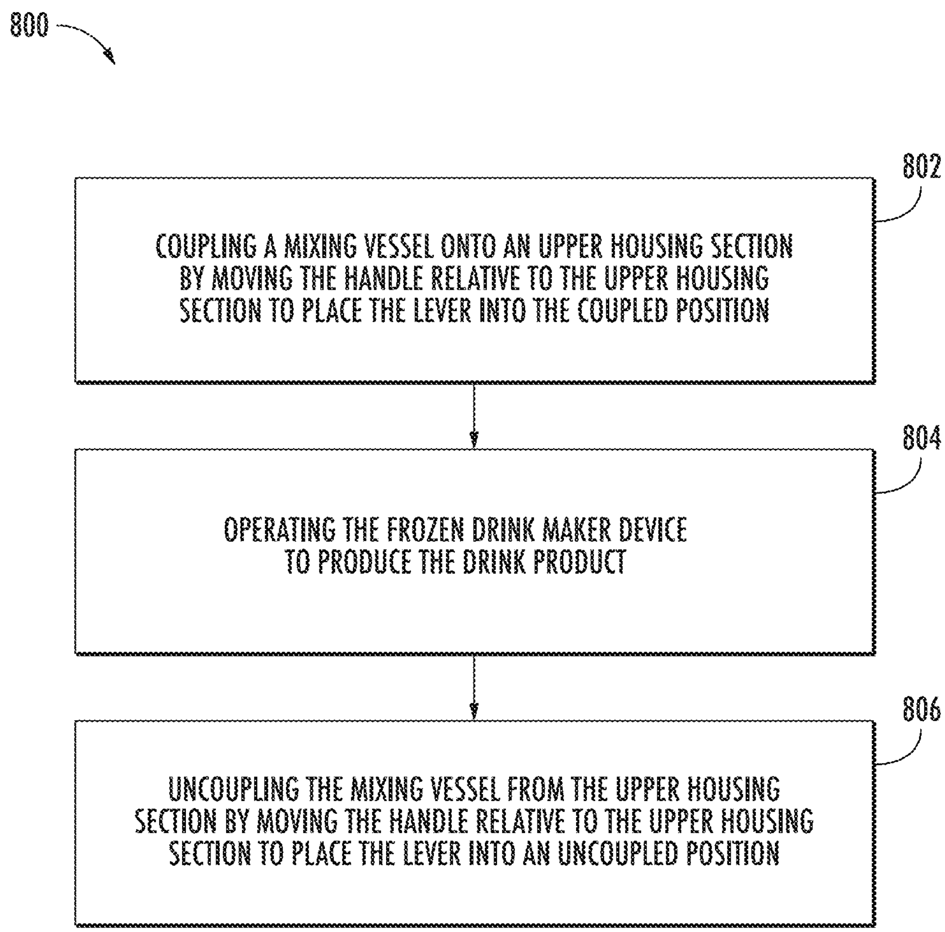

800
802
COUPLING A MIXING VESSEL ONTO AN UPPER HOUSING SECTION BY MOVING THE HANDLE RELATIVE TO THE UPPER HOUSING SECTION TO PLACE THE LEVER INTO THE COUPLED POSITION
804
OPERATING THE FROZEN DRINK MAKER DEVICE TO PRODUCE THE DRINK PRODUCT
806
UNCOUPLING THE MIXING VESSEL FROM THE UPPER HOUSING SECTION BY MOVING THE HANDLE RELATIVE TO THE UPPER HOUSING SECTION TO PLACE THE LEVER INTO AN UNCOUPLED POSITION

FIG. 10

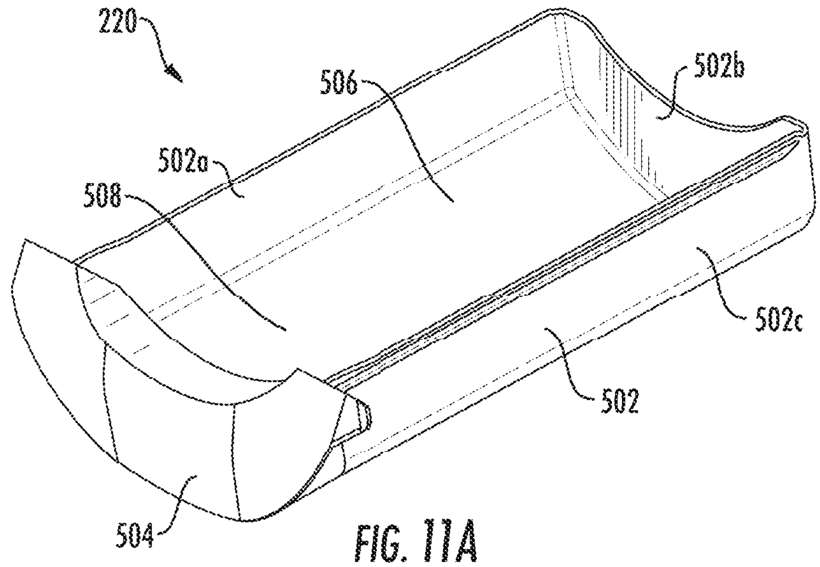


FIG. 11A

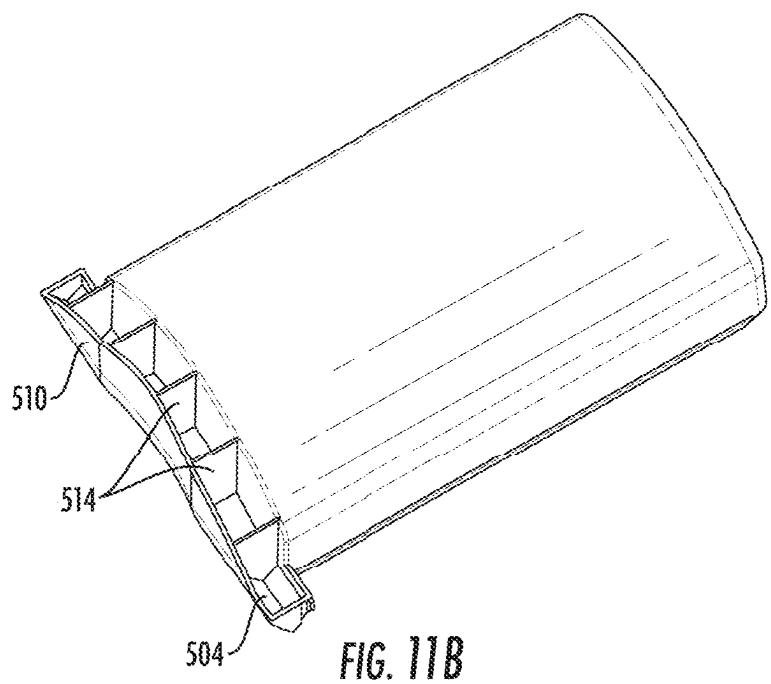


FIG. 11B

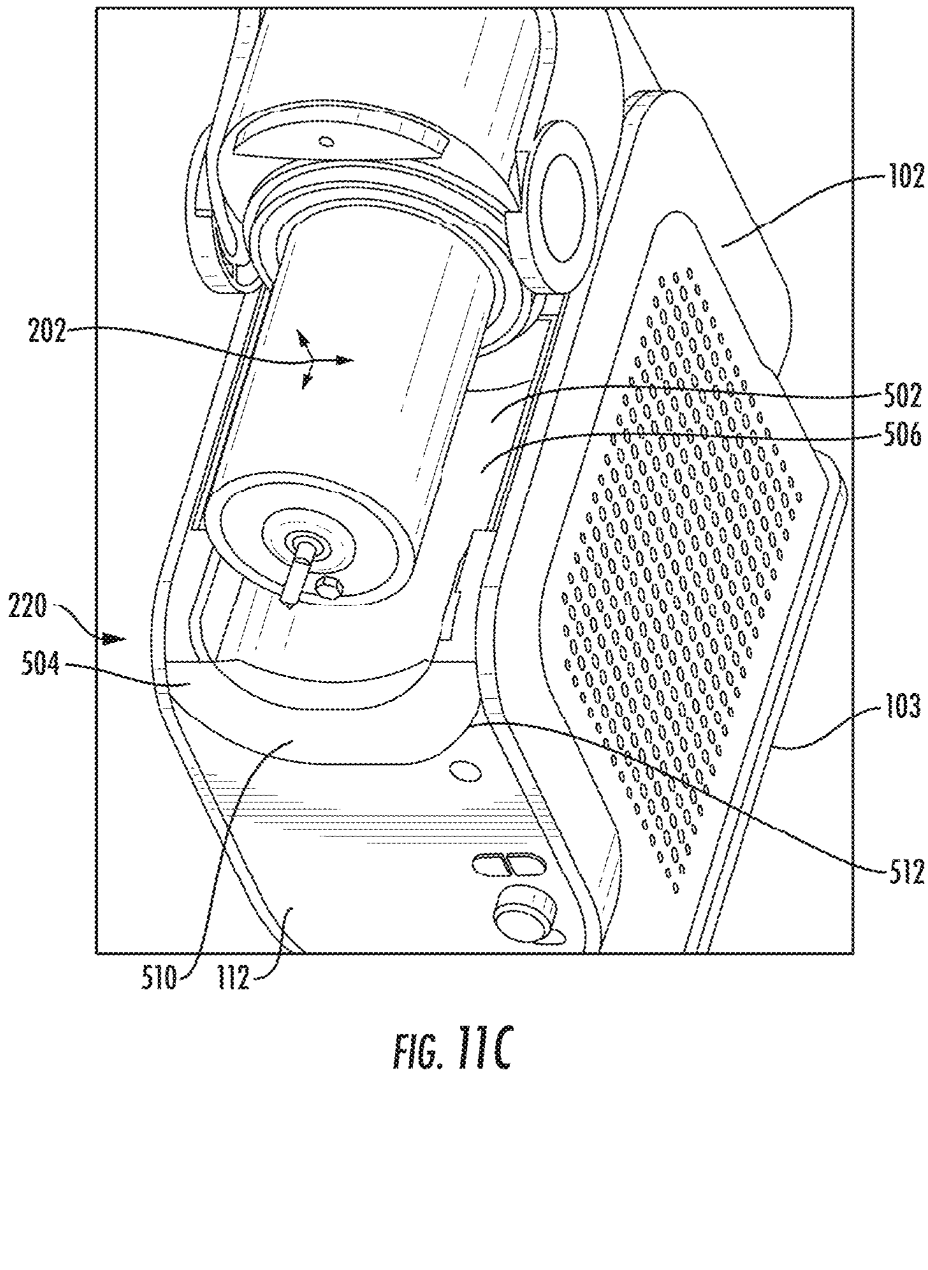


FIG. 11C

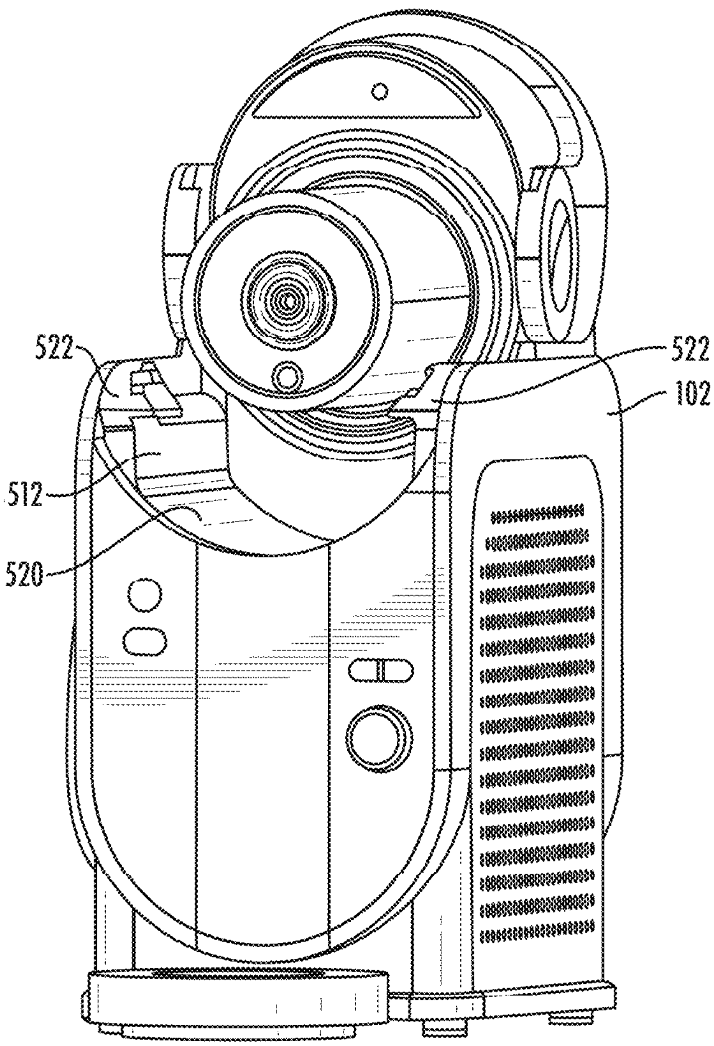
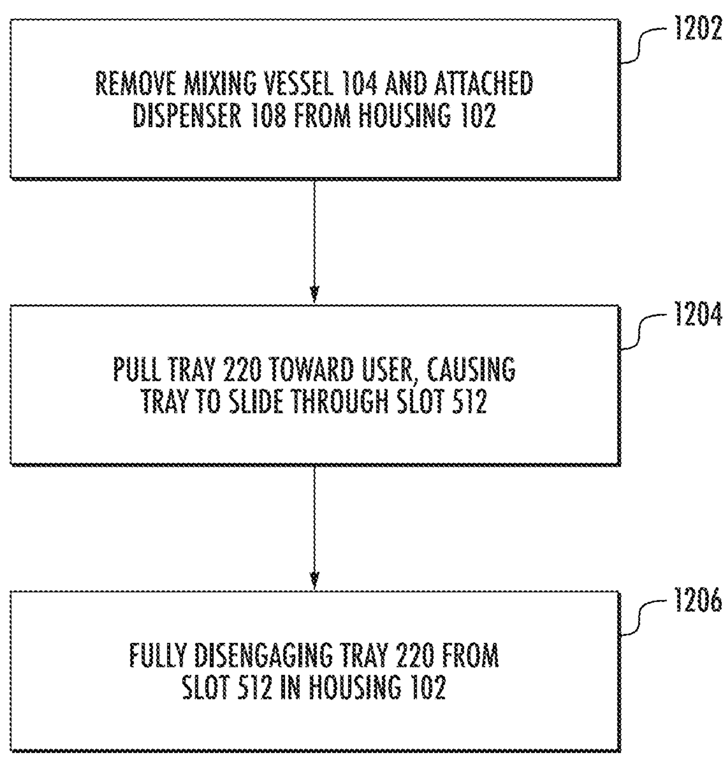


FIG. 11D



REMOVE MIXING VESSEL 104 AND ATTACHED DISPENSER 108 FROM HOUSING 102
1202
PULL TRAY 220 TOWARD USER, CAUSING TRAY TO SLIDE THROUGH SLOT 512
1204
FULLY DISENGAGING TRAY 220 FROM SLOT 512 IN HOUSING 102
1206

FIG. 12

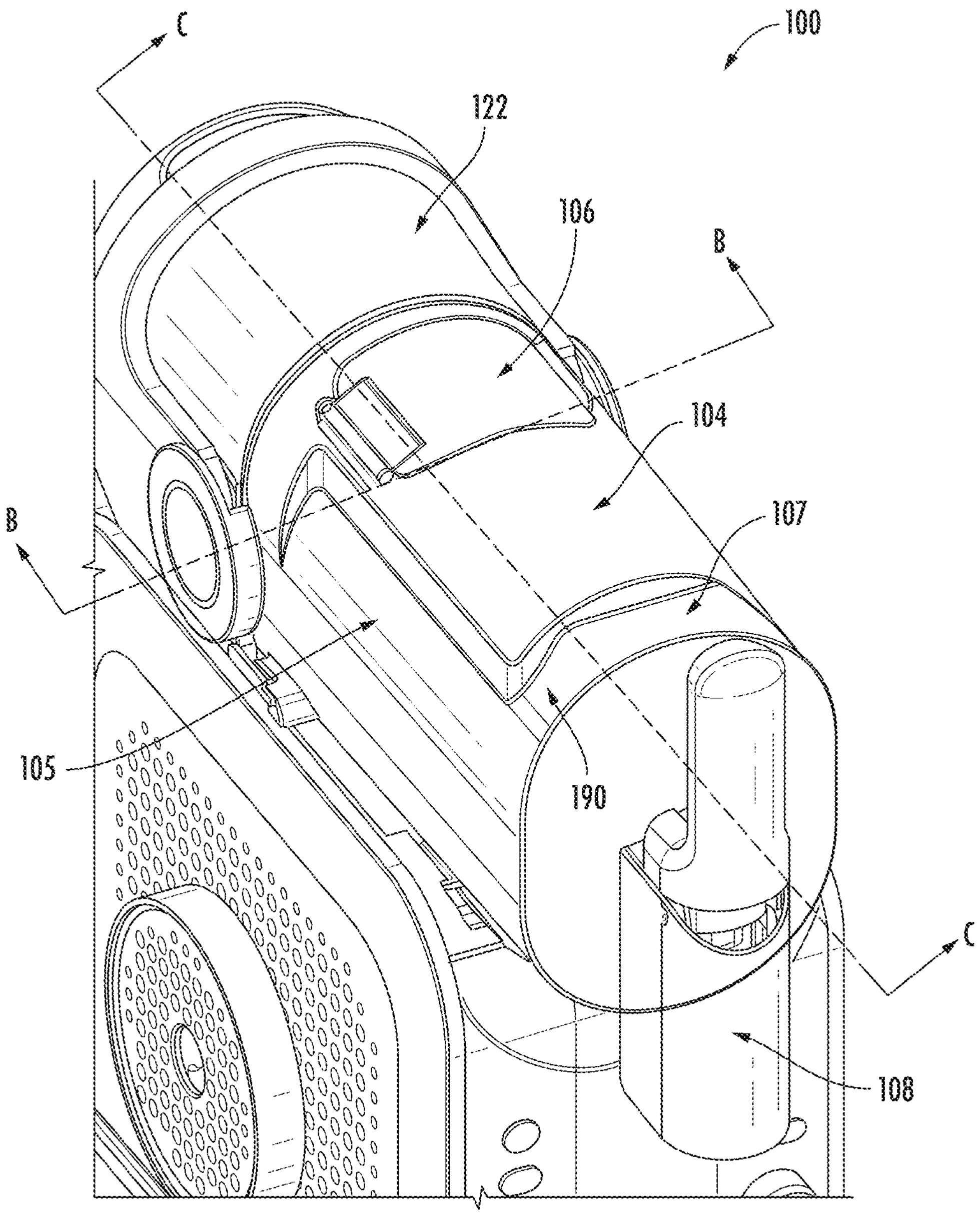


FIG. 13A

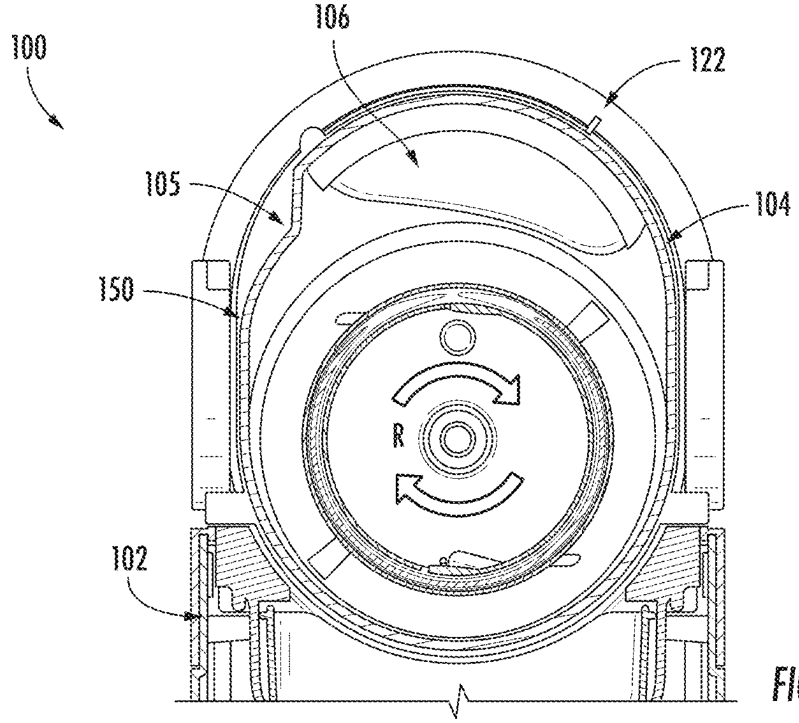


FIG. 13B

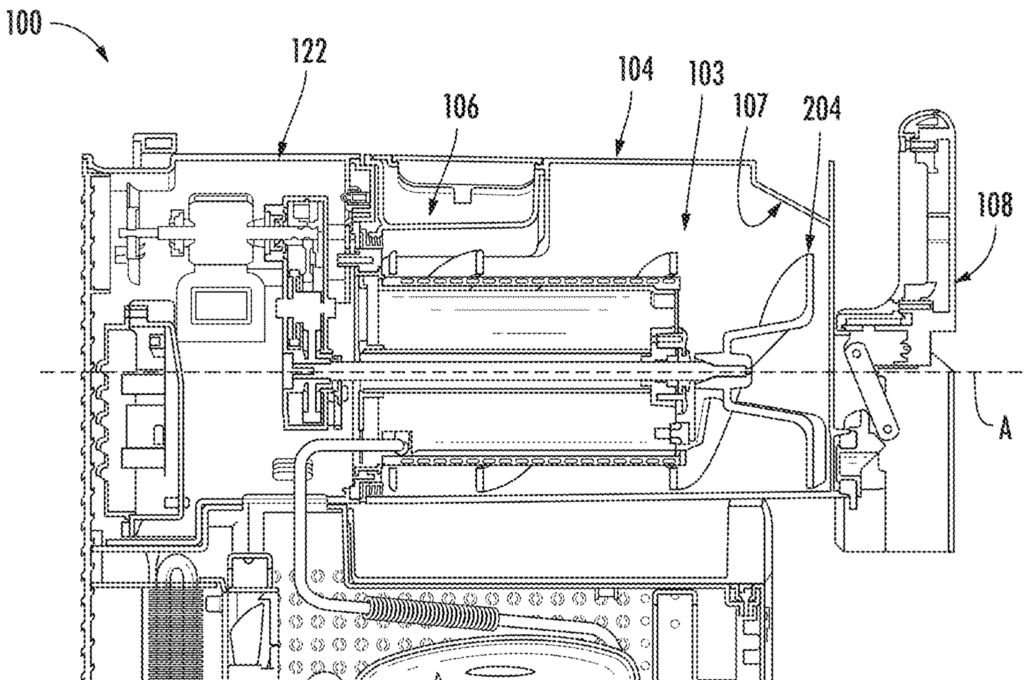


FIG. 13C

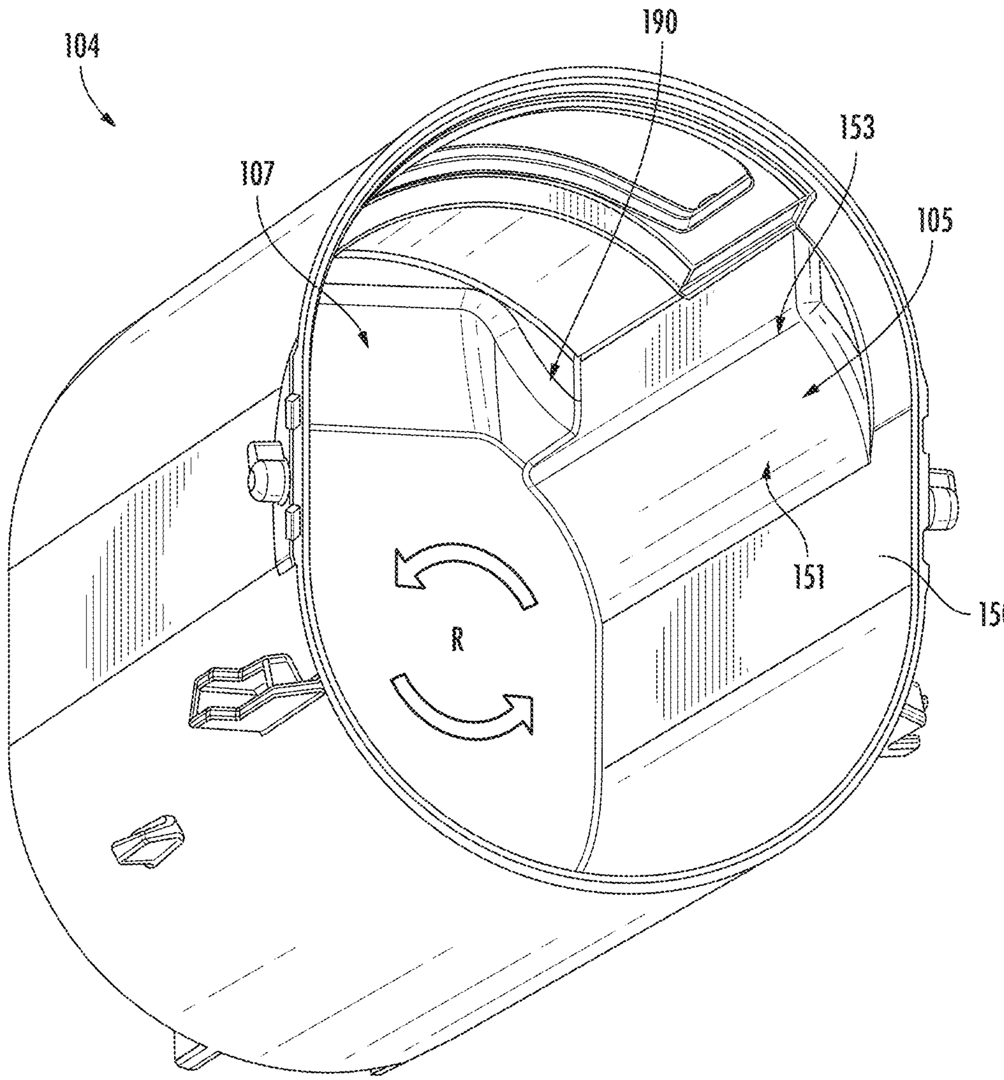


FIG. 14A

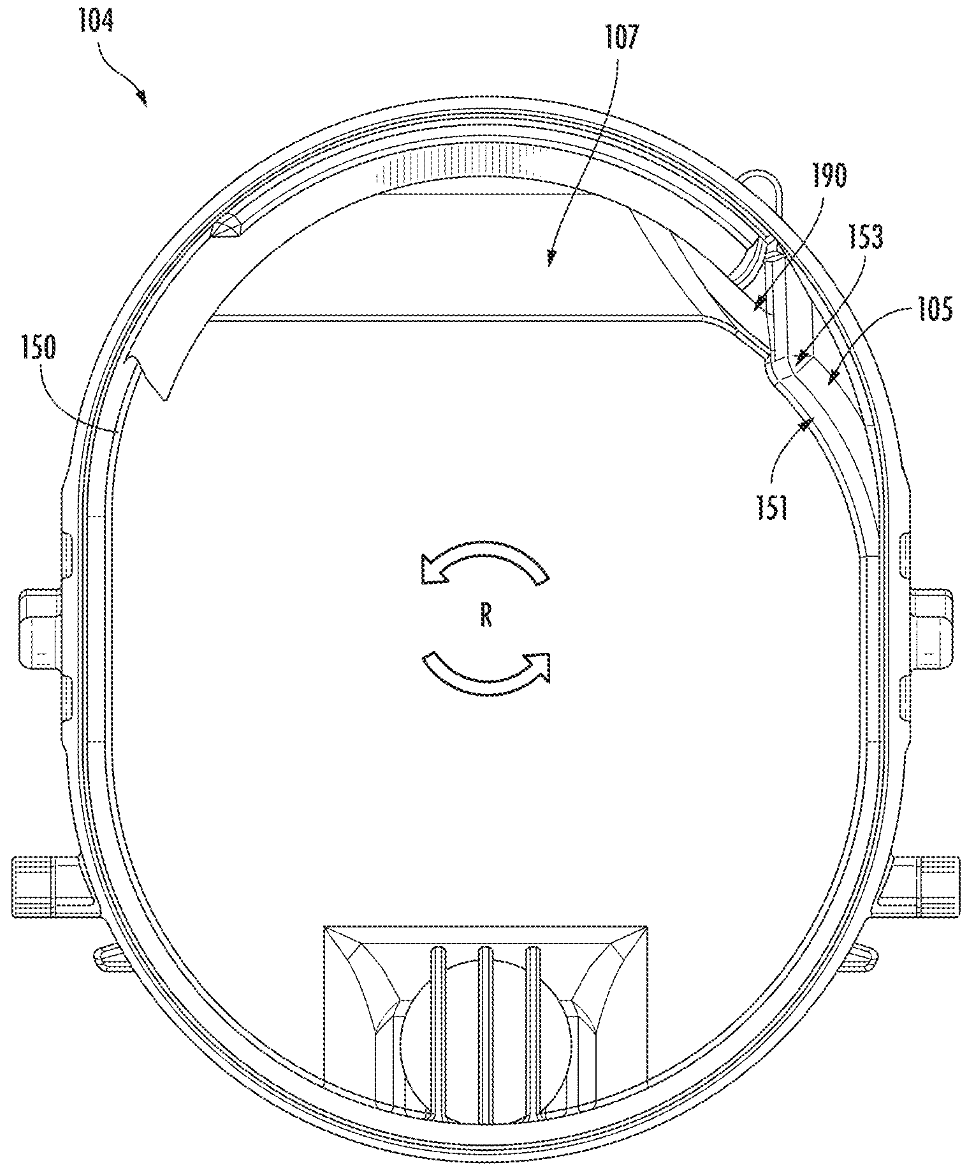


FIG. 14B

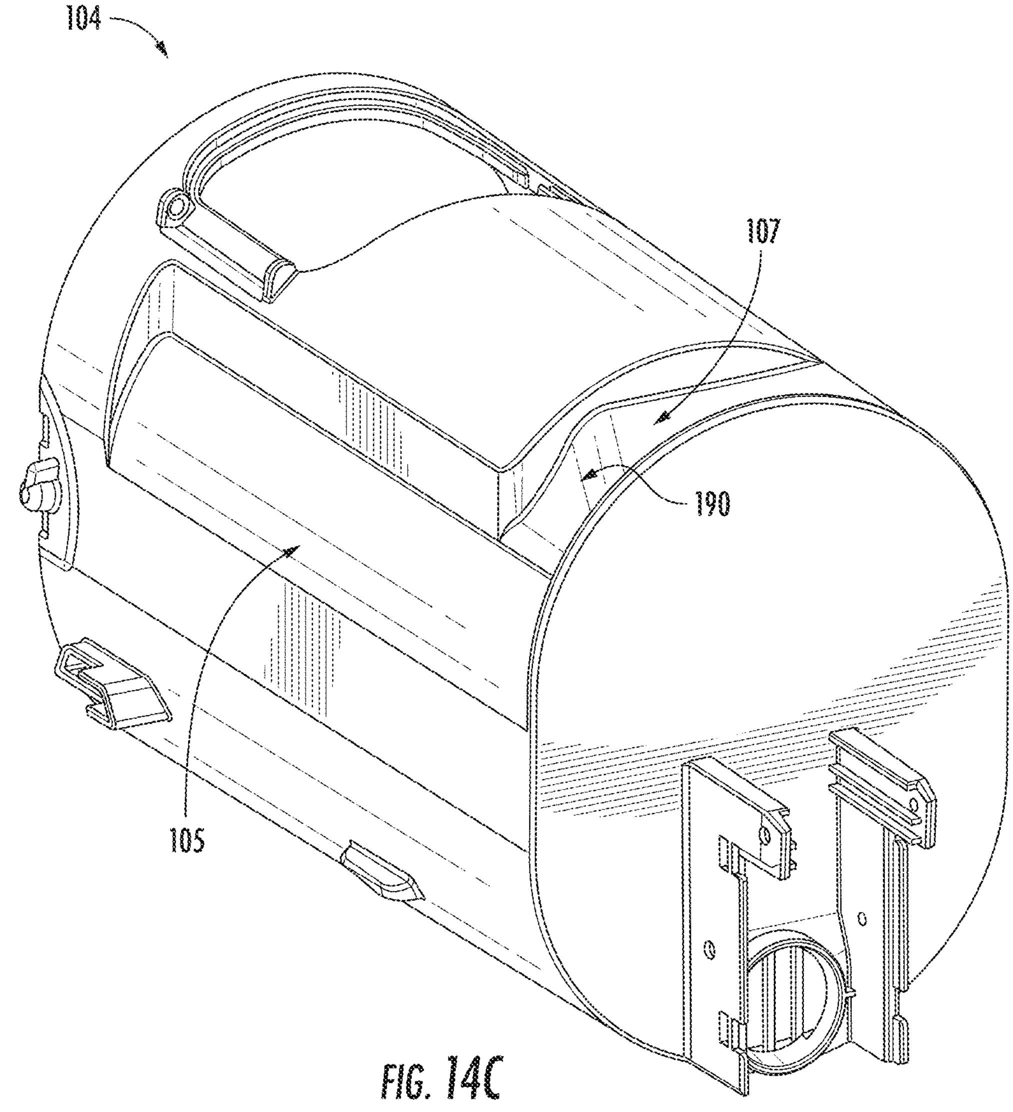


FIG. 14C

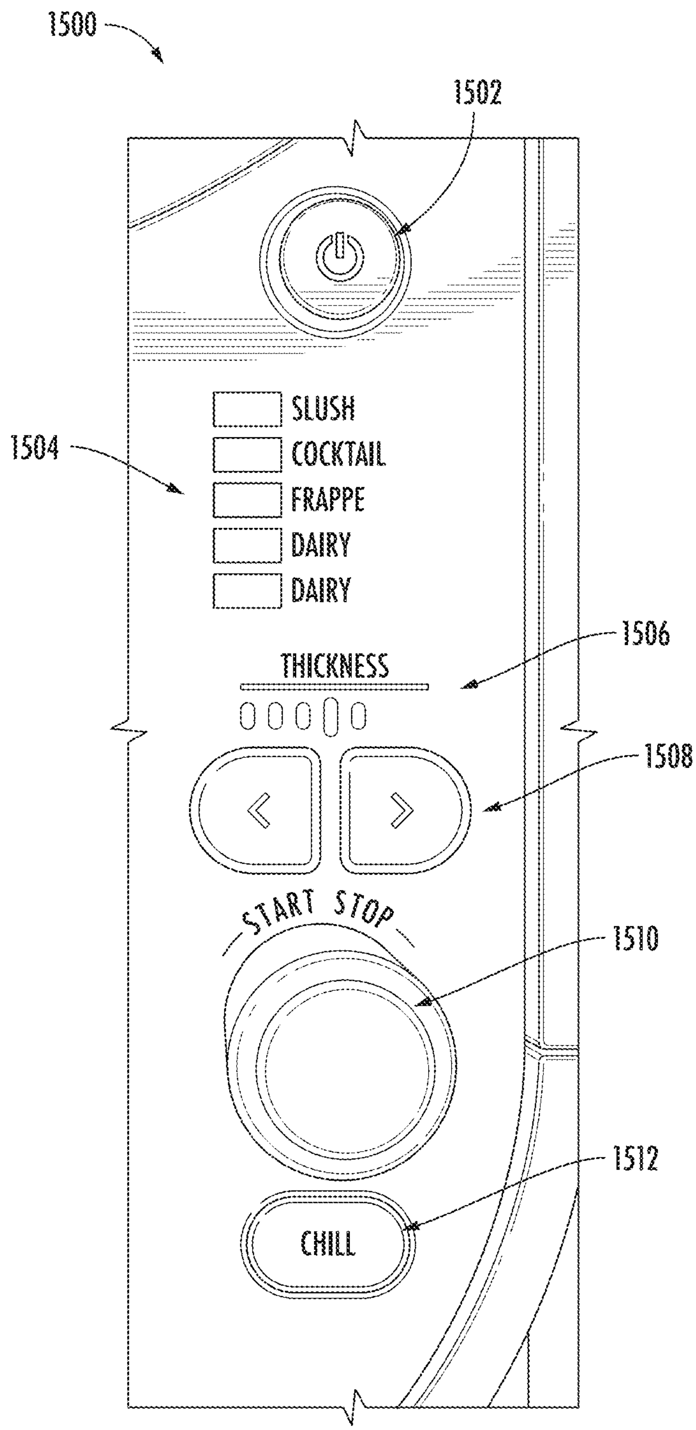

1500
1502
SLUSH
COCKTAIL
FRAPPE
DAIRY
DAIRY
1504
THICKNESS
1506
1508
START STOP
1510
1512
CHILL

FIG. 15

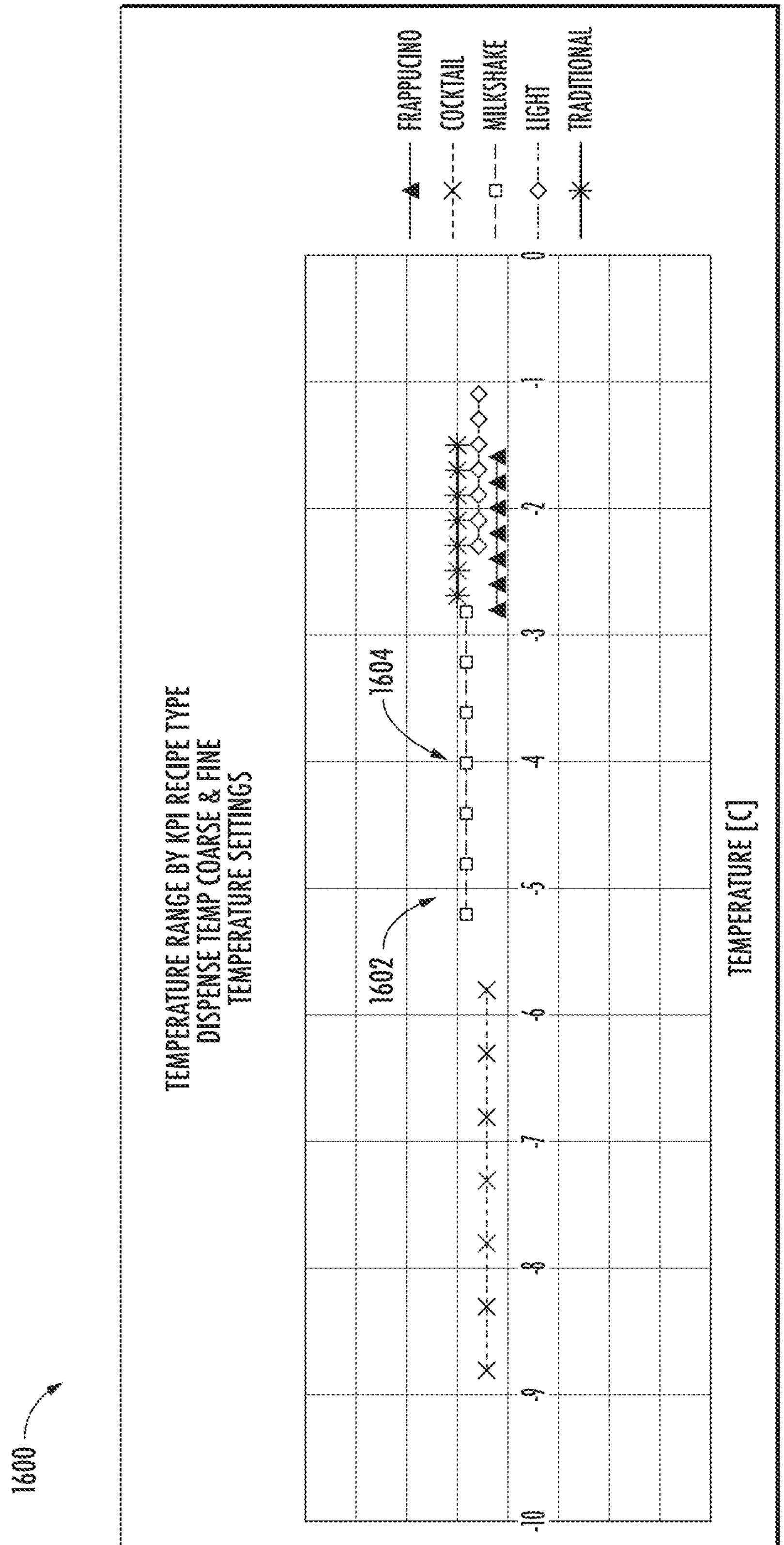

1600
TEMPERATURE RANGE BY KPI RECIPE TYPE
DISPENSE TEMP COARSE & FINE
TEMPERATURE SETTINGS
1602
1604
FRAPPUCINO
COCKTAIL
MILKSHAKE
LIGHT
TRADITIONAL
0
-1
-2
-3
-4
-5
-6
-7
-8
-9
-10
TEMPERATURE [C]

FIG. 16

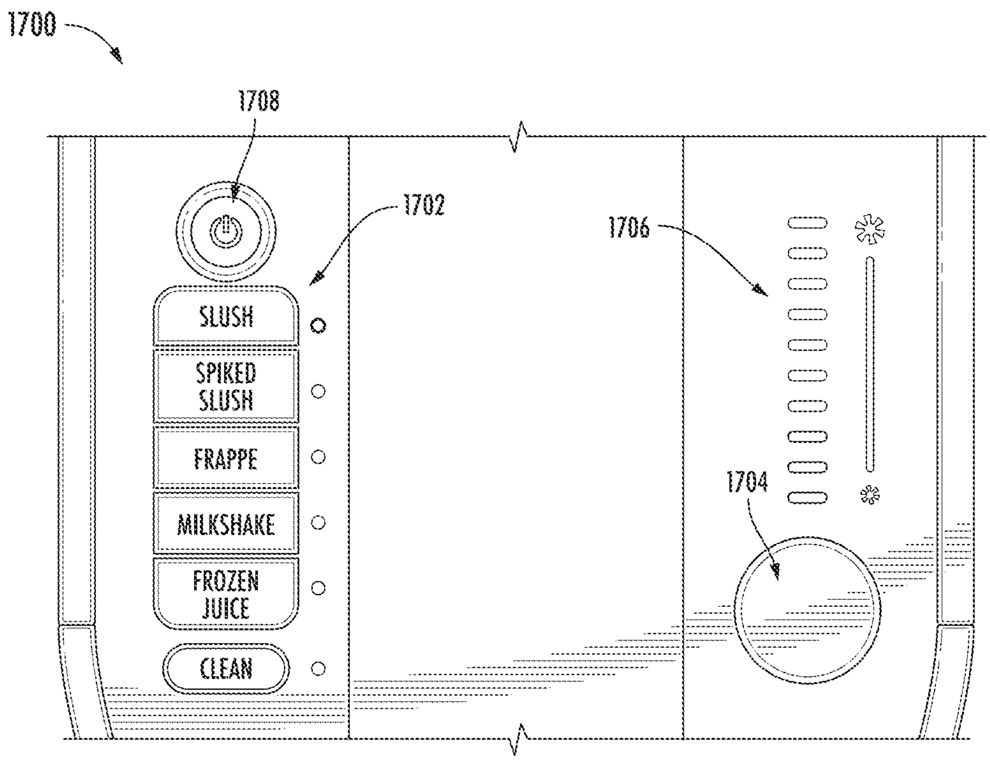

1700
1708
1702
1706
1704
SLUSH
SPIKED SLUSH
FRAPPE
MILKSHAKE
FROZEN JUICE
CLEAN

FIG. 17

1800

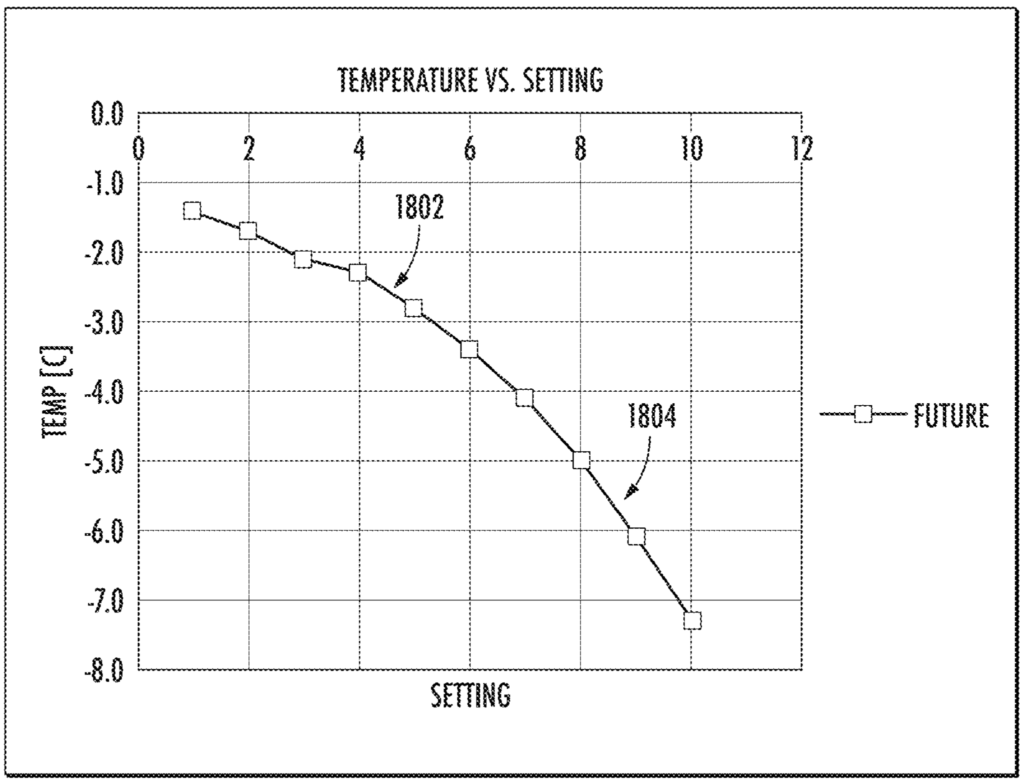

TEMPERATURE VS. SETTING
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
-6.0
-7.0
-8.0
0
2
4
6
8
10
12
TEMP [C]
SETTING
1802
1804
FUTURE

FIG. 18

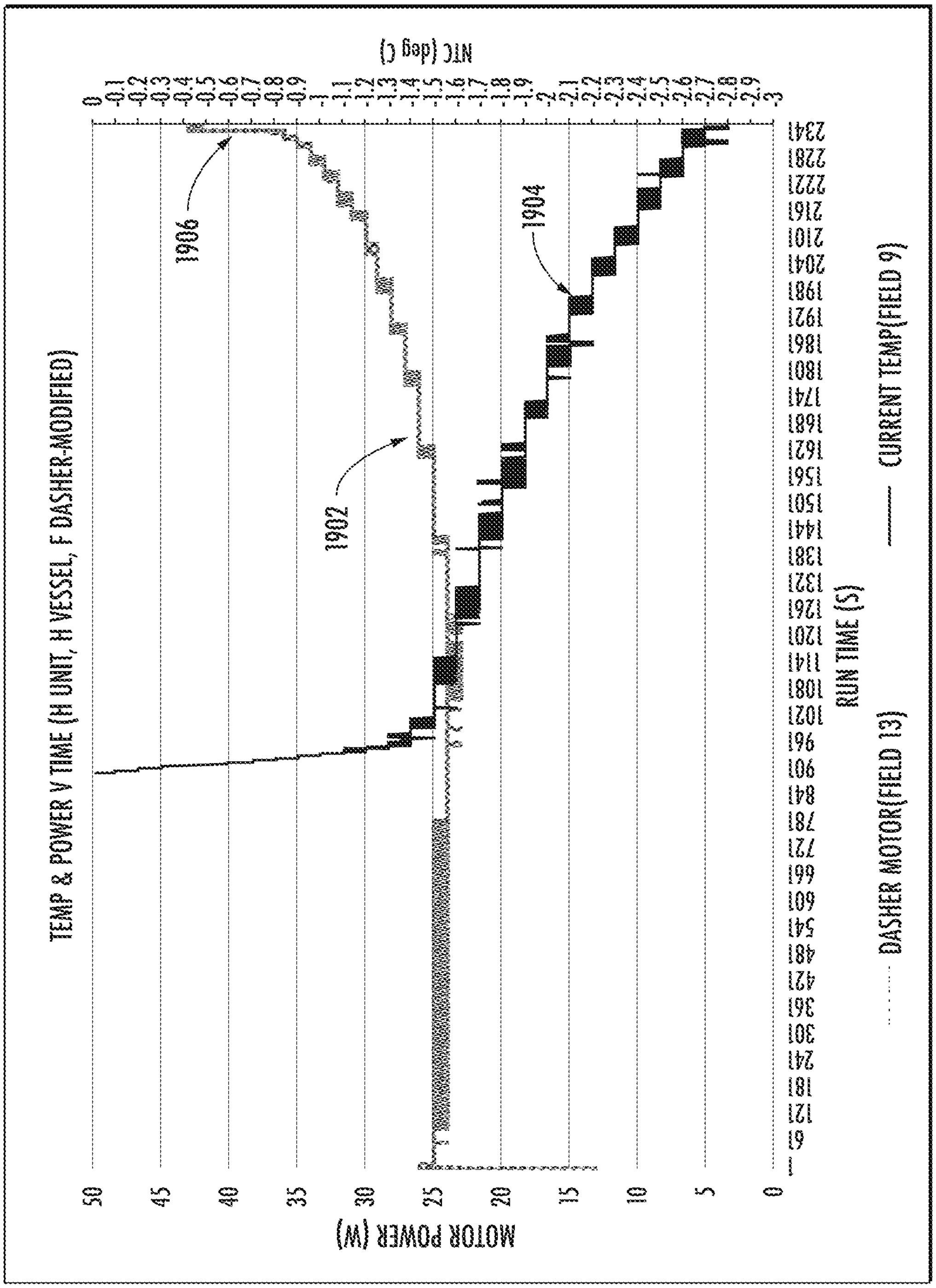

TEMP & POWER V TIME (H UNIT, H VESSEL, F DASHER-MODIFIED)
MOTOR POWER (W)
NTC (deg C)
RUN TIME (S)
1902
1904
1906
DASHER MOTOR(FIELD 13)
CURRENT TEMP(FIELD 9)

FIG. 19

1900

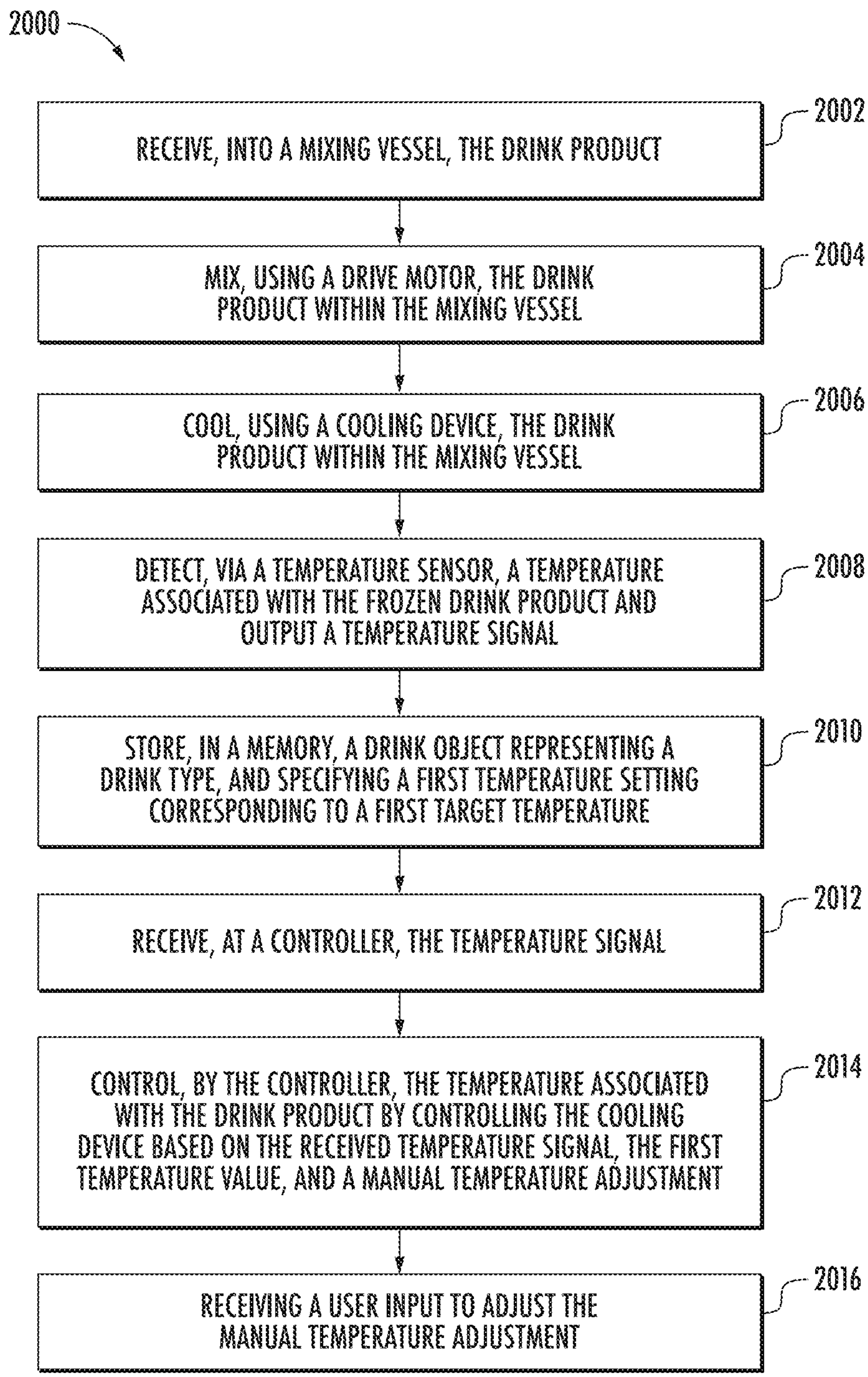

2000
RECEIVE, INTO A MIXING VESSEL, THE DRINK PRODUCT
2002
MIX, USING A DRIVE MOTOR, THE DRINK PRODUCT WITHIN THE MIXING VESSEL
2004
COOL, USING A COOLING DEVICE, THE DRINK PRODUCT WITHIN THE MIXING VESSEL
2006
DETECT, VIA A TEMPERATURE SENSOR, A TEMPERATURE ASSOCIATED WITH THE FROZEN DRINK PRODUCT AND OUTPUT A TEMPERATURE SIGNAL
2008
STORE, IN A MEMORY, A DRINK OBJECT REPRESENTING A DRINK TYPE, AND SPECIFYING A FIRST TEMPERATURE SETTING CORRESPONDING TO A FIRST TARGET TEMPERATURE
2010
RECEIVE, AT A CONTROLLER, THE TEMPERATURE SIGNAL
2012
CONTROL, BY THE CONTROLLER, THE TEMPERATURE ASSOCIATED WITH THE DRINK PRODUCT BY CONTROLLING THE COOLING DEVICE BASED ON THE RECEIVED TEMPERATURE SIGNAL, THE FIRST TEMPERATURE VALUE, AND A MANUAL TEMPERATURE ADJUSTMENT
2014
RECEIVING A USER INPUT TO ADJUST THE MANUAL TEMPERATURE ADJUSTMENT
2016

FIG. 20

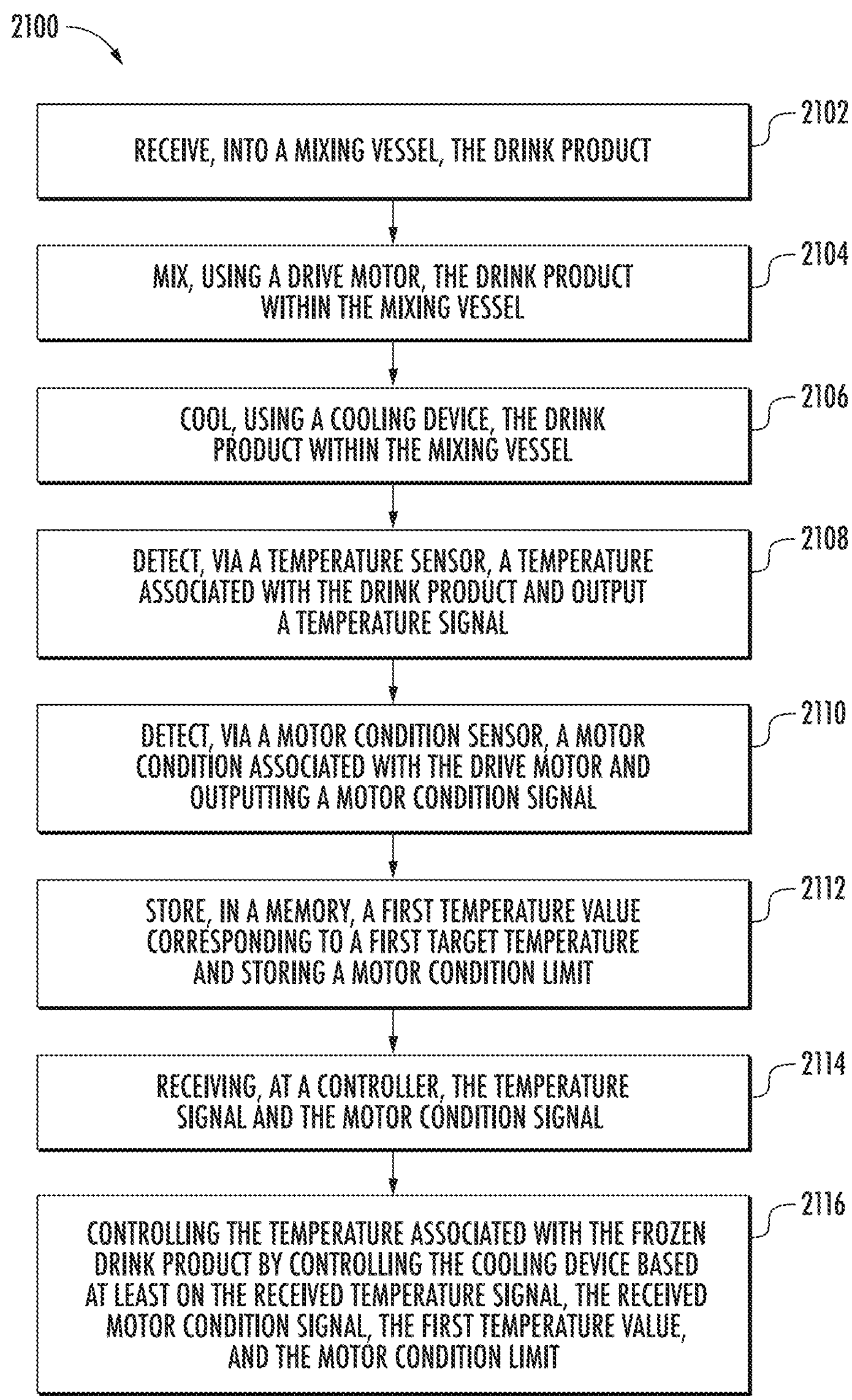

2100
RECEIVE, INTO A MIXING VESSEL, THE DRINK PRODUCT
2102
MIX, USING A DRIVE MOTOR, THE DRINK PRODUCT WITHIN THE MIXING VESSEL
2104
COOL, USING A COOLING DEVICE, THE DRINK PRODUCT WITHIN THE MIXING VESSEL
2106
DETECT, VIA A TEMPERATURE SENSOR, A TEMPERATURE ASSOCIATED WITH THE DRINK PRODUCT AND OUTPUT A TEMPERATURE SIGNAL
2108
DETECT, VIA A MOTOR CONDITION SENSOR, A MOTOR CONDITION ASSOCIATED WITH THE DRIVE MOTOR AND OUTPUTTING A MOTOR CONDITION SIGNAL
2110
STORE, IN A MEMORY, A FIRST TEMPERATURE VALUE CORRESPONDING TO A FIRST TARGET TEMPERATURE AND STORING A MOTOR CONDITION LIMIT
2112
RECEIVING, AT A CONTROLLER, THE TEMPERATURE SIGNAL AND THE MOTOR CONDITION SIGNAL
2114
CONTROLLING THE TEMPERATURE ASSOCIATED WITH THE FROZEN DRINK PRODUCT BY CONTROLLING THE COOLING DEVICE BASED AT LEAST ON THE RECEIVED TEMPERATURE SIGNAL, THE RECEIVED MOTOR CONDITION SIGNAL, THE FIRST TEMPERATURE VALUE, AND THE MOTOR CONDITION LIMIT
2116

FIG. 21

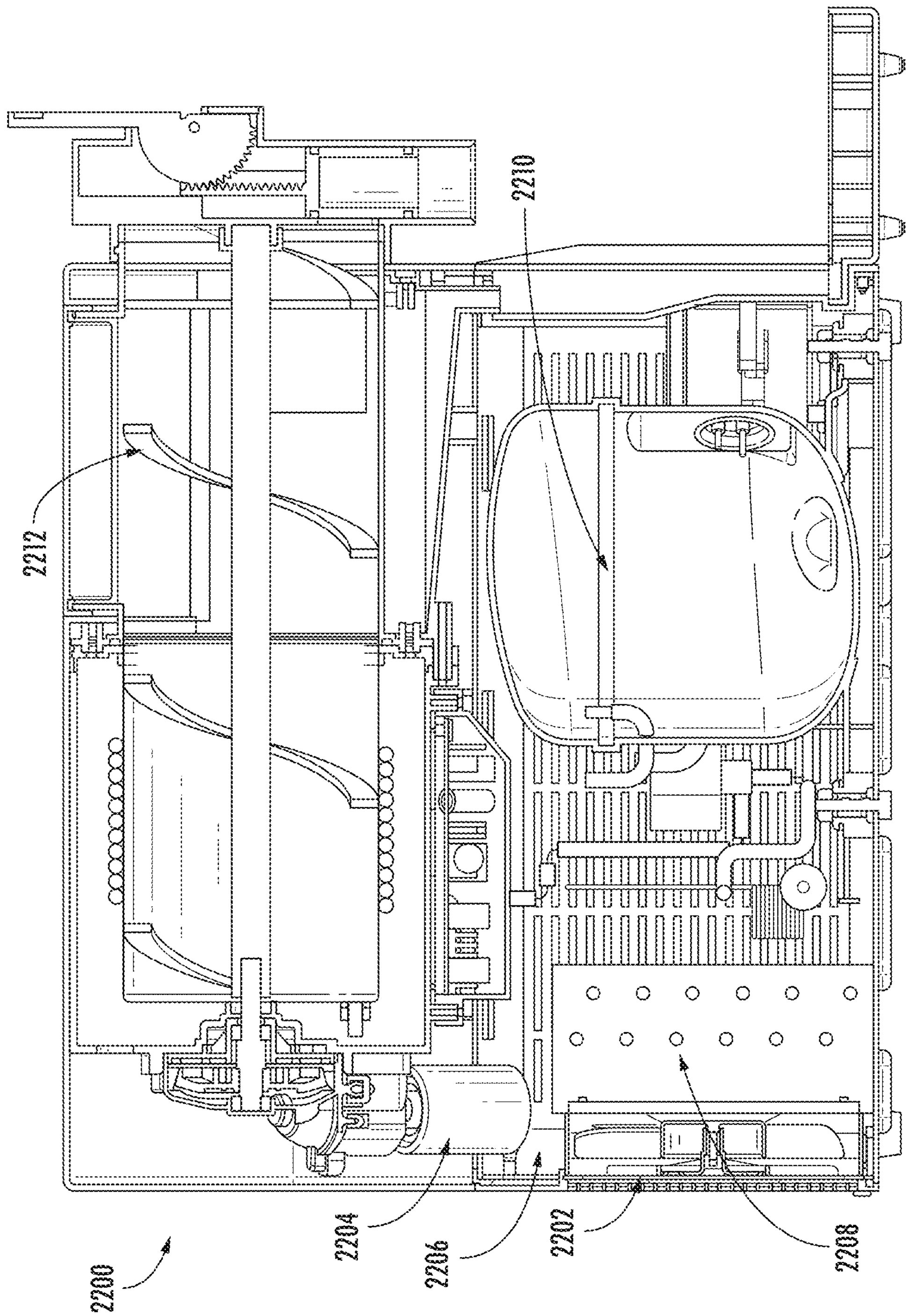


FIG. 22A

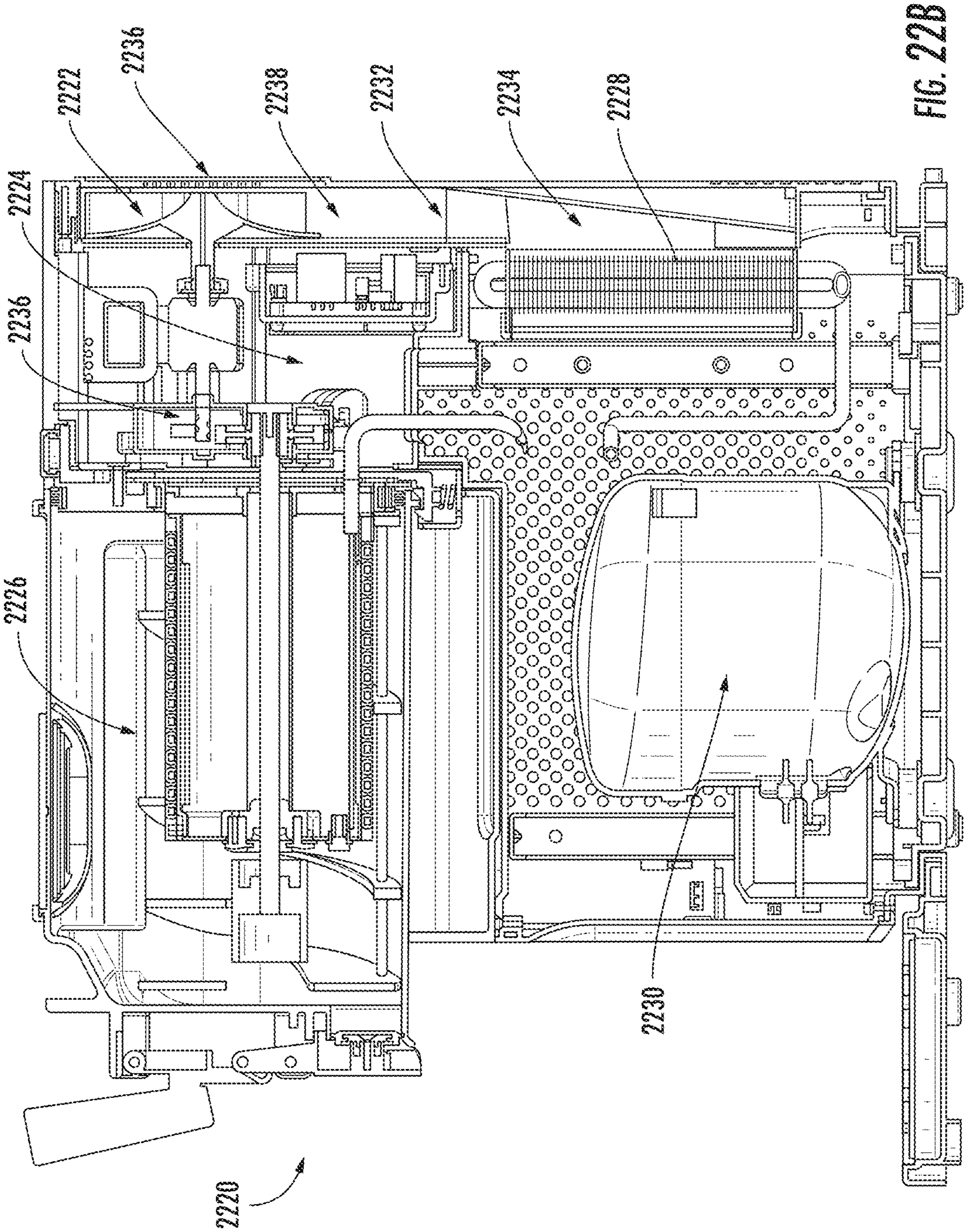


FIG. 22B

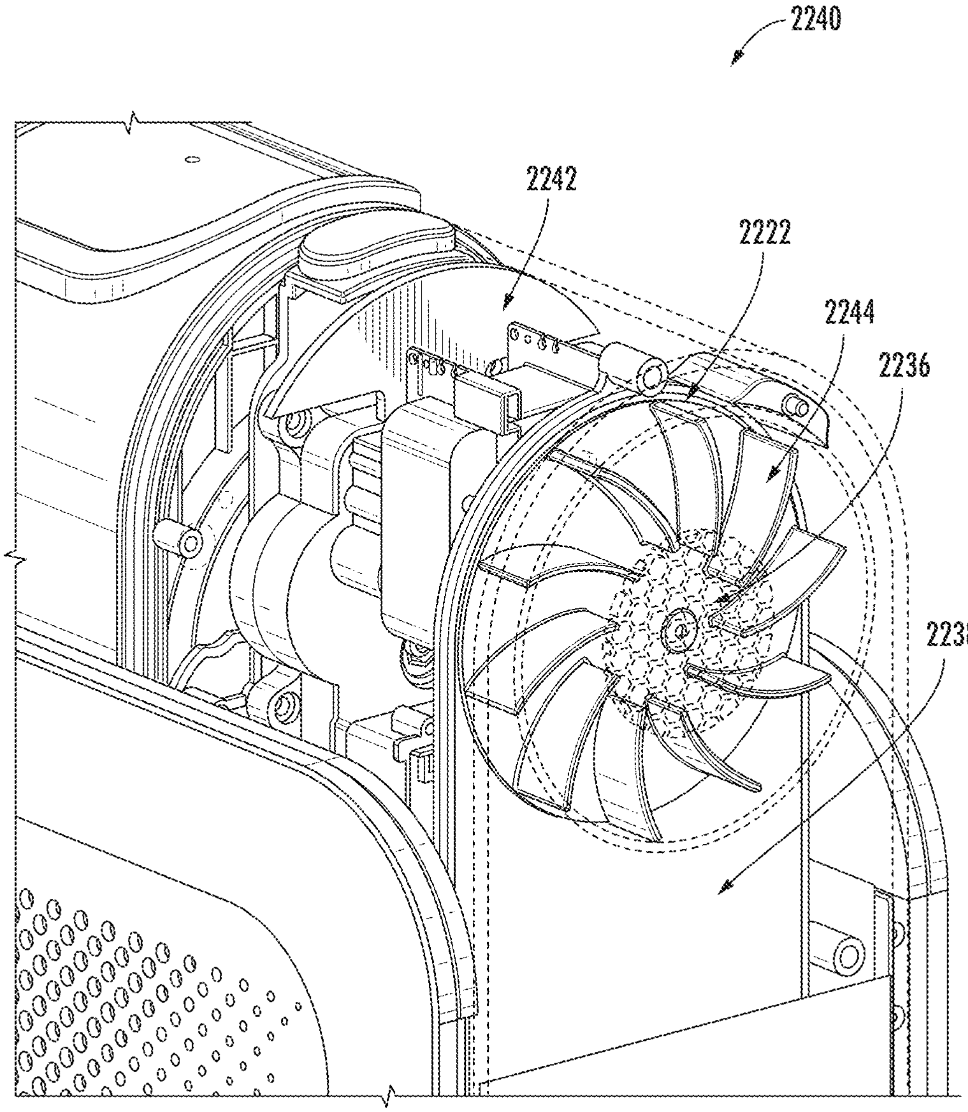


FIG. 22C

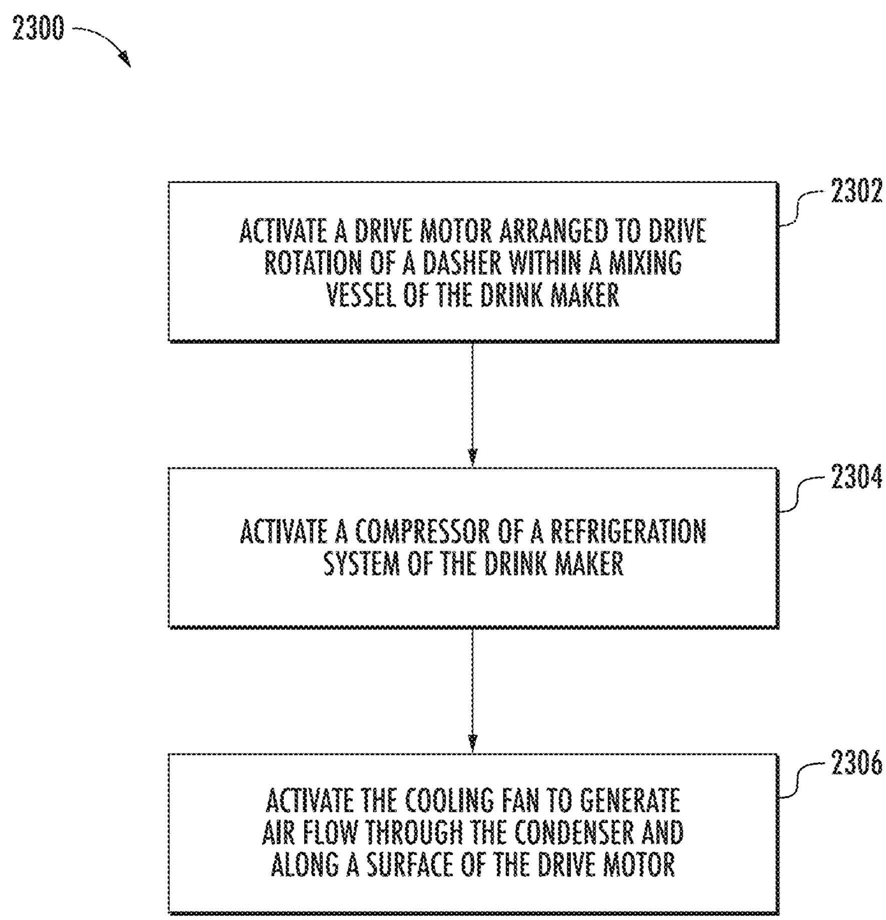

2300
ACTIVATE A DRIVE MOTOR ARRANGED TO DRIVE ROTATION OF A DASHER WITHIN A MIXING VESSEL OF THE DRINK MAKER
2302
ACTIVATE A COMPRESSOR OF A REFRIGERATION SYSTEM OF THE DRINK MAKER
2304
ACTIVATE THE COOLING FAN TO GENERATE AIR FLOW THROUGH THE CONDENSER AND ALONG A SURFACE OF THE DRIVE MOTOR
2306

FIG. 23

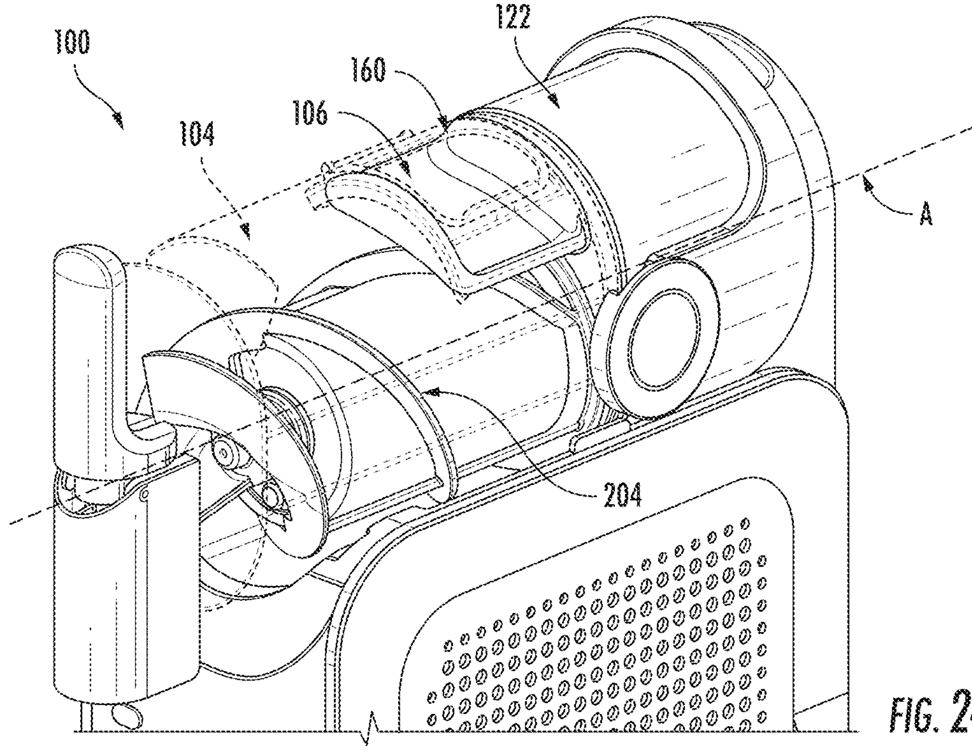


FIG. 24A

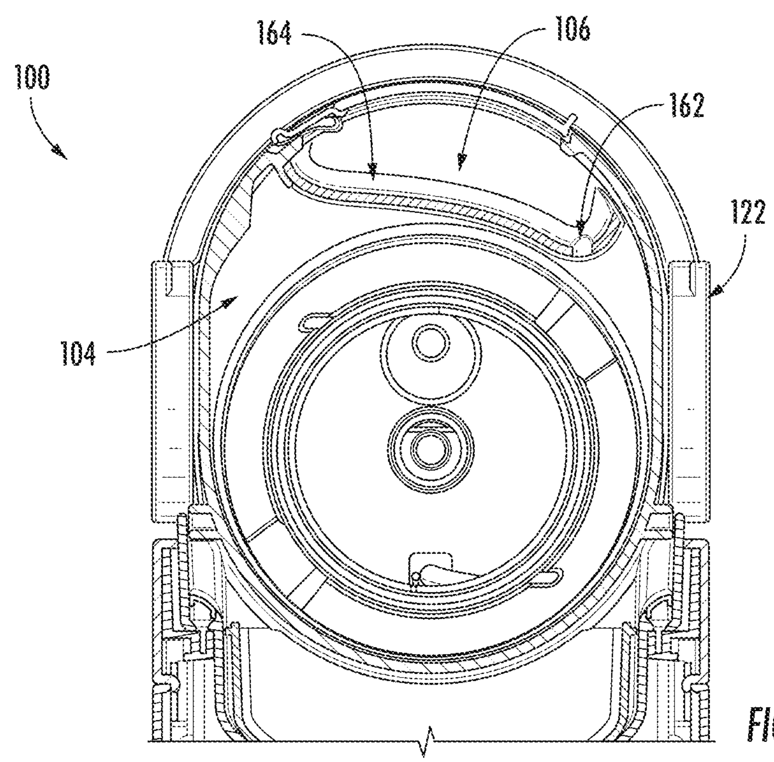


FIG. 24B

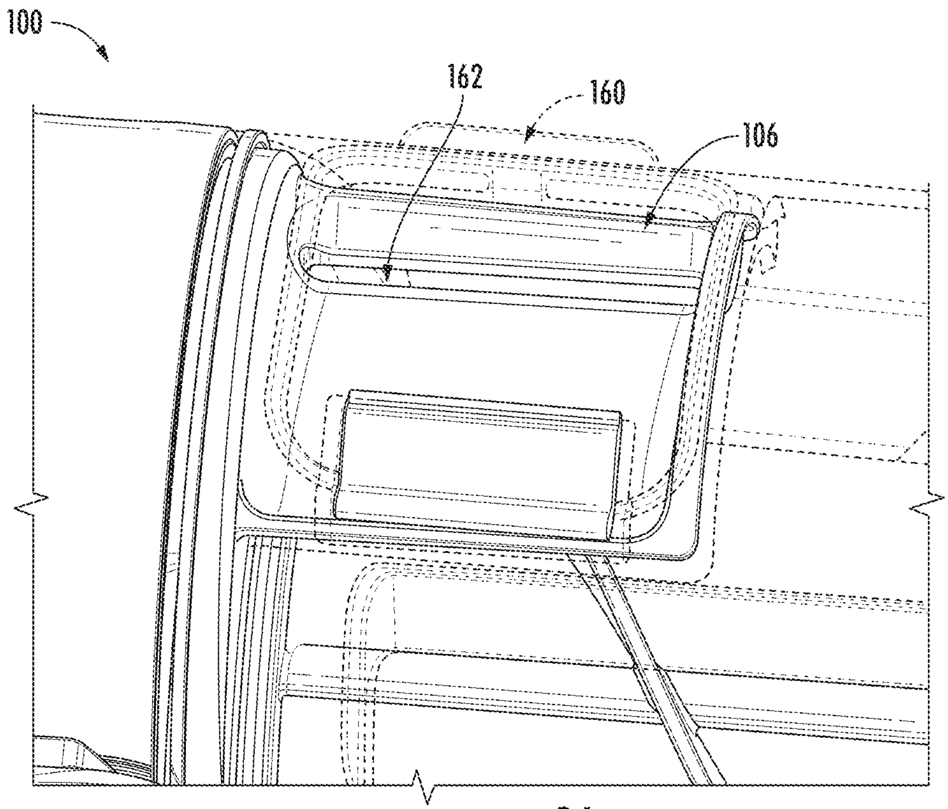


FIG. 24C

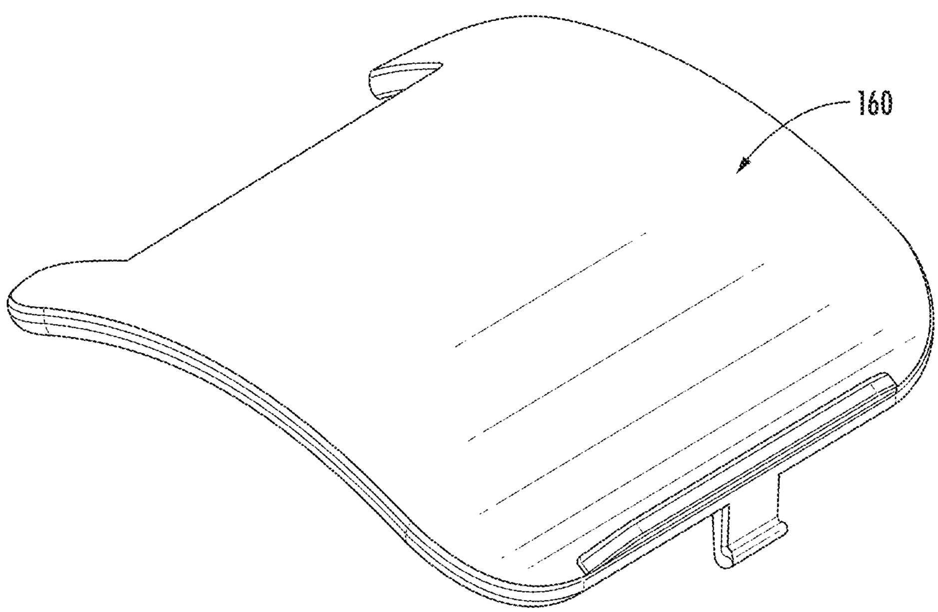


FIG. 25

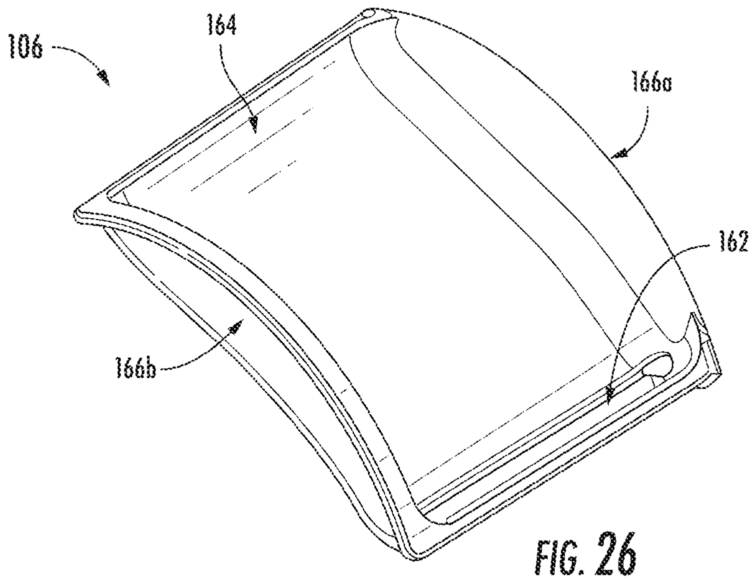


FIG. 26

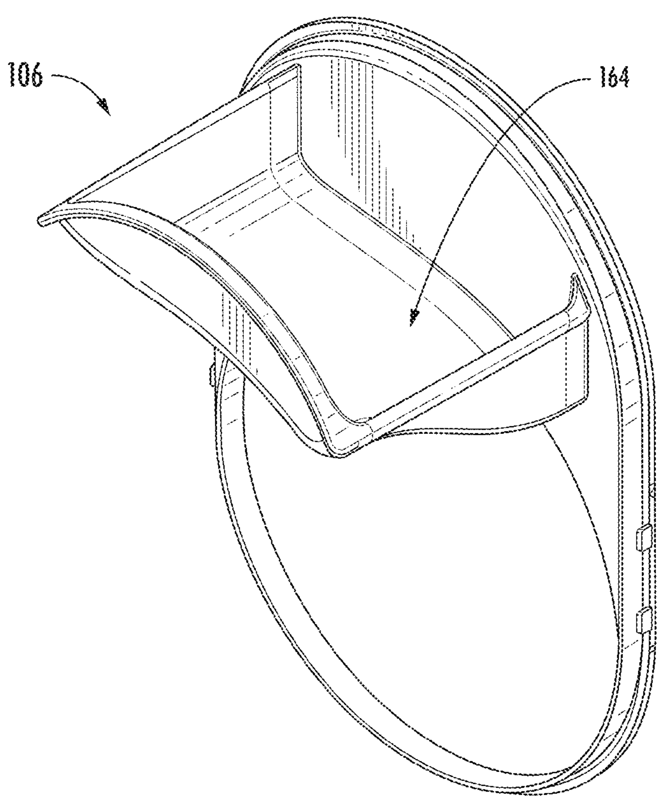


FIG. 27A

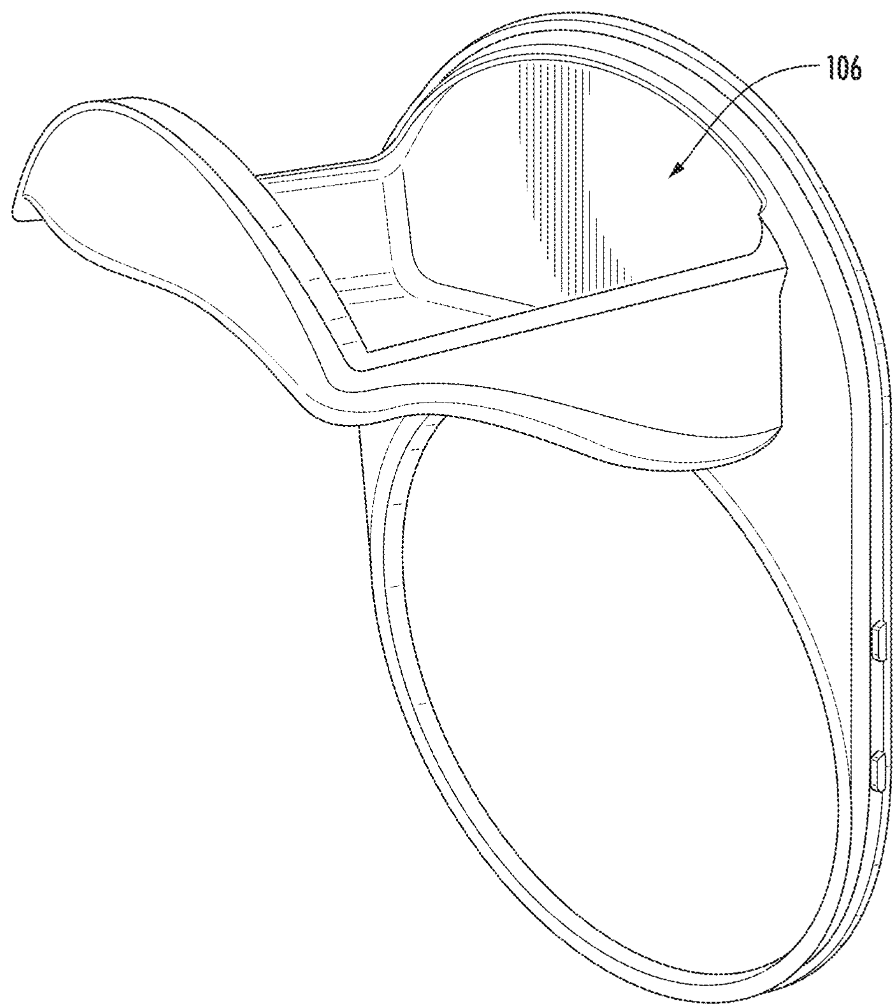


FIG. 27B

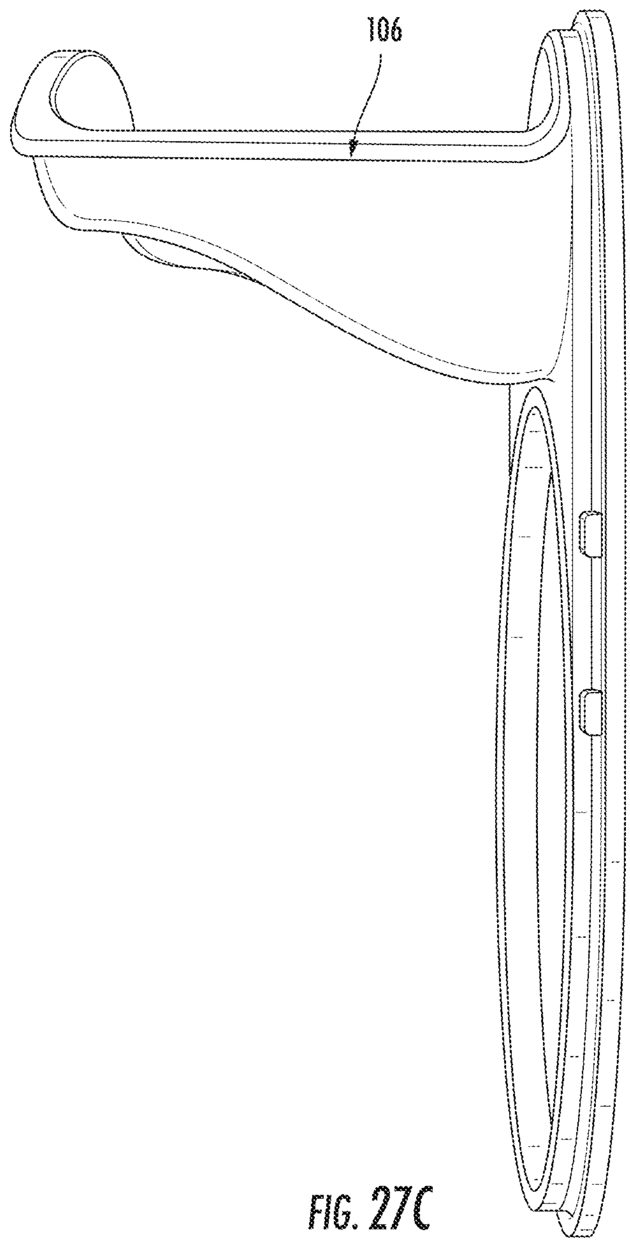


FIG. 27C

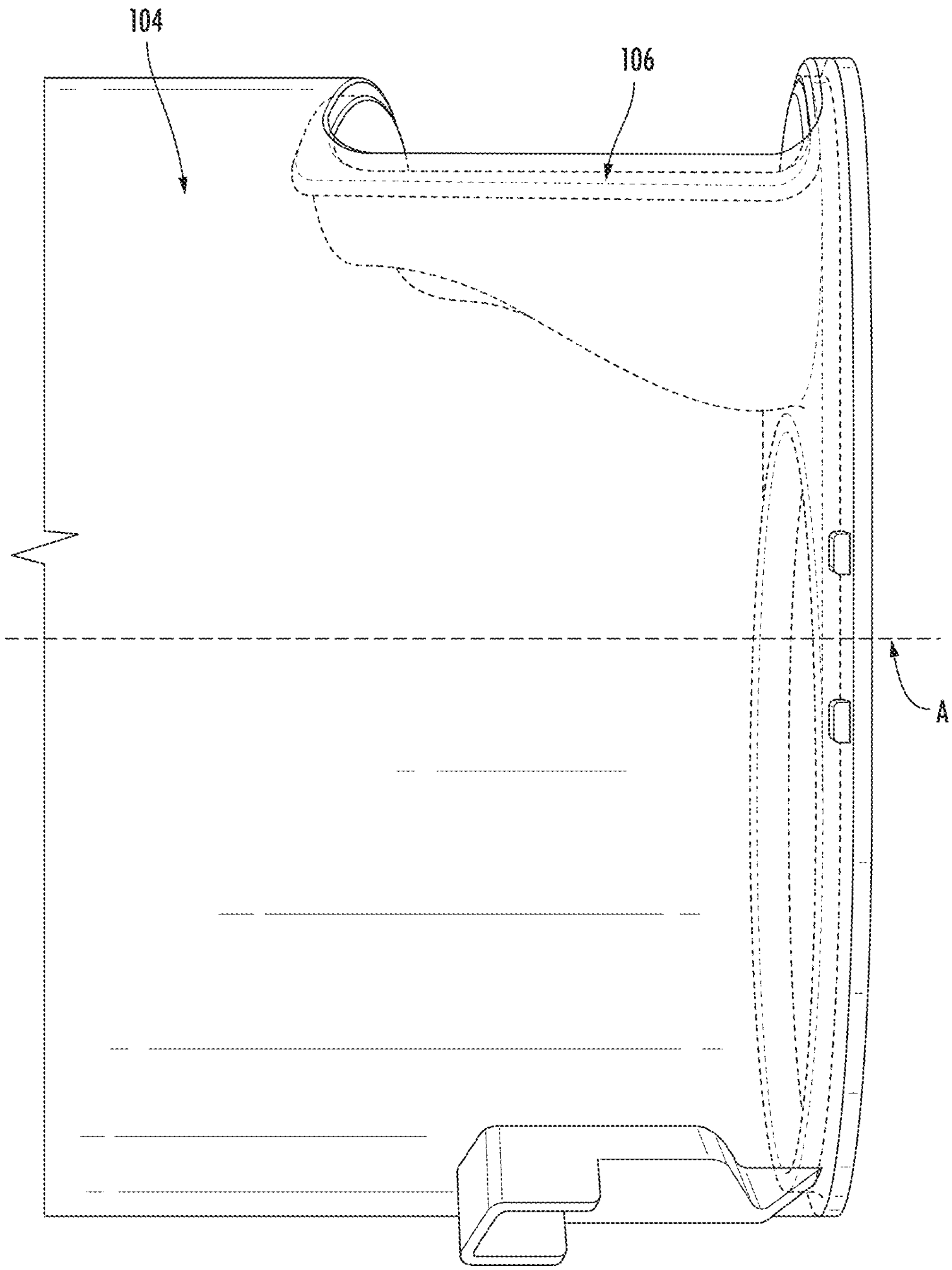


FIG. 27D

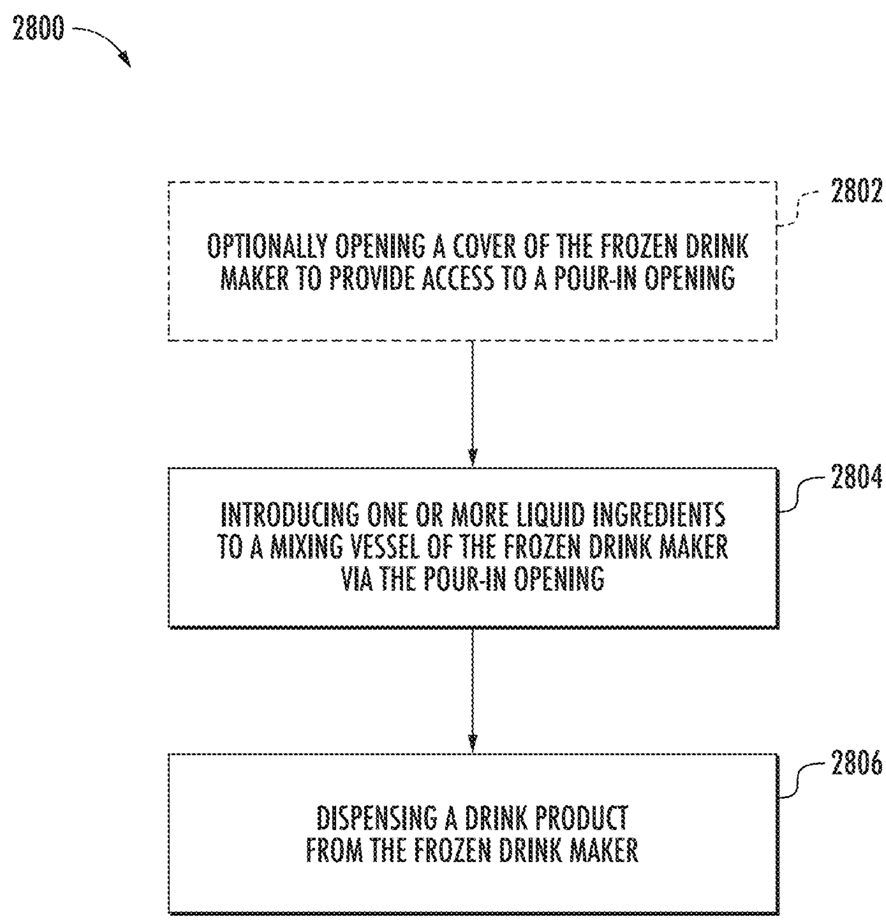

2800
OPTIONALLY OPENING A COVER OF THE FROZEN DRINK MAKER TO PROVIDE ACCESS TO A POUR-IN OPENING
2802
INTRODUCING ONE OR MORE LIQUID INGREDIENTS TO A MIXING VESSEL OF THE FROZEN DRINK MAKER VIA THE POUR-IN OPENING
2804
DISPENSING A DRINK PRODUCT FROM THE FROZEN DRINK MAKER
2806

FIG. 28

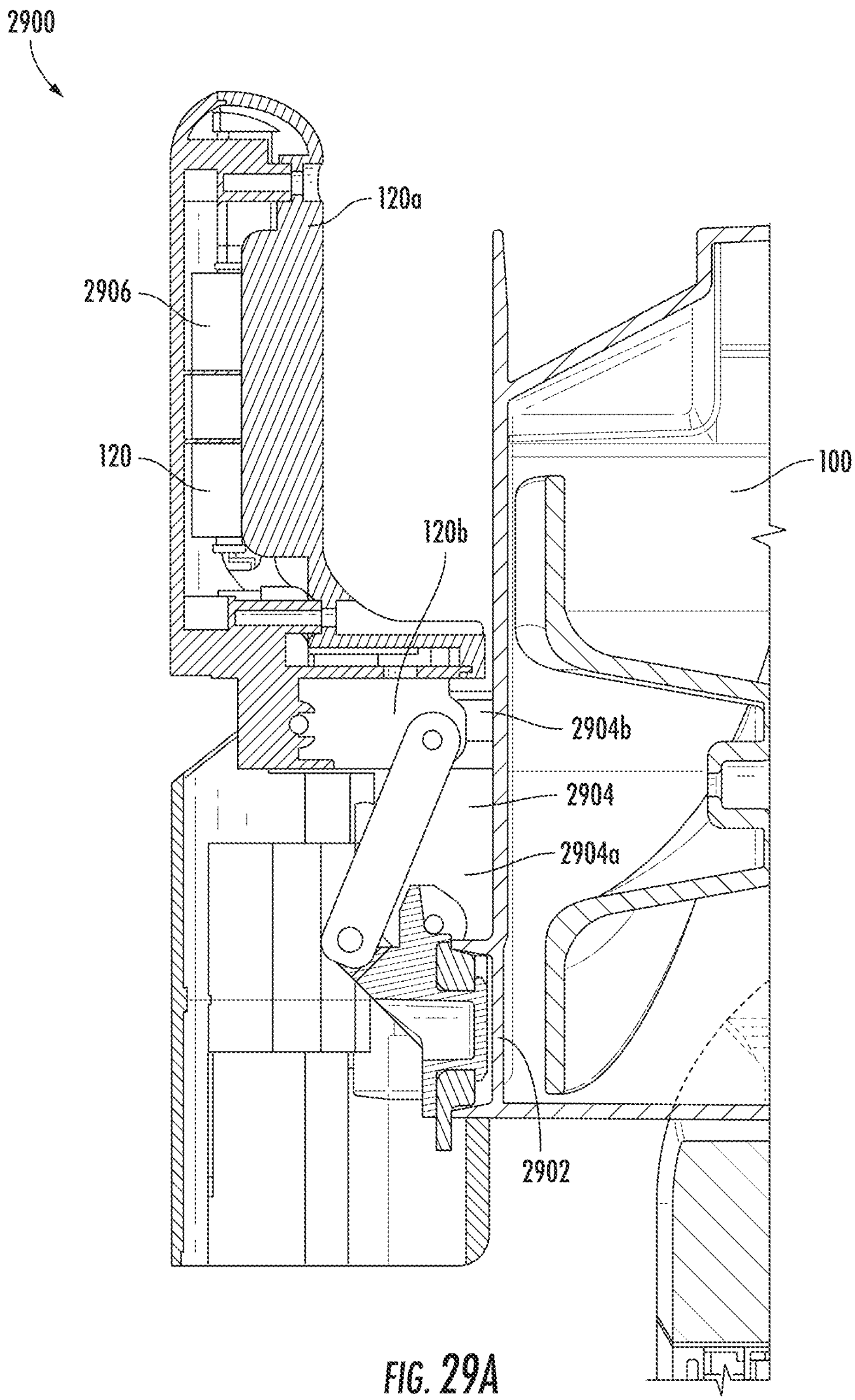
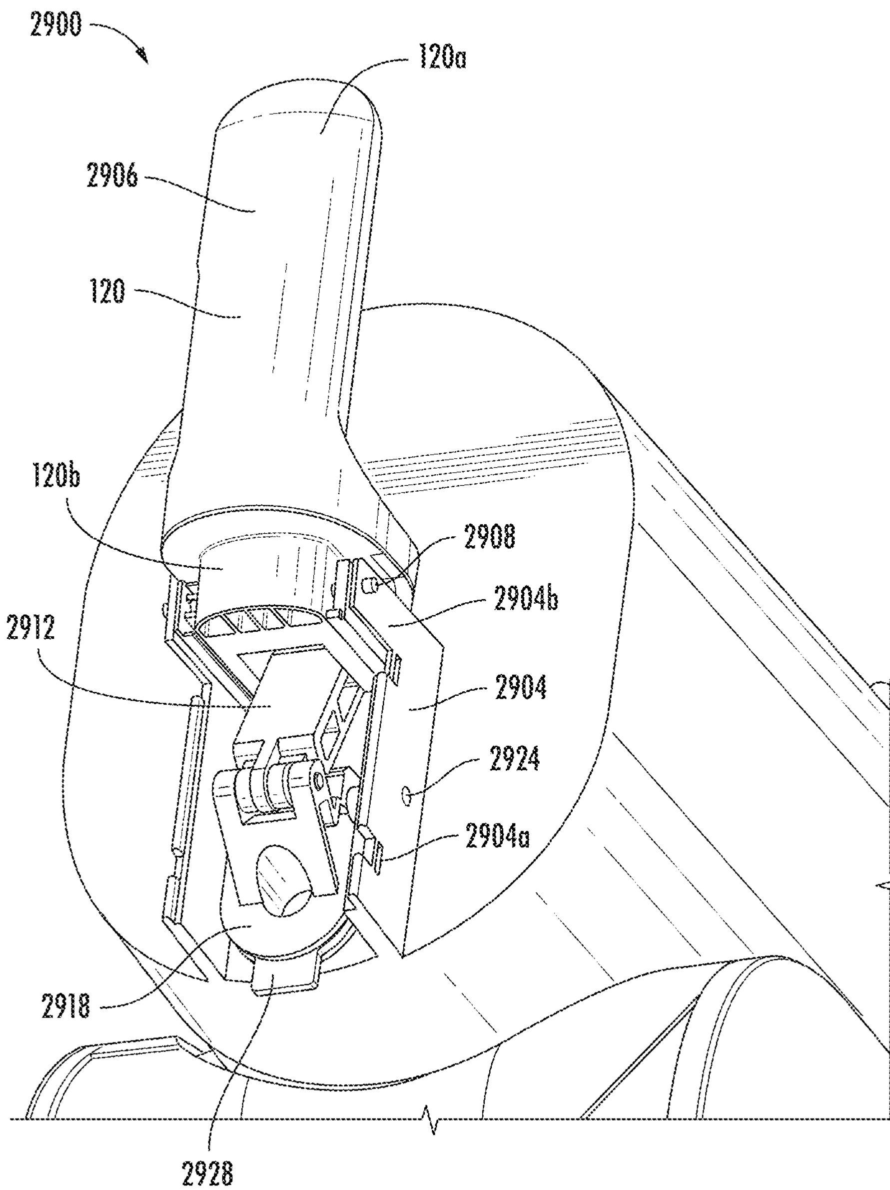


FIG. 29A



2900
120a
2906
120
120b
2908
2904b
2912
2904
2924
2904a
2918
2928

FIG. 29B

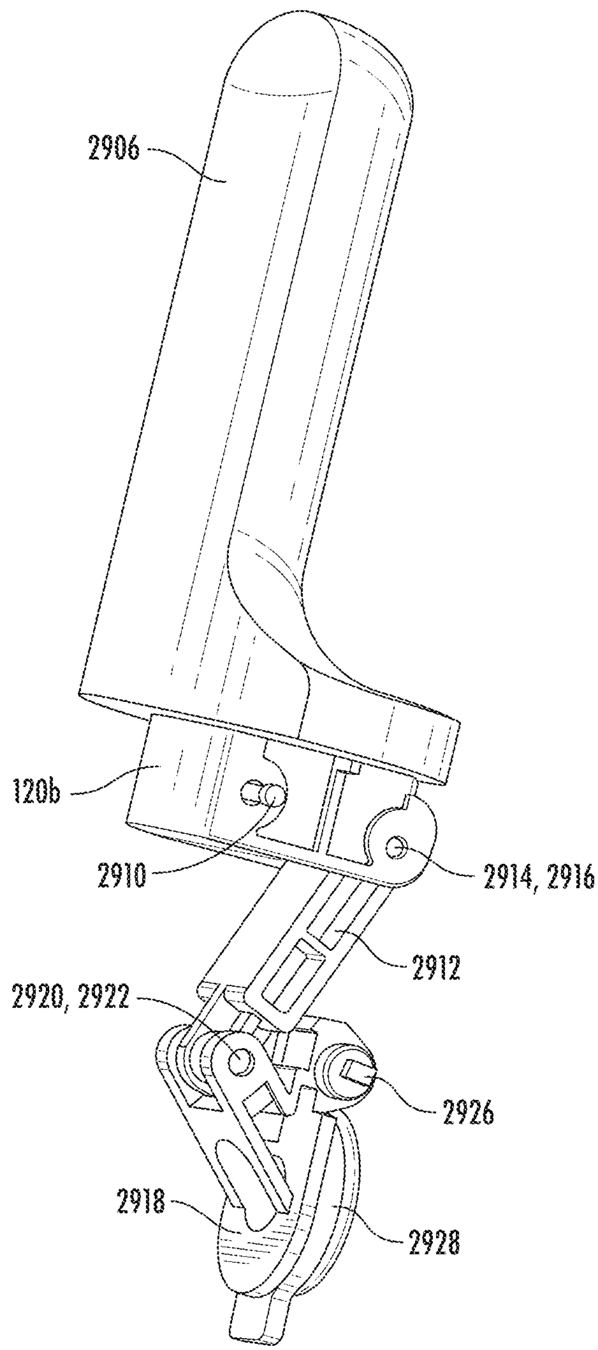


FIG. 29C

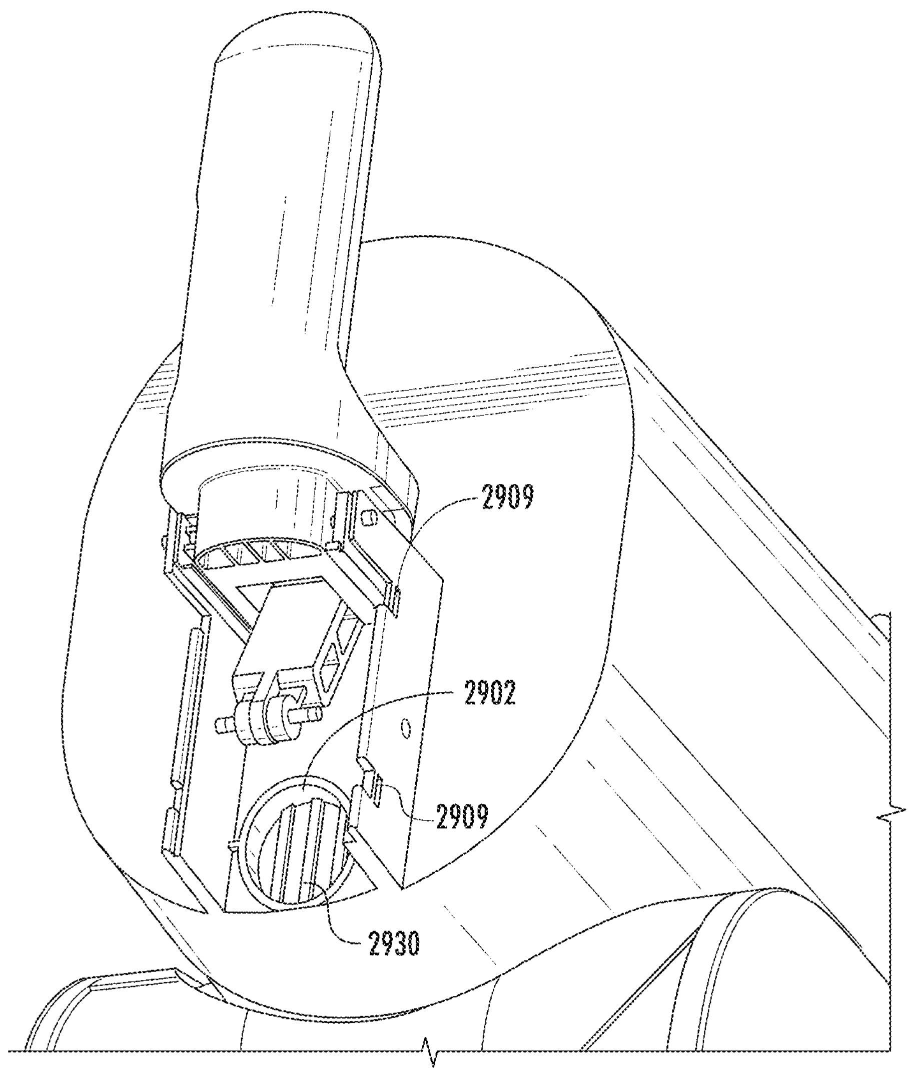
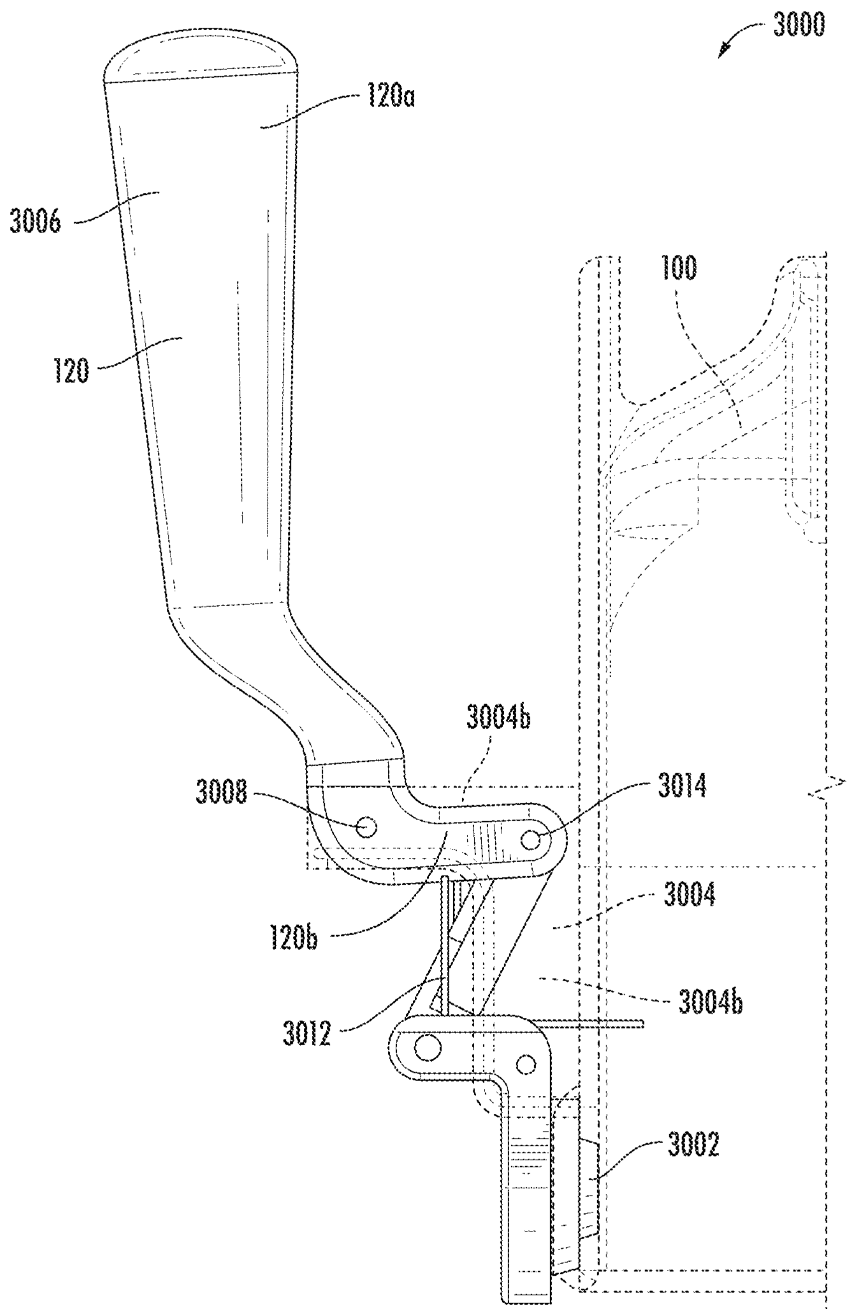


FIG. 29D


3000
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3006
100
120
3004b
3008
3014
3004
120b
3004b
3012
3002

FIG. 30A

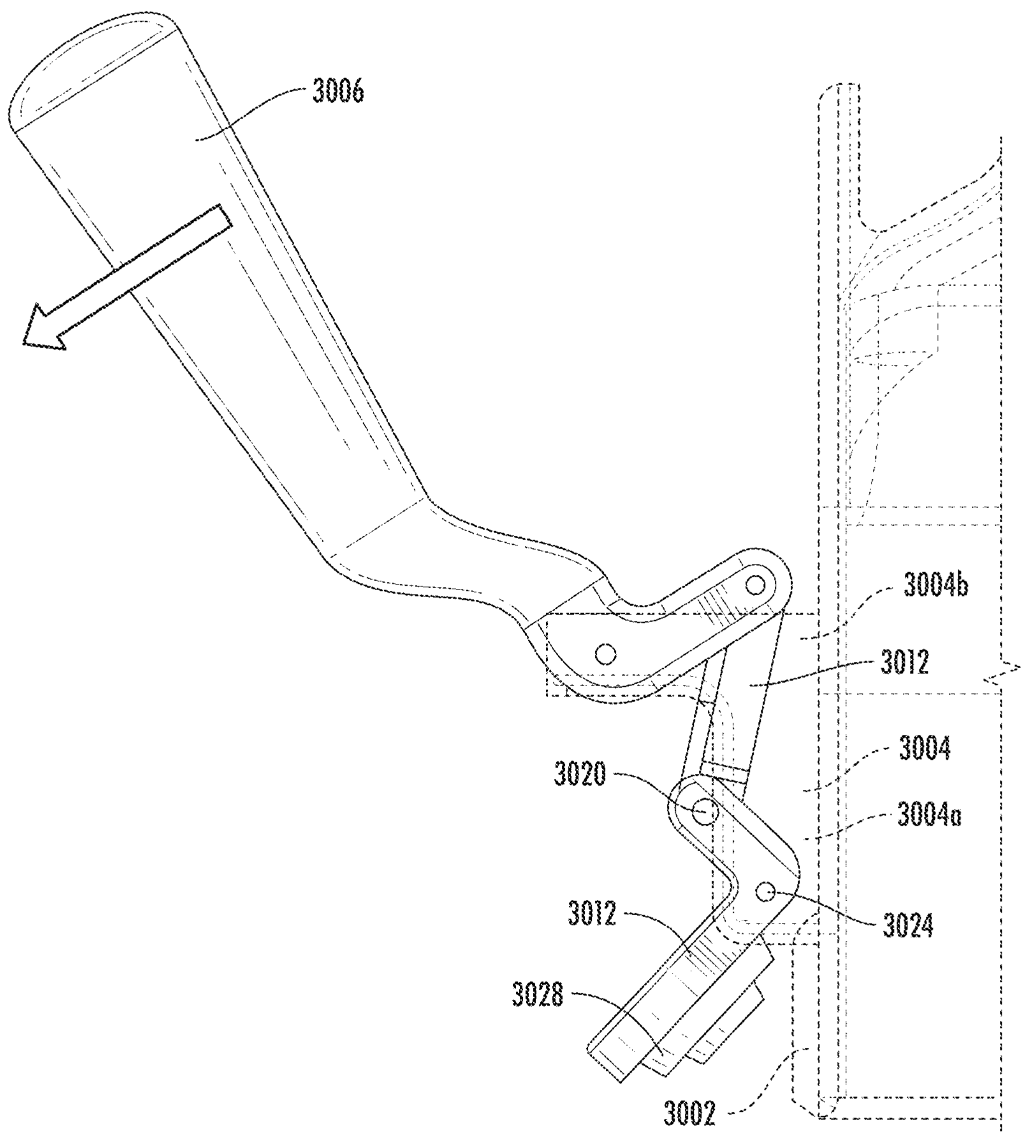


FIG. 30B

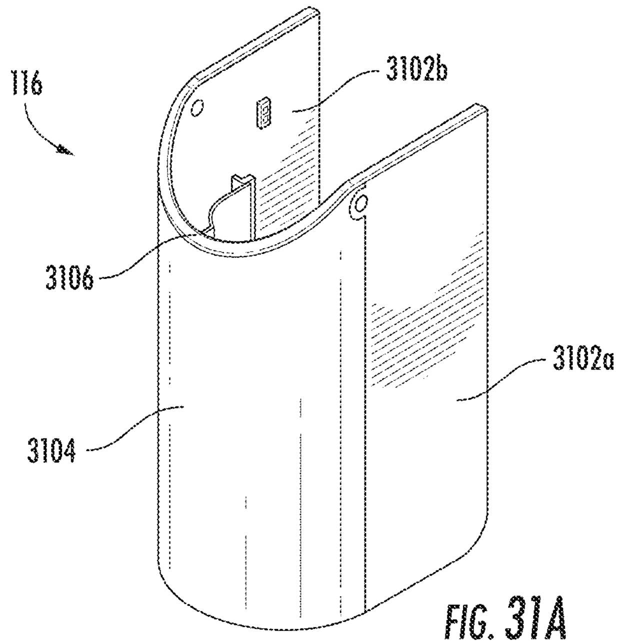


FIG. 31A

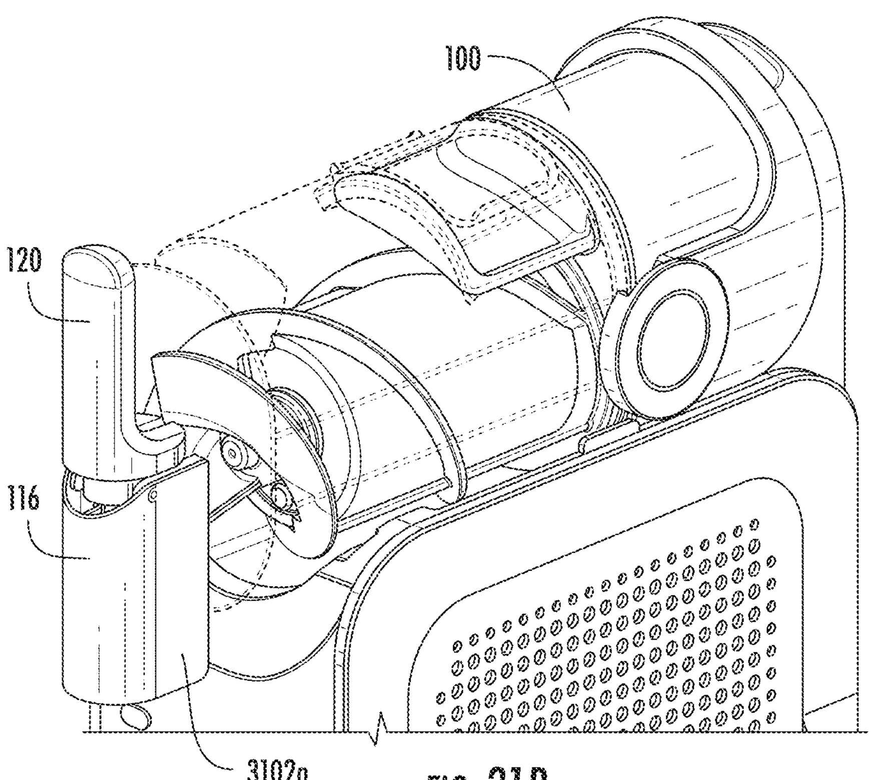


FIG. 31B

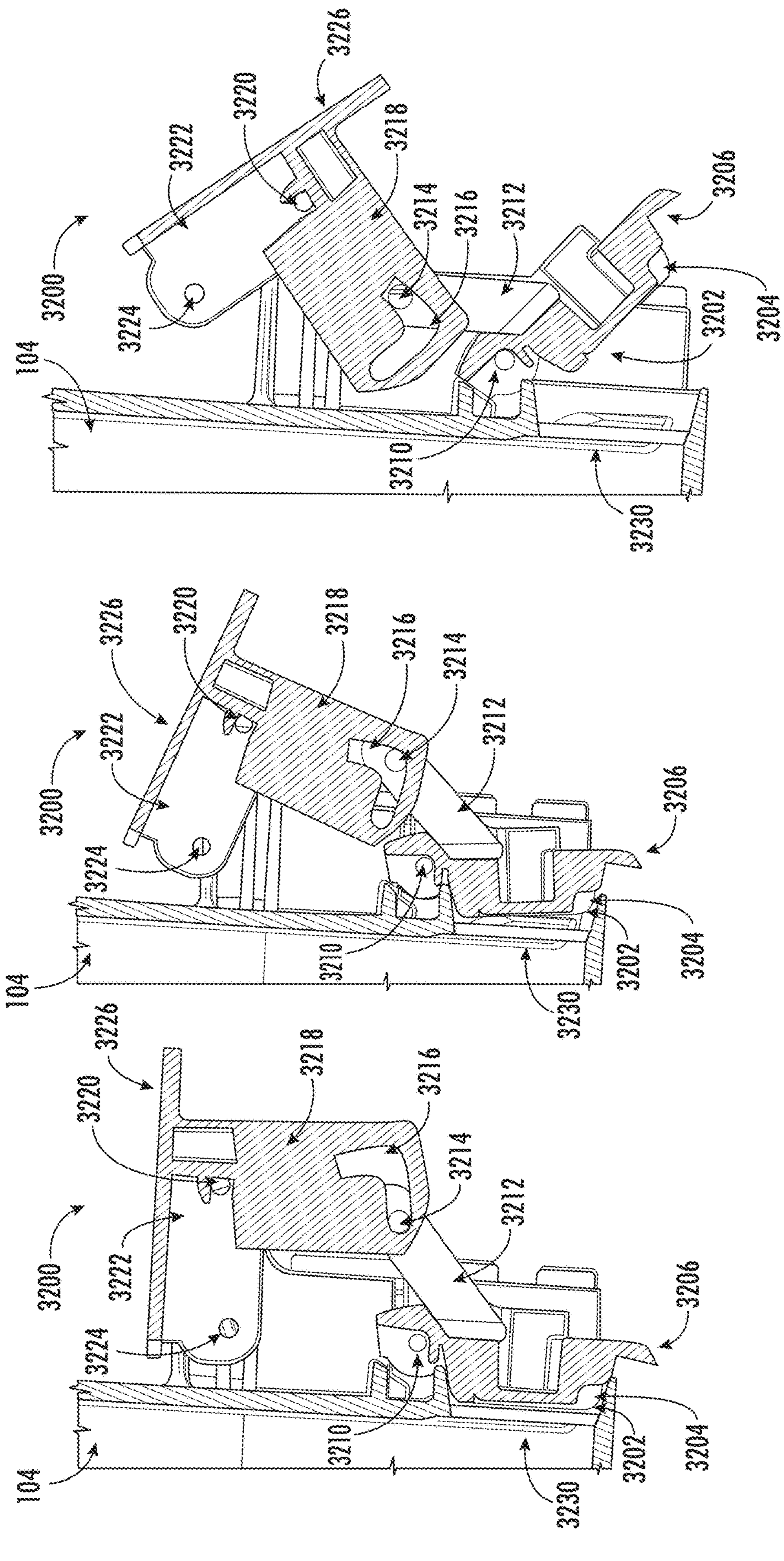


FIG. 32A

FIG. 32B

FIG. 32C

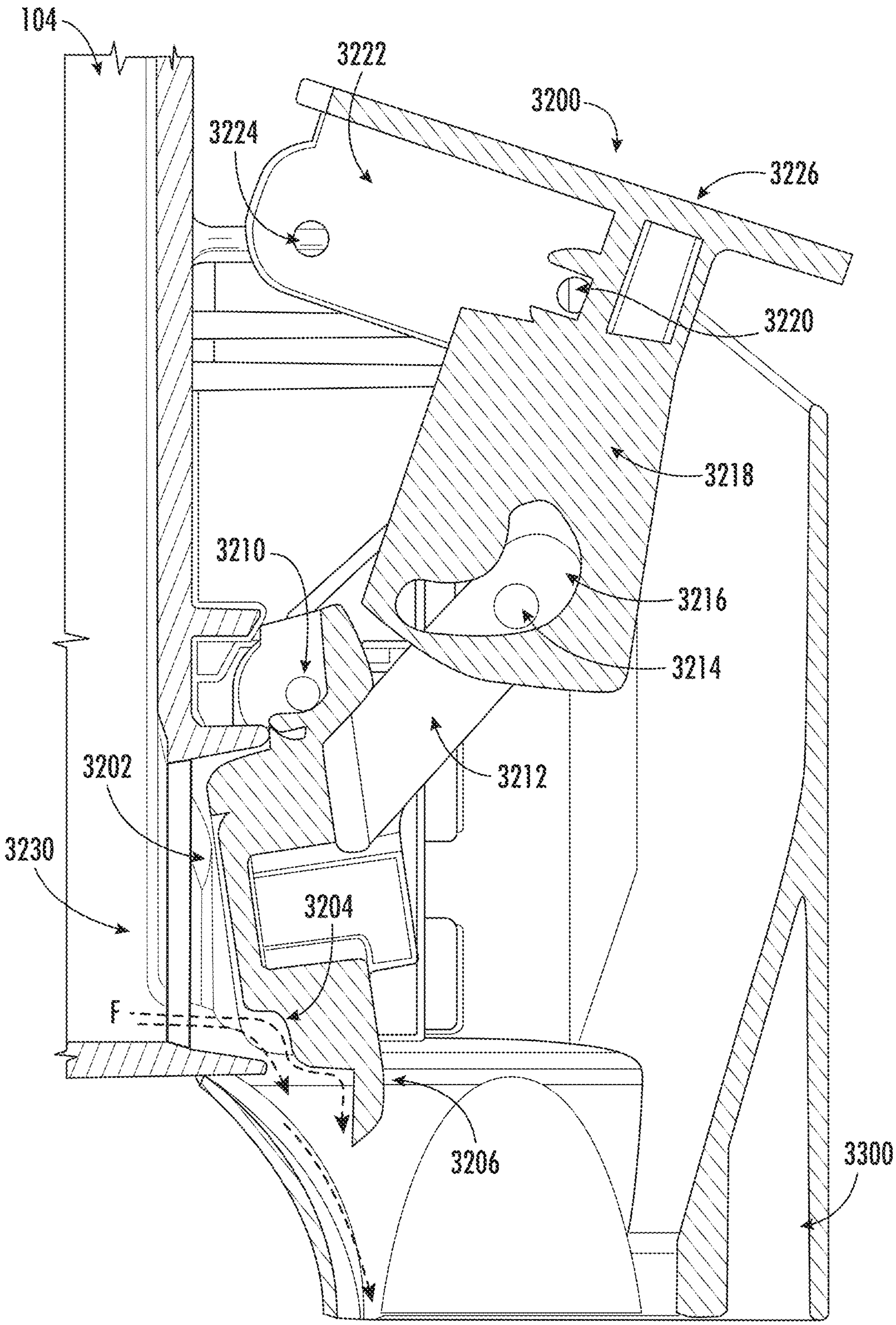

104
3222
3200
3224
3226
3220
3218
3210
3216
3214
3212
3202
3230
3204
F
3206
3300

FIG. 33

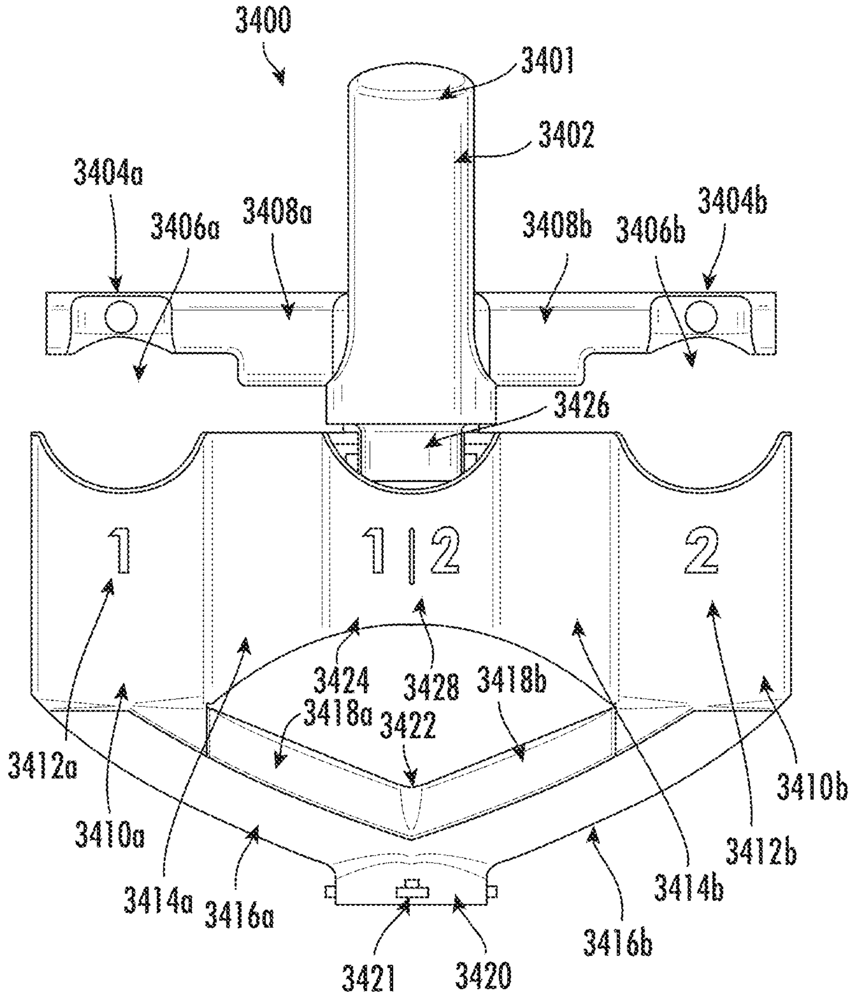
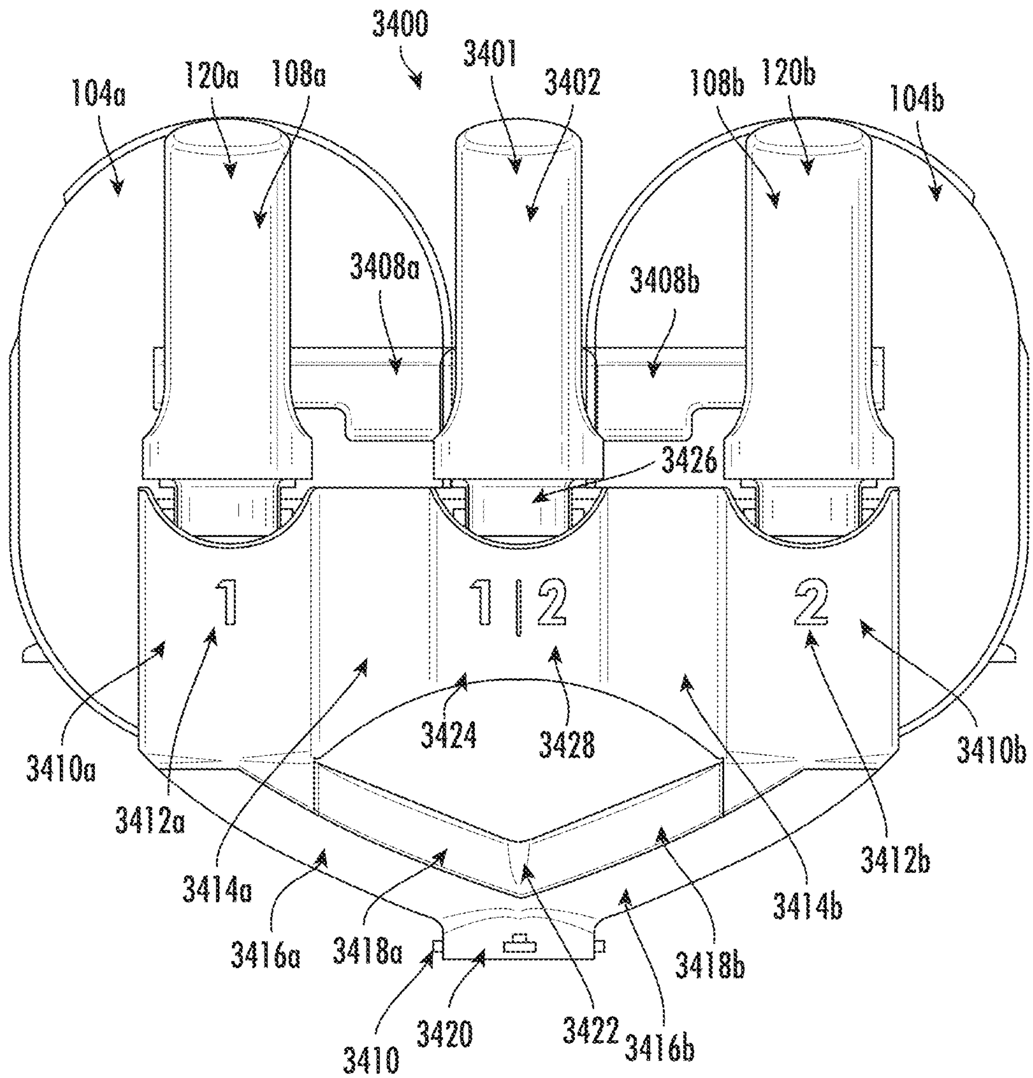

3400
3401
3402
3404a
3406a
3408a
3408b
3406b
3404b
3426
1
1 | 2
2
3424
3428
3418b
3418a
3422
3412a
3410a
3414a
3416a
3421
3420
3416b
3414b
3412b
3410b

FIG. 34


3400
3401
3402
104a
120a
108a
108b
120b
104b
3408a
3408b
3426
1
1|2
2
3410a
3412a
3414a
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3428
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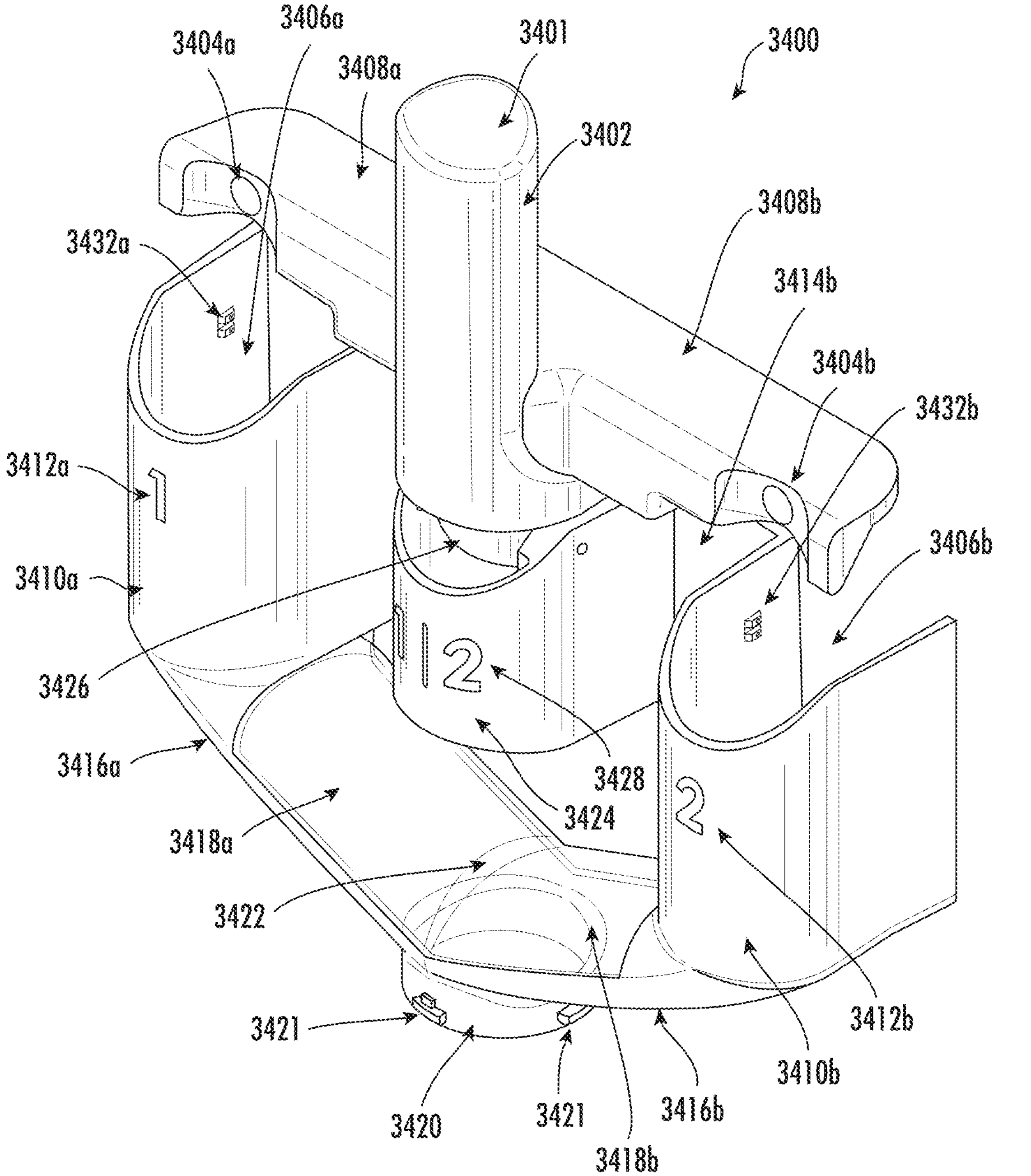
3412b
3414b
3416a
3418a
3418b
3410
3420
3422
3416b

FIG. 35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3408b
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3404b
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3428
3416a
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3418a
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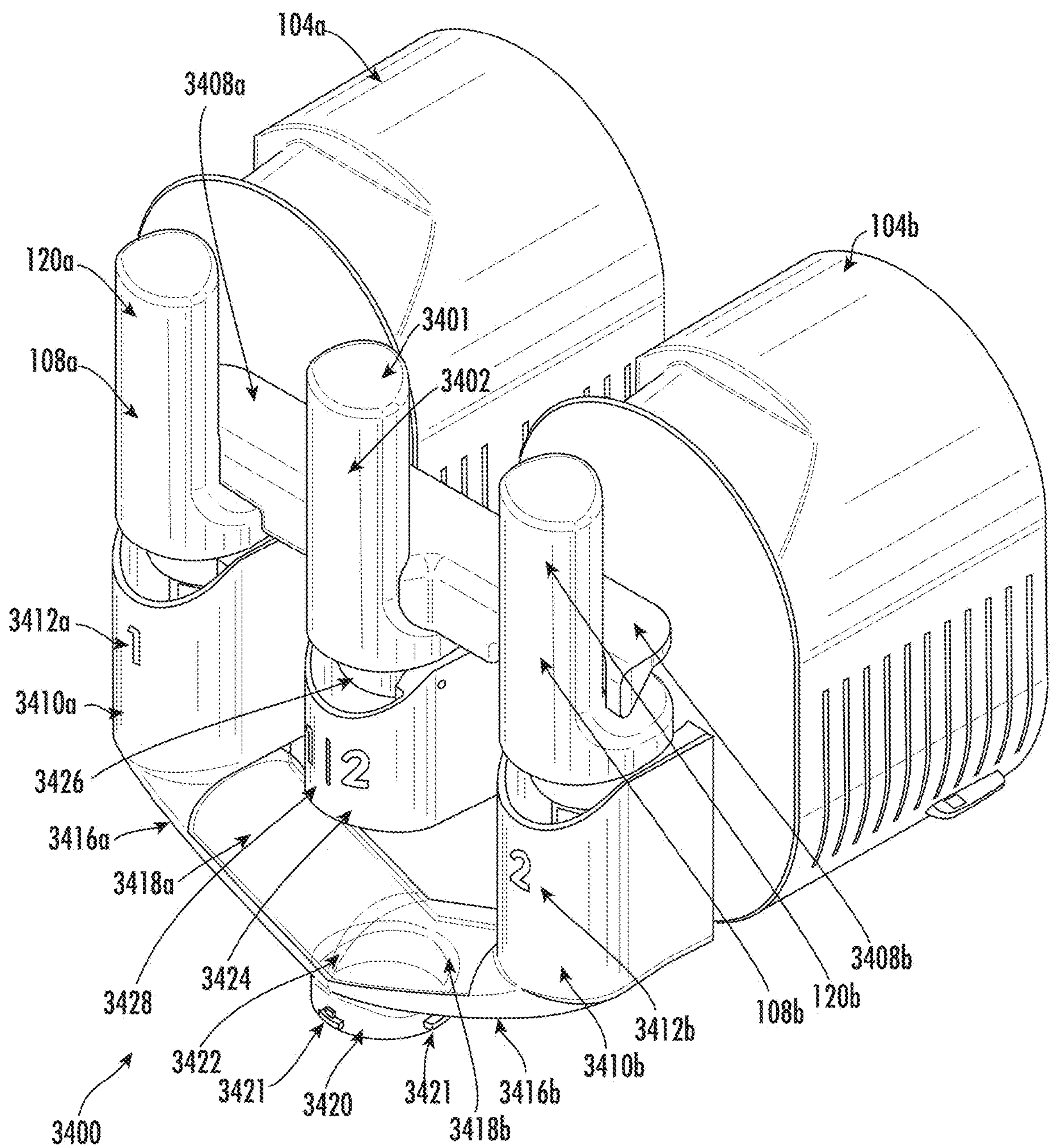
3412b
3421
3410b
3420
3421
3416b
3418b
1
2
2

FIG. 36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120a
108a
3401
3402
104b
3412a
3410a
3426
3416a
3418a
3428
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3422
3421
3420
3421
3416b
3418b
3400
3410b
3412b
108b
120b
3408b
1
12
2

FIG. 37

SHARED OUTPUT CONNECTOR ASSEMBLY FOR TWO DRINK MAKER DISPENSER ASSEMBLIES

BACKGROUND

1. Technical Field

The present disclosure relates to drink makers and, more particularly, to a shared output connector assembly for two dispenser assemblies of one or more drink makers.

2. Technical Considerations

Frozen drink makers, which may also be referred to as semi-frozen beverage makers, or crushed-ice drink makers typically include a transparent tank or mixing vessel in which a drink product is received and processed, including being cooled, often transforming the drink product from a pure liquid (or a combination of a liquid and portions of ice) to a frozen or semi-frozen product, such as, for example, a granita, slush drink, smoothie, ice cream, or other frozen or semi-frozen product, which is then dispensed. The cooled product is typically dispensed through a tap, spigot, or dispenser located at the front and near the bottom of the vessel. Thus, the term "frozen drink maker" as used herein is not limited to a device that only makes drinks or frozen drinks, but includes devices that cool received drink products to produce cooled outputs in any of a variety of frozen and semi-frozen forms. A drink product typically consists of a mixture of water, juice, or milk, and syrup, flavoring powders, or other additives that give the drink product the desired taste and color.

While a single drink maker may be configured to produce a homogenous, mixed drink product therein, a single drink maker may not be adequate to provide variances in the flavor, color, texture, thickness, ingredients, and the like in a dispensed drink product. Though it is possible to individually operate separate drink makers for different types of drink product output, it is inefficient and imprecise to move a receiving container from one drink maker's dispenser assembly to another, if it is desired to mix (e.g., layer, blend, etc.) multiple different drink products in a single container. Moving a receiving container between dispenser assemblies may lead to spilling while shuttling a container back and forth, cause waste from separately operated dispensers that may have slow-flowing drink product, and/or create time delay to allow drink product to fully dispense before moving the container to a new drink maker dispenser.

SUMMARY

Accordingly, provided is a shared output connector assembly for two dispenser assemblies of one or more drink makers.

According to some non-limiting embodiments or aspects, provided is a shared output connector assembly for connecting a first dispenser assembly and a second dispenser assembly. An example shared output connector assembly may include a primary lever and at least one projection configured to mechanically connect the primary lever to a first lever of the first dispenser assembly and a second lever of the second dispenser assembly. When the shared output connector assembly is attached to the first dispenser assembly and the second dispenser assembly, the first lever and the second lever may be independently actuatable. When the

primary lever is actuated, the at least one projection may cause the first lever and the second lever to actuate.

In some non-limiting embodiments or aspects, a first spout cover may be configured to be mounted over at least part of the first dispenser assembly. A second spout cover may be configured to be mounted over at least part of the second dispenser assembly.

In some non-limiting embodiments or aspects, the first spout cover may include a first connector configured to engage with the first dispenser assembly and the second spout cover may include a second connector configured to engage with the second dispenser assembly.

In some non-limiting embodiments or aspects, the first connector and the second connector each may include a snap-fit connector configured to engage with a corresponding slot on the first dispenser assembly and the second dispenser assembly, respectively.

In some non-limiting embodiments or aspects, the first connector and the second connector each may include a magnetic connector configured to engage with a corresponding magnetic connector on the first dispenser assembly and the second dispenser assembly, respectively.

In some non-limiting embodiments or aspects, the first connector and the second connector may be configured to mechanically interlock with the first dispenser assembly and the second dispenser assembly, respectively.

In some non-limiting embodiments or aspects, a pivot may be connected to the primary lever. A pivot cover may be configured to at least partly conceal the pivot.

In some non-limiting embodiments or aspects, a first face panel may extend between the pivot cover and the first spout cover. A second face panel may extend between the pivot cover and the second spout cover.

In some non-limiting embodiments or aspects, a shared outlet may be configured to output drink product released from at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof.

In some non-limiting embodiments or aspects, a first channel may be configured to guide the drink product released from the first dispenser assembly to the shared outlet. A second channel may be configured to guide the drink product released from the second dispenser assembly to the shared outlet.

In some non-limiting embodiments or aspects, the first channel may include a first sloped chute configured to guide the drink product released from the first dispenser assembly to the shared outlet. The second channel may include a second sloped chute configured to guide the drink product released from the second dispenser assembly to the shared outlet.

In some non-limiting embodiments or aspects, a first channel cover may be configured to extend at least partially along the first channel and at least partly enclose a first flow volume of the first channel. A second channel cover may be configured to extend at least partially along the second channel and at least partly enclose a second flow volume of the second channel.

In some non-limiting embodiments or aspects, the first channel cover and the second channel cover may be formed of an at least partially transparent material.

In some non-limiting embodiments or aspects, the first channel cover and the second channel cover may be formed of an at least partially opaque material.

In some non-limiting embodiments or aspects, the at least one projection may include a first projection segment, which may be configured to mechanically connect the primary lever to the first lever of the first dispenser assembly, and a second projection segment, which may be configured to mechanically connect the primary lever to the second lever of the second dispenser assembly.

In some non-limiting embodiments or aspects, the at least one projection may include a horizontal projection extending from the first lever to the second lever. The horizontal projection may include the first projection segment and the second projection segment.

In some non-limiting embodiments or aspects, the first projection segment may include a first recess configured to receive a first handle of the first lever. The second projection segment may include a second recess configured to receive a second handle of the second lever.

In some non-limiting embodiments or aspects, the primary lever may include a primary handle, the first lever may include a first handle, and the second lever may include a second handle. The primary handle, the first handle, and the second handle may be configured to be manually pulled by a user when the shared output connector assembly is attached to the first dispenser assembly and the second dispenser assembly.

In some non-limiting embodiments or aspects, when the user manually pulls the primary handle in a first direction, the at least one projection may be configured to cause the first handle and the second handle to simultaneously move in the first direction.

In some non-limiting embodiments or aspects, when the user manually pulls the primary handle in the first direction, the at least one projection may be configured to cause the first handle and the second handle to simultaneously move a distance in the first direction that is proportional to a distance moved by the primary handle.

In some non-limiting embodiments or aspects, a pivot may be connected to the primary lever. A biasing element may be connected to the primary lever. The biasing element may be configured to provide a biasing force to the primary lever. When the primary lever is released by the user, the biasing element may cause the primary lever to return to a neutral position.

In some non-limiting embodiments or aspects, the biasing element may include a spring.

In some non-limiting embodiments or aspects, a first drink maker may include the first dispenser assembly. A second drink maker may include the second dispenser assembly.

In some non-limiting embodiments or aspects, a first mixing vessel may include the first dispenser assembly. A second mixing vessel may include the second dispenser assembly.

In some non-limiting embodiments or aspects, the first lever and the second lever each may be configured to be actuatable without movement of the primary lever.

According to some non-limiting embodiments or aspects, provided is a shared output connector assembly for connecting a first dispenser assembly and a second dispenser assembly. An example shared output connector assembly may include a shared outlet configured to output drink product released from at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof. A first channel may be configured to guide the drink product released from the first dispenser assembly to the shared outlet. A second channel may be to guide the drink product released from the second dispenser assembly to the shared outlet.

In some non-limiting embodiments or aspects, the shared output connector assembly may be configured to be removably attachable to at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof.

Further non-limiting embodiments or aspects are set forth in the following numbered clauses:

Clause 1: A shared output connector assembly for connecting a first dispenser assembly and a second dispenser assembly, the shared output connector assembly comprising: a primary lever; and at least one projection configured to mechanically connect the primary lever to (i) a first lever of the first dispenser assembly and (ii) a second lever of the second dispenser assembly, wherein, when the shared output connector assembly is attached to the first dispenser assembly and the second dispenser assembly, the first lever and the second lever are independently actuatable, and when the primary lever is actuated, the at least one projection causes the first lever and the second lever to actuate.

Clause 2: The shared output connector assembly of clause 1, further comprising: a first spout cover configured to be mounted over at least part of the first dispenser assembly; a second spout cover configured to be mounted over at least part of the second dispenser assembly.

Clause 3: The shared output connector assembly of clause 1 or clause 2, wherein the first spout cover comprises a first connector configured to engage with the first dispenser assembly and the second spout cover comprises a second connector configured to engage with the second dispenser assembly.

Clause 4: The shared output connector assembly of any of clauses 1-3, wherein the first connector and the second connector each comprise a snap-fit connector configured to engage with a corresponding slot on the first dispenser assembly and the second dispenser assembly, respectively.

Clause 5: The shared output connector assembly of any of clauses 1-4, wherein the first connector and the second connector each comprise a magnetic connector configured to engage with a corresponding magnetic connector on the first dispenser assembly and the second dispenser assembly, respectively.

Clause 6: The shared output connector assembly of any of clauses 1-5, wherein the first connector and the second connector are configured to mechanically interlock with the first dispenser assembly and the second dispenser assembly, respectively.

Clause 7: The shared output connector assembly of any of clauses 1-6, further comprising: a pivot connected to the primary lever; and a pivot cover configured to at least partly conceal the pivot.

Clause 8: The shared output connector assembly of any of clauses 1-7, further comprising: a first face panel extending between the pivot cover and the first spout cover; and a second face panel extending between the pivot cover and the second spout cover.

Clause 9: The shared output connector assembly of any of clauses 1-8, further comprising a shared outlet configured to output drink product released from at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof.

Clause 10: The shared output connector assembly of any of clauses 1-9, further comprising: a first channel configured to guide the drink product released from the first dispenser assembly to the shared outlet; and a second channel configured to guide the drink product released from the second dispenser assembly to the shared outlet.

Clause 11: The shared output connector assembly of any of clauses 1-10, wherein the first channel comprises a first sloped chute configured to guide the drink product released from the first dispenser assembly to the shared outlet, and wherein the second channel comprises a second sloped chute configured to guide the drink product released from the second dispenser assembly to the shared outlet.

Clause 12: The shared output connector assembly of any of clauses 1-11, further comprising: a first channel cover configured to extend at least partially along the first channel and at least partly enclose a first flow volume of the first channel; and a second channel cover configured to extend at least partially along the second channel and at least partly enclose a second flow volume of the second channel.

Clause 13: The shared output connector assembly of any of clauses 1-12, wherein the first channel cover and the second channel cover are formed of an at least partially transparent material.

Clause 14: The shared output connector assembly of any of clauses 1-13, wherein the first channel cover and the second channel cover are formed of an at least partially opaque material.

Clause 15: The shared output connector assembly of any of clauses 1-14, wherein the at least one projection comprises: a first projection segment configured to mechanically connect the primary lever to the first lever of the first dispenser assembly; and a second projection segment configured to mechanically connect the primary lever to the second lever of the second dispenser assembly.

Clause 16: The shared output connector assembly of any of clauses 1-15, wherein the at least one projection comprises a horizontal projection extending from the first lever to the second lever, the horizontal projection comprising the first projection segment and the second projection segment.

Clause 17: The shared output connector assembly of any of clauses 1-16, wherein the first projection segment comprises a first recess configured to receive a first handle of the first lever, and wherein the second projection segment comprises a second recess configured to receive a second handle of the second lever.

Clause 18: The shared output connector assembly of any of clauses 1-17, wherein the primary lever comprises a primary handle, the first lever comprises a first handle, and the second lever comprises a second handle, and wherein the primary handle, the first handle, and the second handle are configured to be manually pulled by a user when the shared output connector assembly is attached to the first dispenser assembly and the second dispenser assembly.

Clause 19: The shared output connector assembly of any of clauses 1-18, wherein, when the user manually pulls the primary handle in a first direction, the at least one projection is configured to cause the first handle and the second handle to simultaneously move in the first direction.

Clause 20: The shared output connector assembly of any of clauses 1-19, wherein, when the user manually pulls the primary handle in the first direction, the at least one projection is configured to cause the first handle and the second handle to simultaneously move a distance in the first direction that is proportional to a distance moved by the primary handle.

Clause 21: The shared output connector assembly of any of clauses 1-20, further comprising: a pivot connected to the primary lever; and a biasing element connected to the primary lever, the biasing element configured to provide a biasing force to the primary lever wherein, when the primary lever is released by the user, the biasing element causes the primary lever to return to a neutral position.

Clause 22: The shared output connector assembly of any of clauses 1-21, wherein the biasing element comprises a spring.

Clause 23: The shared output connector assembly of any of clauses 1-22, wherein a first drink maker comprises the first dispenser assembly, and wherein a second drink maker comprises the second dispenser assembly.

Clause 24: The shared output connector assembly of any of clauses 1-23, wherein a first mixing vessel comprises the first dispenser assembly, and wherein a second mixing vessel comprises the second dispenser assembly.

Clause 25: The shared output connector assembly of any of clauses 1-24, wherein the first lever and the second lever are each configured to be actuatable without movement of the primary lever.

Clause 26: A shared output connector assembly for connecting a first dispenser assembly and a second dispenser assembly, the shared output connector assembly comprising: a shared outlet configured to output drink product released from at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof; a first channel configured to guide the drink product released from the first dispenser assembly to the shared outlet; and a second channel configured to guide the drink product released from the second dispenser assembly to the shared outlet.

Clause 27: The shared output connector assembly of clause 26, wherein the shared output connector assembly is configured to be removably attachable to at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof.

These and other features and characteristics of the present disclosure, as well as the methods of operation and functions of the related elements of structures and the combination of parts and economics of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. It is to be expressly understood, however, that the drawings and appendix are for the purpose of illustration and description only and are not intended as a definition of the limits of the disclosed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

Additional advantages and details are explained in greater detail below with reference to the non-limiting, exemplary embodiments that are illustrated in the accompanying schematic figures, in which:

FIG. **1** shows a perspective view of a frozen drink maker, according to non-limiting embodiments or aspects;

FIG. **2** shows a view of various internal components within the housing and mixing vessel of the frozen drink maker of FIG. **1**, according to non-limiting embodiments or aspects;

FIG. **3** shows a front view of the frozen drink maker of FIG. **1**, according to non-limiting embodiments or aspects;

FIG. **4** is a block diagram of an example of a control system of the frozen drink maker of FIG. **1**, according to non-limiting embodiments or aspects;

FIG. **5A** illustrates a side view of the frozen drink maker of FIG. **1** with the mixing vessel in a coupled position relative to the upper housing section, according to non-limiting embodiments or aspects;

FIG. **5B** illustrates a side view of the frozen drink maker illustrated in FIG. **5A** with some features of the housing and the lever shown in partial cross-section, according to non-limiting embodiments or aspects;

FIG. **6** shows a detailed view of a lever with cams for coupling a mixing vessel to the housing of a frozen drink maker, according to non-limiting embodiments or aspects;

FIG. **7**A shows a rear view of a mixing vessel, according to non-limiting embodiments or aspects;

FIG. **7**B shows a perspective view of the rear of a mixing vessel, according to non-limiting embodiments or aspects;

FIG. **8** shows a perspective view of a flexible seal, according to non-limiting embodiments or aspects;

FIG. **9** shows a cross-sectional view of a flexible seal, according to non-limiting embodiments or aspects;

FIG. **10** illustrates a flow diagram for a method of using the disclosed frozen drink maker, according to non-limiting embodiments or aspects;

FIGS. **11**A and **11**B show perspective views of a condensation collection tray of the frozen drink maker of FIG. **1**, according to non-limiting embodiments or aspects;

FIG. **11**C shows the collection tray of FIGS. **11**A and **11**B inserted into the frozen drink maker of FIG. **1**, according to non-limiting embodiments or aspects;

FIG. **11**D shows the frozen drink maker of FIG. **1** with the collection tray removed, according to non-limiting embodiments or aspects;

FIG. **12** is a flow chart illustrating a method of removing the collection tray of FIGS. **11**A and **11**B from the frozen drink maker of FIG. **1**, according to non-limiting embodiments or aspects;

FIG. **13**A shows an isometric view of the frozen drink maker with a mixing vessel having at least one internal baffle, according to non-limiting embodiments or aspects;

FIG. **13**B shows a cross-sectional view of the frozen drink maker shown in FIG. **13**A, taken along line B-B, according to non-limiting embodiments or aspects;

FIG. **13**C shows a cross-sectional view of the frozen drink maker shown in FIG. **13**A, taken along line C-C, according to non-limiting embodiments or aspects;

FIG. **14**A shows a rear isometric view of a mixing vessel for a frozen drink maker with three internal baffles, according to non-limiting embodiments or aspects;

FIG. **14**B shows a rear view of the mixing vessel shown in FIG. **14**A, according to non-limiting embodiments or aspects;

FIG. **14**C shows a front isometric view of the mixing vessel shown in FIG. **14**A, according to non-limiting embodiments or aspects;

FIG. **15** is a close-up view of a user interface, according to non-limiting embodiments or aspects;

FIG. **16** is a graph of coarse and fine temperature settings, according to non-limiting embodiments or aspects;

FIG. **17** is a close-up view of another user interface, according to non-limiting embodiments or aspects;

FIG. **18** is a graph of temperature values associated with automatic recipe temperature target temperatures and manual temperature adjustments, according to non-limiting embodiments or aspects;

FIG. **19** is a graph of drive motor current and temperature vs. time as a drink product being processing by the frozen drink maker of FIG. **1**, according to non-limiting embodiments or aspects;

FIG. **20** is a flow diagram of a process for making a cooled drink product using a food type for initial or coarse temperature and/or texture control and then using a user input to subsequently fine tune the temperature and/or texture of the drink product, according to non-limiting embodiments or aspects;

FIG. **21** is a flow diagram of a process for automatically detecting when drive motor current is too high and/or a drink product is too thick and, in response, adjusting the temperature of the drink product to reduce drive motor current and/or to increase the temperature of the drink product to reduce a thickness of the drink product, according to non-limiting embodiments or aspects;

FIG. **22**A shows an implementation of a dual-use cooling fan within the housing of a drink maker, according to non-limiting embodiments or aspects;

FIG. **22**B shows another implementation of a dual-use cooling fan within the housing of a drink maker, according to non-limiting embodiments or aspects;

FIG. **22**C shows a perspective view of the dual-use cooling fan of FIG. **22**B, according to non-limiting embodiments or aspects;

FIG. **23** is a flow diagram of a process for operating the dual-use cooling fan, according to non-limiting embodiments or aspects;

FIG. **24**A shows a perspective view of a sample pour-in opening for a frozen drink maker, according to non-limiting embodiments or aspects;

FIG. **24**B shows a front view of the pour-in opening shown in FIG. **24**A, according to non-limiting embodiments or aspects;

FIG. **24**C shows a left perspective view of the pour-in opening shown in FIG. **24**A, according to non-limiting embodiments or aspects;

FIG. **25** shows a perspective view of a sample cover for a pour-in opening, according to non-limiting embodiments or aspects;

FIG. **26** shows a perspective view of a sample pour-in opening, according to non-limiting embodiments or aspects;

FIG. **27**A shows a perspective view of a sample pour-in opening, according to non-limiting embodiments or aspects;

FIG. **27**B shows an isometric view of a protype of the pour-in opening of FIG. **27**A, according to non-limiting embodiments or aspects;

FIG. **27**C shows a side view of the pour-in opening protype of FIG. **27**B, according to non-limiting embodiments or aspects;

FIG. **27**D shows the pour-in opening protype shown in FIG. **27**B affixed to a mixing vessel, according to non-limiting embodiments or aspects;

FIG. **28** shows a sample method of using a pour-in opening, according to non-limiting embodiments or aspects;

FIGS. **29**A-**29**D show a dispenser assembly for dispensing a drink product from the frozen drink maker, according to non-limiting embodiments or aspects;

FIGS. **30**A and **30**B show a dispenser assembly for dispensing a drink product from the frozen drink maker, according to non-limiting embodiments or aspects;

FIGS. **31**A and **31**B show a shroud for covering the dispenser assembly of FIGS. **29**A-**29**D and FIGS. **30**A and **30**B, according to non-limiting embodiments or aspects;

FIGS. **32**A-**32**C show a cross-sectional profile view of a dispenser assembly, according to non-limiting embodiments or aspects;

FIG. **33** shows a cross-sectional profile view of a dispenser assembly with a shroud, according to non-limiting embodiments or aspects;

FIG. **34** shows a front view of a shared output connector assembly, according to non-limiting embodiments or aspects;

FIG. **35** shows a front view of a shared output connector assembly attached to two dispenser assemblies, according to non-limiting embodiments or aspects;

FIG. **36** shows a perspective view of a shared output connector assembly, according to non-limiting embodiments or aspects; and

FIG. **37** shows a perspective view of a shared output connector assembly attached to two dispenser assemblies, according to non-limiting embodiments or aspects.

DETAILED DESCRIPTION

In the following description, like components have the same reference numerals, regardless of different illustrated implementations. To illustrate implementations clearly and concisely, the drawings may not necessarily reflect appropriate scale and may have certain structures shown in somewhat schematic form. The disclosure may describe and/or illustrate structures in one implementation, and in the same way or in a similar way in one or more other implementations, and/or combined with or instead of the structures of the other implementations.

For purposes of the description hereinafter, the terms "end," "upper," "lower," "right," "left," "vertical," "horizontal," "top," "bottom," "lateral," "longitudinal," "front," "rear," and derivatives thereof shall relate to the embodiments as they are oriented in the drawing figures. However, it is to be understood that the present disclosure may assume various alternative variations and step sequences, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary and non-limiting embodiments or aspects of the disclosed subject matter. Hence, specific dimensions and other physical characteristics related to the embodiments or aspects disclosed herein are not to be considered as limiting.

In the specification and claims, for the purposes of describing and defining the disclosure, the terms "about" and "substantially" represent the inherent degree of uncertainty attributed to any quantitative comparison, value, measurement, or other representation. The terms "about" and "substantially" moreover represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue. Open-ended terms, such as "comprise," "include," and/or plural forms of each, include the listed parts and can include additional parts not listed, while terms such as "and/or" include one or more of the listed parts and combinations of the listed parts. Use of the terms "top," "bottom," "above," "below" and the like helps only in the clear description of the disclosure and does not limit the structure, positioning and/or operation of the disclosure in any manner.

The application, in various implementations, addresses deficiencies associated with prior commercial slush machines. Unfortunately, the architecture of prior commercial slush machines usually requires a significant amount of force to seat the vessel over a large radial seal, making it challenging for a user to install and uninstall the vessel from the device. In many prior commercial slush machines, the vessel is installed by engaging a catch to retain the vessel, which strains the plastic to properly position the vessel and requires significant user effort. Accordingly, there is a need for a more user-friendly architecture to install and uninstall the vessel of a frozen drink maker, such as, for example, a lever that can be used to couple and decouple the vessel to a housing of the frozen drink maker with minimal force and/or that only requires one hand to use.

FIG. **1** shows a perspective view of a frozen drink maker **100**, according to an illustrative implementation of the disclosure. Frozen drink maker **100** includes a housing **102** and mixing vessel **104**. Housing **102** may include user interface **112** for receiving user inputs to control frozen drink maker **100** and/or to output or display information. User interface **112** may include one or more buttons, dials, switches, touchscreens, indicators, LEDs, and the like. User interface **112** may display status information including, for example, a temperature of a drink product within mixing vessel **104**, an indicator of a recipe and/or program currently being implemented, and a timer associated with the progress of a recipe and/or program in progress and/or currently being implemented. User interface **112** may provide indicators and/or warnings to users regarding, for example, when a recipe is complete or when a user is expected to perform an action associated with processing a drink product. User interface **112** may include a selectable menu of drink types (e.g., recipes) and/or programs for different types of drink products such as, without limitation, granita, smoothie, margarita, daiquiri, pina colada, slush, cocktail, frappe, juice, diary, milk shake, cool drink, semi-frozen drink, frozen drink, and the like.

Housing **102** may include a panel (e.g., a removable panel) **114** along a side of housing **102**. Panel **114** may include a plurality of openings that facilitate air flow to aid in cooling components within housing **102**. Housing **102** may include upper housing section **122** that is arranged to couple with a rear end of mixing vessel **104** when mixing vessel **104** is attached to housing **102**. Mixing vessel **104** may include walls, or a portion thereof, that are transparent to enable a viewer to see a drink product within mixing vessel **104** during processing. Mixing vessel **104** may include pour-in opening **106** whereby mixing vessel **104** can receive ingredients for processing a drink product within mixing vessel **104**. FIG. **1** shows pour-in opening **106** in a closed configuration with a cover sealing pour-in opening **106**. The cover may be detachably removable or moveable to open or close pour-in opening **106**. Pour-in opening **106** may include a grate to inhibit a user from reaching into mixing vessel **104** when pour-in opening **106** is open, e.g., the cover is not installed. Mixing vessel **104** may include a dispenser assembly **108** having a user handle **120**, a spout (not shown), and a spout shroud and/or cover **116**. Dispenser assembly **108** enables a user, by pulling down on handle **120**, to open a spout, connected to a wall of mixing vessel **104**, to dispense a processed (e.g., cooled) drink product from mixing vessel **104**. The user can close the spout by pushing handle **120** back to its upright position (shown in FIG. **1**) and, thereby, stop the dispensing of the processed drink product.

Frozen drink maker **100** may include a coupling mechanism that enables a secure coupling of mixing vessel **104** to housing **102**, including upper housing section **122**. In some non-limiting embodiments or aspects, the coupling mechanism is a lever **110** rotatably coupled to upper housing section **122**. FIG. **1** shows lever **110** in the coupled, locked, and/or closed position whereby mixing vessel **104** is coupled to (e.g., attached to, latched to, and/or locked to) housing **102** and upper housing section **122**. In the coupled position, lever **110** ensures that there is a water-tight seal to prevent leakage of drink product from mixing vessel **104**. Lever **110** may be placed in the coupled position by sliding mixing vessel **104** against upper housing section **122** and then rotating lever **110** in a clockwise direction until its handle rests on or about the top surface of upper housing section **122**. Mixing vessel **104** can be disengaged and/or decoupled from housing **102** and upper housing section **122** by pulling and/or rotating lever **110** in a counter-clockwise direction (from the perspective of FIG. **1**) toward the front of mixing vessel **104**, which causes lever **110** to release mixing vessel **104**. Once released and/or decoupled, mixing vessel **104** may slide in a forward direction (away from upper housing section **122**) to be fully detached and/or removed from housing **102**.

A flexible seal (illustrated in FIG. **8**) may be positioned between mixing vessel **104** and upper housing section **122**. The flexible seal may include a face seal portion and/or a radial seal portion. If present, the face seal portion may provide an improved seal based on compression provided by lever **110** pushing mixing vessel **104** laterally against a wall of upper housing section **122**. Mixing vessel **104** may have a substantially cylindrical shape with a base having an opening formed therein, and the opening is sealed by the flexible seal when lever **110** is in the coupled position. An interlock switch may be implemented at upper housing section **122** that is activated when mixing vessel **104** is coupled to upper housing section **122** that prevents activation of drive motor **208** unless mixing vessel **104** is coupled to upper housing section **122**. This ensures that a user is not exposed to a moving dasher **204**. Frozen drink maker **100** may also include drip tray **118** being positioned below dispenser assembly **108** and arranged to collect any drink product that is not properly dispensed from mixing vessel **104** to, for example, a user cup. Drip tray **118** may be attachably removable from its operational position shown in FIG. **1**. For example, drip tray **118** may be mounted and/or stored on a side panel of housing **102**, as illustrated in FIG. **3** as water tray **304**.

FIG. **2** shows a view **200** of various internal components within housing **102** and mixing vessel **104** of frozen drink maker **100** of FIG. **1**. Frozen drink maker **100** includes a cylindrical evaporator **202** that is surrounded by an auger and/or dasher **204**. Dasher **204** may include one or more mixing blades and/or protrusions that extend helically around evaporator and/or chiller **202**. Dasher **204** may be driven to rotate by a central drive shaft (not shown) within mixing vessel **104**. The drive shaft may be surrounded by evaporator **202**. However, in various implementations, evaporator **202** does not rotate. The drive shaft may be coupled via a gear assembly **210** to a drive motor **208**. In some non-limiting embodiments or aspects, drive motor **208** is an AC motor, but another type of motor may be used such as, without limitation, a DC motor. Drive motor **208** may include a motor fan **212** arranged to provide air cooling for drive motor **208**. While FIG. **2** shows an implementation where drive motor **208** is not coaxially aligned with the drive shaft used to rotate dasher **204**, in other implementations, drive motor **208** can be aligned coaxially with the drive shaft. During processing of a drink product, drive motor **208** may be continuously operated at one or more speeds to drive continuous rotation of dasher **204** and, thereby, provide continuous mixing of the drink product within mixing vessel **104**. In some non-limiting embodiments or aspects, the rotation of dasher **204** causes the helically arranged blades to push the cooling drink product to the front of mixing vessel **104**. During the processing, portions of the drink product may freeze against the surface of evaporator **202** as a result of being cooled by evaporator **202**. In some non-limiting embodiments or aspects, the blades of the rotating dasher **204** scrape frozen portions of the drink product from the surface evaporator **202** while concurrently mixing and pushing the cooling drink product towards the front of mixing vessel **104**.

Frozen drink maker **100** may include a refrigeration circuit and/or system to provide cooling of a drink product and/or to control the temperature of a drink product within mixing vessel **104**. The refrigeration circuit may include a compressor **214**, an evaporator **202**, a condenser **216**, a condenser fan **218**, a bypass valve, and conduit that carries refrigerant in a closed loop among the refrigeration circuit components to facilitate cooling and/or temperature control of a drink product in mixing vessel **104**. Operations of the refrigeration circuit may be controlled by a controller, such as controller **402**, as described further with respect to FIG. **4** later herein. Frozen drink maker **100** may also include a condensation collection tray **220** arranged to collect any liquid condensation caused by cooling from evaporator **202**. FIG. **2** shows condensation collection tray **220** in the inserted position. Condensation collection tray **220** may be insertably removable from a slot within housing **102** to enable collection of condensed liquid when inserted into the slot and then efficient removal to empty condensation collection tray **220**, and then re-insertion into the slot for subsequent liquid collection.

FIG. **3** shows a front view **300** of frozen drink maker **100** of FIG. **1**. Frozen drink maker **100** may include user interface **112** on a front surface of housing **102**. In other implementations, user interface **112** may be located on a side, top, or back of housing **102**. Frozen drink maker **100** may include a mount **302** on a side of housing **102** where drip tray **118** can be mounted when not in use (shown as drip tray **304** in FIG. **3**), such as during transport of frozen drink maker **100**. Frozen drink maker **100** may include a power interface arranged to receive AC power from a power outlet (not shown). In some non-limiting embodiments or aspects, frozen drink maker **100** may include one or more batteries housed within housing **102** and arranged to provide power to various components of frozen drink maker **100**. Frozen drink maker **100** may also include a printed circuit board assembly (PCBA) **222** within housing **102**. As will be explained with respect to FIG. **4**, PCBA **222** may include a control system **400** arranged to automatically control certain operations of frozen drink maker **100**.

FIG. **4** is a block diagram illustrating an example of a control system **400** of frozen drink maker **100**, according to some implementations of the disclosure. Control system **400** may include a microcontroller, a processor, a system-on-a-chip (SoC), a client device, and/or a physical computing device and may include hardware and/or virtual processor(s). In some non-limiting embodiments or aspects, control system **400** and its elements, as shown in FIG. **4**, each relate to physical hardware, while in some implementations one, more, or all of the elements could be implemented using emulators or virtual machines. Regardless, electronic control system **400** may be implemented on physical hardware, such as in frozen drink maker **100**.

As also shown in FIG. **4**, control system **400** may include a user interface **212** and/or **112**, having, for example, a keyboard, keypad, one or more buttons, dials, touchpad, or sensor readout (e.g., biometric scanner) and one or more output components, such as displays, speakers for audio, LED indicators, and/or light indicators. Control system **400** may also include communications interfaces **410**, such as a network communication unit that could include a wired communication component and/or a wireless communications component, which may be communicatively coupled to controller and/or processor **402**. The network communication unit may utilize any of a variety of proprietary or standardized network protocols, such as Ethernet, TCP/IP, to name a few of many protocols, to effect communications between processor **402** and another device, network, or system. Network communication units may also comprise one or more transceivers that utilize the Ethernet, power line communication (PLC), Wi-Fi®, cellular, and/or other communication methods. For example, control system **400** may send one or more communications associated with a status of frozen drink maker **100** to a mobile device of a user, e.g., send an alert to the mobile device when a recipe is complete and/or a drink product is ready for dispensing, or to indicate that the mixing vessel is low or out of a drink product.

Control system **400** may include a processing element, such as controller and/or processor **402**, that contains one or more hardware processors, where each hardware processor may have a single or multiple processor cores. In one implementation, processor **402** includes at least one shared cache that stores data (e.g., computing instructions) that are utilized by one or more other components of processor **402**. For example, the shared cache may be a locally cached data stored in a memory for faster access by components of the processing elements that make up processor **402**. Examples of processors include, but are not limited to, a central processing unit (CPU) and/or microprocessor. Controller and/or processor **402** may utilize a computer architecture base on, without limitation, the Intel® 8051 architecture, Motorola® 68HCX, Intel® 80X86, and the like. The processor **402** may include, without limitation, an 8-bit, 12-bit, 16-bit, 32-bit, or 64-bit architecture. Although not illustrated in FIG. **4**, the processing elements that make up processor **402** may also include one or more other types of hardware processing components, such as graphics processing units (GPUs), application specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and/or digital signal processors (DSPs).

FIG. **4** also illustrates that memory **404** may be operatively and communicatively coupled to controller **402**. Memory **404** may be a non-transitory medium configured to store various types of data. For example, memory **404** may include one or more storage devices **408** that include a non-volatile storage device and/or volatile memory. Volatile memory, such as random-access memory (RAM), can be any suitable non-permanent storage device. The non-volatile storage devices **408** may include one or more disk drives, optical drives, solid-state drives (SSDs), tape drives, flash memory, read-only memory (ROM), and/or any other type of memory designed to maintain data for a duration time after a power loss or shut down operation. In certain configurations, the non-volatile storage devices **408** may be used to store overflow data if allocated RAM is not large enough to hold all working data. The non-volatile storage devices **408** may also be used to store programs that are loaded into the RAM when such programs are selected for execution. Data store and/or storage devices **408** may be arranged to store a plurality of drink product making and/or processing instruction programs associated with a plurality of drink product processing sequences, e.g., recipes. Such drink product making and/or processing instruction programs may include instruction for controller and/or processor **402** to: start or stop one or more motors and/or compressors **414** (e.g., such as motor **208** and/or compressor **214**), start or stop compressor **214** to regulate a temperature of a drink product being processed within mixing vessel **104**, operate the one or more motors **414** (e.g., motor **208** and/or compressor **214**) at certain periods during a particular drink product processing sequence, operate motor **208** at certain speeds during certain periods of time of a recipe, issue one or more cue instructions to user interface **412** and/or **112** that are output to a user to illicit a response, action, and/or input from the user.

Persons of ordinary skill in the art are aware that software programs may be developed, encoded, and compiled in a variety of computing languages for a variety of software platforms and/or operating systems and subsequently loaded and executed by processor **402**. In one implementation, the compiling process of the software program may transform program code written in a programming language to another computer language such that the processor **402** is able to execute the programming code. For example, the compiling process of the software program may generate an executable program that provides encoded instructions (e.g., machine code instructions) for processor **402** to accomplish specific, non-generic, particular computing functions.

After the compiling process, the encoded instructions may be loaded as computer executable instructions or process steps to processor **402** from storage device **408**, from memory **404**, and/or embedded within processor **402** (e.g., via a cache or on-board ROM). Processor **402** may be configured to execute the stored instructions or process steps in order to perform instructions or process steps to transform the electronic control system **400** into a non-generic, particular, specially programmed machine or apparatus. Stored data, e.g., data stored by a data store and/or storage device **408**, may be accessed by processor **402** during the execution of computer executable instructions or process steps to instruct one or more components within control system **400** and/or other components or devices external to control system **400**. For example, the recipes may be arranged in a lookup table and/or database within storage device **408** and be accessed by processor **402** when executing a particular recipe selected by a user via user interface **412** and/or **112**.

User interface **412** and/or **112** can include a display, positional input component (such as a mouse, touchpad, touchscreen, or the like), keyboard, keypad, one or more buttons, one or more dials, a microphone, speaker, or other forms of user input and output components. The user interface components may be communicatively coupled to processor **402**. When the user interface output component is or includes a display, the display can be implemented in various ways, including by a liquid crystal display (LCD) or a cathode-ray tube (CRT) or light emitting diode (LED) display, such as an OLED display.

Sensors **406** may include one or more sensors that detect and/or monitor conditions of a drink product within mixing vessel **104**, conditions associated with a component of frozen drink maker **100**, and/or conditions of a refrigerant within the refrigeration system. Conditions may include, without limitation, rotation, speed of rotation, and/or movement of a device or component (e.g., a motor), rate of such movement, frequency of such movement, direction of such movements, motor current, motor voltage, motor power, motor torque, temperature, pressure, fluid level in mixing vessel **104**, position of a device or component (e.g., whether pour-in opening **106** is open or closed), and/or the presence of a device or component (e.g., whether shroud **116** is installed or not). Types of sensors may include, for example, electrical metering chips, Hall sensors, pressure sensors, temperature sensors, optical sensors, current sensors, torque sensors, voltage sensors, cameras, other types of sensors, or any suitable combination of the foregoing. Frozen drink maker **100** may include one or more temperature sensors positioned in various locations within mixing vessel **104** such as, for example, on or about the lower front area within mixing vessel **104**, on or about the upper front area within mixing vessel **104**, on or about the upper rear area within mixing vessel **104**, within one or more coils of evaporator **202**, and/or within housing **102**.

Sensors **406** may also include one or more safety and/or interlock switches that prevent or enable operation of certain components, e.g., a motor, when certain conditions are met (e.g., enabling activation of motor **208** and/or compressor **414** when a lid or cover for opening **106** is attached or closed and/or when a sufficient level of drink product is in mixing vessel **104**). Persons of ordinary skill in the art are aware that electronic control system **400** may include other components well known in the art, such as power sources and/or analog-to-digital converters, not explicitly shown in FIG. **4**.

In some non-limiting embodiments or aspects, control system **400** and/or processor **402** includes an SoC having multiple hardware components including, but not limited to: a microcontroller, microprocessor or digital signal processor (DSP) core and/or multiprocessor SoCs (MPSoC) having more than one processor cores; memory blocks including a selection of read-only memory (ROM), random access memory (RAM), electronically crasable programmable read-only memory (EEPROM) and flash memory; timing sources including oscillators and phase-docked loops; peripherals including counter-timers, real-time timers and power-on reset generators; external interfaces, including industry standards such as universal serial bus (USB), Fire Wire, Ethernet, universal synchronous/asynchronous receiver/transmitter (USART), serial peripheral interface (SPI); analog interfaces including analog-to-digital converters (ADCs) and digital-to-analog converters (DACs); and voltage regulators and power management circuits.

A SoC includes both the hardware, described above, and software controlling the microcontroller, microprocessor and/or DSP cores, peripherals and interfaces. Most SoCs are developed from pre-qualified hardware blocks for the hardware elements (e.g., referred to as modules or components which represent an IP core or IP block), together with software drivers that control their operation. The above listing of hardware elements is not exhaustive. A SoC may include protocol stacks that drive industry-standard interfaces like a universal serial bus (USB).

Once the overall architecture of the SoC has been defined, individual hardware elements may be described in an abstract language called RTL which stands for register-transfer level. RTL is used to define the circuit behavior. Hardware elements are connected together in the same RTL language to create the full SoC design. In digital circuit design, RTL is a design abstraction which models a synchronous digital circuit in terms of the flow of digital signals (data) between hardware registers, and the logical operations performed on those signals. RTL abstraction is used in hardware description languages (HDLs) like Verilog and VHDL to create high-level representations of a circuit, from which lower-level representations and ultimately actual wiring can be derived. Design at the RTL level is typical practice in modern digital design. Verilog is standardized as Institute of Electrical and Electronic Engineers (IEEE) 1364 and is an HDL used to model electronic systems. Verilog is most commonly used in the design and verification of digital circuits at the RTL level of abstraction. Verilog may also be used in the verification of analog circuits and mixed-signal circuits, as well as in the design of genetic circuits. In some non-limiting embodiments or aspects, various components of control system **400** are implemented on a PCBA such as PCBA **222**.

In operation in certain implementations, a user fills mixing vessel **104** via pour-in opening **106** with ingredients associated with a drink product. The user selects the type of drink product to be processed via user interface **112**, e.g., the user selects the recipe for "margarita." In some non-limiting embodiments or aspects, the user selects the product type and/or recipe before filling mixing vessel **104** and user interface **112** provides one or more indicators or queues (visible and/or audible) that instruct the user to add ingredients to mixing vessel **104**. Mixing vessel **104** may include one or more fill sensors that detect when a sufficient amount or level of ingredients and/or fluid is within mixing vessel **104**. The one or more fill sensors may provide a signal to processor **402** that indicates when mixing vessel **104** is sufficiently filled or not filled. Processor **402** may prevent operations of frozen drink maker **100** (e.g., prevent activation of motor **208** and/or other components) if the fill sensor(s) **406** indicate(s) that mixing vessel **104** is not sufficiently filled. A lid sensor may be associated with opening **106**, whereby the lid sensor sends an open and/or closed signal to processor **402** that indicates whether opening **106** is open or closed. Processor **402** may prevent operations of frozen drink maker **100** if the lid sensor indicates that opening **106** is open and/or not closed. Depending on the sensed condition, user interface **112** may provide an indication regarding the condition, e.g., that mixing vessel **104** is sufficiently filled or not sufficiently filled and/or that opening **106** is not closed, to enable a user to take appropriate action(s).

Once mixing vessel **104** is filled with ingredients, the user may provide an input, e.g., a button press, to start processing of the drink product based on the selected recipe. Processing may include activation of motor **208** to drive rotation of dasher **204** and/or blade **206** to effect mixing of the ingredients of the drink product. Processing may also include activation of the refrigeration system including activation of compressor **214** and condenser fan **218**. Compressor **214** facilitates refrigerant flow through one or more coils of evaporator **202** and through condenser **216** to provide cooling and/or temperature control of the drink product within mixing vessel **104**. Processor **402** may control operations of various components such as motor **208** and compressor **214**. To regulate temperature at a particular setting associated with a recipe, processor **402** may activate/start and/or deactivate/stop compressor **214** to start and/or stop refrigerant flow through the coil(s) of evaporator **202** and, thereby, start or stop cooling of the drink product within mixing vessel **104**.

By cooling a drink product to a particular temperature, slush and/or ice particles may be formed within the drink product. Typically, the amount of particles and/or texture of a drink product corresponds to a temperature of the drink product, e.g., the cooler the temperature—the larger the amount of particles (and/or the larger the size of particles) and/or the more slush-like the drink product. User interface **112** may enable a user to fine tune and/or adjust a preset temperature associated with a recipe to enable a user to adjust the temperature and/or texture of a drink product to a more desirable temperature and/or texture.

Processor **402** may perform processing of the drink product for a set period of time in one or more phases and/or until a desired temperature and/or texture is determined. Processor **402** may receive one or more temperature signals from one or more temperature sensors **406** within mixing vessel **104** to determine the temperature of the drink product. Processor **402** may determine the temperature of the drink product by determining an average temperature among temperatures detected by multiple temperature sensors **406**. Processor **402** may determine the temperature of the drink product based on the detected temperature from one sensor **406** within mixing vessel **104** and/or based on a temperature of the refrigerant detected by a refrigerant temperature sensor **406**. Once a phase and/or sequence of a recipe is determined to be completed by processor **402**, processor **402** may, via user interface **112**, provide a visual and/or audio indication that the recipe is complete and ready for dispensing. In response, a user may place a cup or container below dispenser assembly **108** and pull handle **120** rotationally downward towards the user to open a spout located at the lower front wall of mixing vessel **104**, resulting in dispensing of the drink product into the cup or container. Once filled, the user can close the spout by pushing handle **120** back rotationally upward away from the user to its upright position shown in FIG. **2**. In implementations where handle **120** is spring-biased to the closed position, the user can release their hold of handle **120** and, thereby, allow a spring force to move handle **120** back rotational upward away from the user to the upright and closed position.

As previously mentioned, frozen drink maker **100** includes upper housing section **122** arranged to couple with a rear end of mixing vessel **104** when mixing vessel **104** is attached to housing **102**. Frozen drink maker **100** also includes lever **110** that enables mixing vessel **104** to be coupled (e.g., locked, attached to, and/or affixed to) to housing **102** (e.g., upper housing section **122**). Lever **110** also enables mixing vessel **104** to be unlocked and decoupled from housing **102** (e.g., upper housing section **122**). Features of lever **110** are shown in FIGS. **5**A and **5**B. FIG. **5**A illustrates a side view of frozen drink maker **100**, with mixing vessel **104** in a coupled position relative to upper housing section **122**. FIG. **5**B illustrates a side view of frozen drink maker **100** illustrated in FIG. **5**A, with some features of housing **102** and lever **110** shown in partial cross-section.

As shown in FIGS. **5**A and **5**B, lever **110** includes handle **111** that can be gripped by a user and moved relative to upper housing section **122**. Handle **111** can be moved into the position shown in FIGS. **5**A and **5**B to couple mixing vessel **104** into place on frozen drink maker **100** and can be moved away from upper housing section **122** and/or toward a front of housing **102** to decouple mixing vessel **104** from frozen drink maker **100**. When handle **111** is moved relative to upper housing section **122**, it activates cam **113**, which engages mating features on mixing vessel **104** to either couple or uncouple mixing vessel **104** relative to upper housing section **122**. In some non-limiting embodiments or aspects, handle **111** moves less than 90° relative to upper housing section **122** when moving between the coupled position and the uncoupled position.

FIG. **6** shows a detailed view of handle **111** with two cams **113***a*, **113***b* positioned on opposing sides. Handle **111** may include one, two, three, four, or more cams **113**, if desired. As handle **111** is moved, cams **113***a*, **113***b* rotate with respect to upper housing section **122**. FIG. **7**A shows a rear view of mixing vessel **104**. Mixing vessel **104** includes protrusions **115***a*, **115***b* on opposing outer sides, near the rear bottom of mixing vessel **104**. Protrusions **115***a*, **115***b* are shaped and positioned to engage with cams **113***a*, **113***b* on handle **111**. In particular, cams **113***a*, **113***b* have channels and/or cam paths **109***a*, **109***b* through which protrusions **115***a*, **115***b* slide respectively. As cams **113***a*, **113***b* rotate toward the back of housing **102**, protrusions **115***a*, **115***b* slide along cam paths **109***a*, **109***b* and are pulled toward upper housing section **122** and the rear of housing **102**, causing mixing vessel **104** to press against upper housing section **122** and form a water-tight seal with housing **102**. When cams **113***a*, **113***b* are rotated toward the front of frozen drink maker **100**, protrusions **115***a*, **115***b* are pushed away from upper housing section **122**, causing mixing vessel **104** to be decoupled from contact with upper housing section **122**.

Cam **113** may be an over-center cam, as shown in FIG. **5**B and FIG. **6**, or cam **113** may have alternative geometry. In the disclosed frozen drink maker **100**, cam **113** retains mixing vessel **104** on housing **102** when lever **110** is in the coupled position. As previously discussed, mixing vessel **104** may have an overall cylindrical or approximately cylindrical shape and may include opening **117** (shown in FIG. **7**B) at its rear end that couples to upper housing section **122**. As shown in FIG. **7**B, opening **117** may be in a rear panel **119** (e.g., rear vessel cap) of mixing vessel **104**. Opening **117** may be positioned to face horizontally when mixing vessel **104** is in the coupled position on upper housing section **122**.

To move lever **110** into a coupled position, handle **111** is moved toward upper housing section **122**. When mixing vessel **104** is in a coupled position on upper housing section **122**, lever **110**, in cooperation with flexible seal **121**, seals opening **117**. FIG. **8** shows flexible seal **121** configured in accordance with an implementation of the present disclosure. Flexible seal **121** may be formed of any elastomeric material, such as natural or synthetic rubber, silicone, neoprene, chloroprene, polyisoprene, polybutadiene, or combinations thereof. Flexible seal **121** may be independent of housing **102**. If desired, flexible seal **121** may be affixed to upper housing section **122**. Flexible seal **121** may be a single member including face seal portion **123** and/or radial seal portion **125**, as shown in FIG. **8**. However, in other embodiments, face seal portion **123** and radial seal portion **125** may be implemented with distinct flexible seals **121**.

Face seal portion **123** has an annular shape with a primary dimension that is vertically aligned to form a vertically aligned seal between a horizontal face of upper housing section **122** and a horizontal edge of mixing vessel **104**. When in the coupled position, face seal portion **125** interfaces a vertically aligned surface of upper housing section **122** to a vertically aligned side of mixing vessel **104**. Radial seal portion **125** includes multiple flexible annular ribs, as shown in FIG. **8**. Radial seal portion **125** forms a radial seal relative to the horizontal axis of mixing vessel **104**, scaling against an inside (e.g., cylindrical) surface of mixing vessel **104**. Flexible seal **121** may include at least one of radial seal portion **125** and face seal portion **123**.

Previously known frozen drink makers do not include both a face seal and a radial seal for a mixing vessel. If present, face seal portion **123** of flexible seal **121** may provide an improved seal based on compression provided by handle **111** pushing mixing vessel **104** laterally against a wall of upper housing section **122**. Cam **113** also allows high force on face seal portion **123** to be easily achieved and maintained. Since face seal portion **123** serves as the primary seal, in some non-limiting embodiments or aspects, radial seal portion **125** size can be reduced, thereby lowering mixing vessel's **104** resistance to seating and improving case of use.

In some non-limiting embodiments or aspects, flexible seal **121** may serve as the seal for mixing vessel **104** and/or evaporator **202**. For example, FIG. **9** shows a cross-sectional view of a sample flexible seal **121** having vessel seal portion **127** and evaporator seal portion **129**. Vessel seal portion **127** of flexible seal **121** creates a watertight seal between mixing vessel **104** and upper housing section **122**. Evaporator seal portion **129** of flexible seal **121** seals evaporator **202** within mixing vessel **104**.

To move lever **110** from a coupled position to an uncoupled position, handle **111** is moved away from upper housing section **122** and/or toward a front of housing **102**, which causes mixing vessel **104** to slide in a forward direction (away from upper housing section **122**) to be fully detached and/or removed from housing **102**. If desired, cam **113** may include an ejection feature to apply an ejection force to mixing vessel **104** to eject past radial seal portion **125**.

In other aspects, methods of using frozen drink maker **100** as disclosed here are described. FIG. **10** illustrates method **800** of producing a frozen drink using a frozen drink maker device. The frozen drink maker device includes a housing having an upper housing section and a lever configured to move relative to the upper housing section between a coupled position and an uncoupled position, and a mixing vessel arranged to couple to the upper housing section. The lever includes a handle that is moveable to place the lever into the coupled position and/or the uncoupled position. As shown in FIG. **10**, method **800** includes coupling the mixing vessel onto the upper housing section by moving the handle relative to the upper housing section to place the lever into the coupled position (block **802**). When in the coupled position, at least one of a face seal and a radial seal are formed between the mixing vessel and the upper housing section. Method **800** also includes operating the frozen drink maker device to produce the frozen drink (block **804**). Method **800** further includes uncoupling the mixing vessel from the upper housing section by moving the handle relative to the upper housing section to place the lever into an uncoupled position (block **806**).

In some non-limiting embodiments or aspects, coupling the mixing vessel onto the upper housing section involves moving the handle toward the upper housing section. In these and other implementations, uncoupling the mixing vessel from the upper housing section involves moving the handle away from the upper housing section and/or toward a front of the housing. Moving the handle relative to the upper housing section to place the lever into the coupled position may be accomplished by a user with only one hand. In these and other implementations, moving the handle relative to the upper housing section to place the lever into the uncoupled position may be accomplished by a user with only one hand. In select implementations, moving the handle relative to the upper housing section to position the lever from the coupled position to the uncoupled position requires moving the handle less than 90° relative to the upper housing section.

FIGS. **11**A and **11**B show perspective views of collection tray **220**, according to an illustrative implementation of the disclosure. Collection tray **220** may generally comprise collection portion **502** and handle **504** that may be used to insert collection tray **220** into and remove collection tray **220** from housing **102**. Collection portion **502** may comprise three walls **502***a*, **502***b*, **502***c* extending generally upwards from evaporator-facing surface **506**. Together with handle **504**, walls **502***a*, **502***b*, **502***c* and surface **506** may define chamber **508** for collecting liquid, including condensation falling from evaporator **202**, spills, and water poured into housing **102** to clean the inside of housing **102**. A shape of collection portion **502**, including evaporator-facing surface **506**, may correspond to an outer shape of evaporator **202**. For example, the shape of evaporator-facing surface **506** may be semi-cylindrical to correspond to the cylindrical shape of evaporator **202**, as shown in FIG. **11**A. However, the disclosure contemplates other suitable shapes, such as rectangular, of collection portion **502**. Chamber **508** may have a liquid volume capacity of about 16 ounces. However, the disclosure contemplates a liquid volume capacity of more or fewer than 16 ounces. As shown in FIG. **11**B, an underside of handle **504** may define one or more ribs **514** for adding structural integrity between a user-facing surface **510** of handle **504** and the main body of collection tray **220**. Collection tray **220** may be made from dishwasher-safe materials for easy cleaning.

FIG. **11**C shows collection tray **220** inserted into housing **102** of frozen drink maker **100**, according to an illustrative implementation of the disclosure. For case of illustration, housing **102** is shown with mixing vessel **104** and attached dispenser assembly **108** removed. When fully inserted, user-facing surface **510** of handle **504** may sit flush with user interface **112** of housing **102**. In the inserted position, collection tray **220** may be spaced vertically above bottom side **103** of housing **102**. Once liquid is collected in chamber **508**, the user may remove collection tray **220** for disposal of the collected liquid and cleaning of collection tray **220**.

FIG. **11**D shows housing **102** with collection tray **220** removed, according to an implementation of the disclosure. As shown in FIG. **11**D, housing **102** may include top surface **520** for supporting collection tray **220** when collection tray **220** is inserted into housing **102**. A shape of top surface **520** may be semi-cylindrical to correspond to the semi-cylindrical shape of evaporator-facing surface **506**. Housing **102** may also include one or more rails **522** defining one or more slots **512** between rails **522** and top surface **520**. Rails **522** may help guide the user in inserting collection tray **220** into slots **512** when installing collection tray **220** to housing **102**.

In some non-limiting embodiments or aspects, to remove collection tray **220** (e.g., for emptying and/or cleaning collection tray **220**), the user must first remove mixing vessel **104** and attached dispenser **108** (FIG. **1**). The user may then remove collection tray **220** by pulling collection tray **220** toward the user. This movement may cause collection tray **220** to slide along slots **512** until it is completely disengaged from housing **102**. Conversely, to insert collection tray **220** into housing **102**, the user may insert collection tray **220** into housing **102** by inserting collection portion **502** into slots **512** underneath evaporator **202** (FIG. **2**) such that evaporator-facing surface **506** faces evaporator **202**. In some non-limiting embodiments or aspects, after collection tray **220** has been inserted into housing **102**, mixing vessel **104** with attached dispenser **108** may be inserted onto housing **102** and fastened and sealed against housing **102**. FIG. **12** is a flow chart illustrating a method of removing collection tray **220** from housing **102**, as described above. FIG. **12** includes removing mixing vessel **104** and attached dispenser **108** from housing **102** (block **1202**), pulling collection tray **220** toward the user, causing collection tray **220** to slide through slot **512** (block **1204**), and fully disengaging collection tray **220** from slot **512** in housing **102** (block **1206**).

FIGS. **13**A-**13**C show a sample frozen drink maker **100** with mixing vessel **104** coupled to housing **102** (specifically, upper housing section **122**) and dispenser assembly **108**, according to some implementations. Mixing vessel **104** has a curved sidewall defining a substantially cylindrical chamber within. In select implementations, mixing vessel **104** is shaped as an ovoid or approximately as an ovoid (e.g., a cylinder with an ovular cross-section), or as an elliptic cylinder (e.g., a cylinder with an elliptic cross-section), or an approximate elliptic cylinder. When coupled to housing **102**, the front of mixing vessel **104** contacts dispenser assembly **108** and the rear of mixing vessel **104** abuts upper housing section **122**. Within mixing vessel **104**, the front face of chamber **508** may have a substantially ovular shape or a substantially circular shape. The rear of mixing vessel **104** chamber **508** may include an opening configured to form a seal with upper housing section **122**. The opening at the rear of mixing vessel **104** may have a substantially circular shape or a substantially ovular shape. Mixing vessel **104** is sized to accommodate dasher **204** that rotates about a center axis (shown as center axis "A" in FIG. **13**C). FIG. **13**B shows a possible direction of dasher **204** rotation ("R"). Mixing vessel **104** may be shaped such that a distance from the center axis (A) of dasher **204** to the top of the vessel chamber is less than 6 inches, less than 8 inches, less than 10 inches, less than 12 inches, less than 14 inches, or less than 16 inches.

FIGS. **14**A-**14**C show an example of mixing vessel **104** with at least one internal baffle configured to control slush flow within mixing vessel **104**. As shown in FIGS. **13**A, **13**B, and **14**A-**14**B, mixing vessel **104** includes side baffle **105** extending laterally along sidewall **150** of mixing vessel **104** chamber **508**. In some non-limiting embodiments or aspects, side baffle **105** extends from the front of mixing vessel **104** chamber **508** (or approximate thereto) to the rear of mixing vessel **104** chamber **508** (or approximate thereto). In some non-limiting embodiments or aspects, side baffle **105** extends along chamber sidewall **150** in a direction parallel to the center axis (A) of dasher **204**. In some non-limiting embodiments or aspects, side baffle **105** is positioned on a left side (when viewed from the front) of chamber sidewall **150** (e.g., in embodiments in which dasher **204** rotates in a clockwise direction). FIGS. **14**A and **14**C illustrate a clockwise direction of dasher rotation (R) when viewed from the front. Side baffle **105** may be positioned slightly above the center axis (A) of dasher **204**, in some implementations.

Side baffle **105** may include curved surface **151** that conforms to the pathway of dasher **204**, as shown in FIGS. **14**A and **14**B. For example, when viewed along the center axis (A) of dasher **204**, side baffle **105** may protrude inwardly relative the ovular (e.g., elliptical) cross-section of chamber sidewall **150**, where, starting from a bottom end of side baffle **105** at which curved surface **151** of side baffle **105** is vertical or substantially vertical, curved surface **151** may slope gradually inward until reaching inflection point **153**. After reaching inflection point **153**, curved surface **151** may slope more sharply vertically until the top end of side baffle **105** is reached and, thereafter, curved surface **151** of side baffle **105** returns to a curvature in conformance with the ovular cross-section of chamber sidewall **150**. The radial direction of curved surface **151** of side baffle **105** from its bottom to inflection point **153** is generally aligned with the radial movement of dasher **204** and thus the contents of mixing vessel **104** chamber **508**. The cross-sectional geometry of side baffle **105** described above directs the contents of mixing vessel **104** away from a top of mixing vessel **104** chamber **508** (e.g., at a lower radial trajectory than if side baffle **105** was not present, such as the right side of mixing vessel **104** chamber **508** as shown in FIG. **13**B). If side baffle **105** was not present, contents of mixing vessel **104** chamber **508** could flow unimpeded up sidewall **150** to a top interior surface of mixing vessel **104** chamber **508**, which would leave these contents excluded from mixing and/or allow them to escape from mixing vessel **104**. Side baffle **105** thus reduces the amount of frozen material that could otherwise form on the top interior surface of mixing vessel **104** as a result of its contents being rotated upwards.

As shown in FIGS. **13**B, **13**C, and **14**A-**14**C, mixing vessel **104** may include front baffle **107**. If present, front baffle **107** may be positioned at a front top portion of vessel chamber **103** (illustrated in FIG. **13**B). In some non-limiting embodiments or aspects, front baffle **107** extends along the front face of vessel chamber **103** between the right sidewall and the left sidewall of vessel chamber **103**. The rotation of dasher **204** pushes vessel contents towards the front of vessel chamber **103**, where, if left unchecked, contents could build up near the top front, perhaps even creating a frozen mass detrimental to the mixing process. Viewing from the cross-section of FIG. **13**C, front baffle **107** may form an angle relative the front face of vessel chamber **103** (e.g., 100°-150°, 100°-125°, or 105°-120°), which redirects vessel contents that have been forced into the top front of mixing vessel **104** towards the rear of vessel chamber **103**. In some non-limiting embodiments or aspects, front baffle **107** may include a curved surface extending upwardly from the front face of vessel chamber **103** toward a top of vessel chamber **103**. In some such implementations, the angle front baffle **107** forms relative to the front face of vessel chamber **103** varies from a lower angle (e.g., 5°-20°) at a section of front baffle **107** proximate to the front face of vessel chamber **103** to a higher angle (e.g., 75°-90°) at a section of front baffle proximate to the top of vessel chamber **103**.

Front baffle **107** is configured to urge contents away from the top surface of vessel chamber **103** to avoid buildup and overflow on the top of mixing vessel **104**. Front baffle **107** thus reduces the amount of frozen material that could otherwise form on the top front interior surface of mixing vessel **104** as a result of the action of dasher **204**.

As shown in FIGS. **13**C and **14**A-**14**C, mixing vessel **104** may include corner baffle **190**. Corner baffle **190** may be positioned at a front top side of vessel chamber **103**. Corner baffle **190** joins or connects side baffle **105** and front baffle **107**. Thus, if side baffle **105**, front baffle **107**, and corner baffle **190** are each present, corner baffle **190** physically joins side baffle **105** to front baffle **107**. As shown in FIGS. **14**A-**14**B, side baffle **105** and front baffle **107** are orthogonal to each other and if these baffles terminated in a hard corner without corner baffle **190**, slush may not be properly directed. Connecting side baffle **105** and front baffle **107** with corner baffle **190** allows slush to easily flow out of the corner between side baffle **105** and front baffle **107**.

Corner baffle **190** has curved surface **155** that extends from side baffle **105** to front baffle **107**. Curved surface **155** may be convex, as shown in FIG. **14**A. Along its length, corner baffle **190** extends into vessel chamber **103** at a relatively constant distance. In other words, the depth of corner baffle **190** may be relatively constant along the length of corner baffle **190**. The side of vessel chamber **103** in which corner baffle **190** is positioned (e.g., the left side or the right side) can be selected based on the direction in which dasher **204** rotates within mixing vessel **104**. In particular, corner baffle **190** may be positioned such that dasher **204** is directed toward corner baffle **190** while moving upwardly within vessel chamber **103**. For example, in select implementations, corner baffle **190** is positioned at the left top front of vessel chamber **103** when dasher **204** is arranged to rotate in a clockwise direction. This positioning may advantageously force slush downward toward dasher **204** when it contacts corner baffle **190** as the slush moves upwardly with dasher **204**, thereby reducing slush buildup on sidewall **150** and the top of mixing vessel **104**.

It should be understood that, in some non-limiting embodiments or aspects, the disclosed mixing vessel **104** includes one, two, three, or more internal baffles positioned within vessel chamber **103**. In other words, mixing vessel **104** may include side baffle **105**, front baffle **107**, and/or corner baffle **190**. Side baffle **105**, front baffle **107**, and/or corner baffle **190** can reduce slush buildup on the sidewalls and top of vessel chamber **150**, which is important for commercial frozen drink makers as well as household frozen drink makers with significantly less headspace than commercial units.

FIG. **15** is a close-up view **1500** of a user interface such as user interface **112**. According to view **1500**, user interface **112** may include power button **1502**, drink type indicator panel **1504**, manual temperature adjustment and/or temperature offset indicator **1506**, manual temperature adjustment interface **1508**, drink type control dial **1510**, and chill button **1512**. A user may turn frozen drink maker **100** on or off using power button **1502**. A user may select a drink type to process a type of drink product by turning dial **1510** until a selected drink type is indicated via panel **1504**. The user may select, for example, a slush, cocktail, a frappe, a juice, or a dairy/milkshake drink type. Dial **1510** may also include a push button feature that enables a user to start or stop processing of a drink type by pressing dial **1510**. Manual temperature adjustment interface **1508** may include left and right buttons that enable a user to adjust a temperature within a temperature offset band. such as temperature offset band **1602** of FIG. **16** for a milkshake recipe. A user may select chill button **1512** to initiate a chill program and/or recipe whereby frozen drink maker **100** and/or controller **402** maintains the drink product within mixing vessel **104** at a cool temperature without forming a frozen or semi-frozen drink product. In some non-limiting embodiments or aspects, the same cool temperature is maintained for any drink type. For example, controller **402** may receive a signal indicative of the selection of chill button **1512**, and reduce the temperature to, and maintain the temperature at or near, a predefined temperature (e.g., in a range) that should not result in any drink type freezing. In another embodiment, controller **402** may receive a signal indicative of the selection of chill button **1512** and a selection of a drink type from drink type control dial **1510**, and reduce the temperature to, and maintain the temperature at or near, a predefined temperature (e.g., in a range) defined for that particular drink type (e.g., as specified by a drink type object in memory) that should not result in that drink type freezing.

FIG. **16** is a graph **1600** of coarse and fine temperature settings associated with processing a drink product, where such temperature settings may be stored as temperature values in memory, as described elsewhere herein. For example, when a user selects a dairy and/or milkshake recipe and starts a frozen drink processing sequence and/or recipe using dial **1510**, controller **402** will control processes of the dairy/milkshake recipe to adjust the temperature of the drink product to a coarse temperature setting **1604** at −4 degrees Celsius in graph **1600**. A user before, during, or after coarse temperature setting **1604** is reached, may fine tune or adjust the coarse target temperature of the drink type by setting a temperature offset using manual temperature adjustment interface **1508**. The user may push the left arrow button to decrease the recipe target temperature in increments of about 0.4 degrees Celsius to about-5.2 degrees Celsius. As the temperature decreases, the thickness and/or amount of frozen drink particles increases. Hence, manual temperature adjustment indicator **1506** may include a “thickness” label. But different labels may be used such as “temperature offset” or “temperature adjust”, and the like.

The user may push the right arrow button to increase the recipe target temperature in increments of about 0.4 degrees Celsius to about −2.8 degrees Celsius. As the temperature increases, the thickness and/or amount of frozen drink particles decreases. Manual temperature adjustment indicator **1506** may include one or more light indicators that are illuminated in a configuration corresponding to the selected temperature offset. For example, manual temperature adjustment indicator **1506** may have a center light indicator that indicates that a 0-degree Celsius offset is selected (e.g., no offset). Manual temperature adjustment indicator **1506** may include light indicators corresponding to each increment of offset selected above or below the coarse setting (e.g., the 0 degree Celsius offset point). FIG. **16** also shows temperature offset and/or manual adjustment bands associated with various types of drink products, such as Milkshake, Frappuccino, Cocktail, Light, and Traditional. Each of the temperature bands may include a center, coarse, and/or target drink type temperature and user-selectable fine tune offset temperatures above and below the drink type target temperature. In some non-limiting embodiments or aspects, the temperature offset band associated with one recipe is different than that temperature offset band of a different recipe, resulting in the temperature offset increments being different between the different recipes.

FIG. **17** is a close-up view **1700** of another user interface, according to an implementation of the disclosure. According to view **1700**, user interface **112** may include power button **1708**, drink type selector/indicator panel **1702**, manual temperature adjustment and/or temperature offset indicator **1706**, and manual temperature adjustment dial **1704**. A user may turn frozen drink maker **100** on or off using power button **1708**. A user may select a drink type to process a type of drink product by pressing a button associated with a selected drink type, e.g., SLUSHI. The selection of a particular drink type may be indicated by illumination of a light indicator associated with the selected drink type button. For example, FIG. **17** shows that the SLUSHI drink type has been selected by illumination of the white LED indicator next to the SLUSHI button. The user may select, for example, a slush drink, spiked slush or cocktail, a frappe, a frozen juice, or a dairy/milkshake drink type. Manual temperature adjustment dial **1704** may be rotated clockwise or counter-clockwise to set the temperature value and/or target temperature setting within a universal range of drink product temperature values. For example, manual temperature adjustment indicator **1706** may include 10 temperature values or settings corresponding to target temperatures such as illustrated in FIG. **18**.

FIG. **18** is a graph **1800** of temperature values associated with automatic recipe temperature target temperatures and manual temperature adjustments. Graph **1800** shows temperature values 1 through 10 where setting #1 is at −1.3 degrees Celsius and setting #10 is at −7.2 degree Celsius. The ten temperature settings of graph **1800** correspond to the ten light indicators of manual temperature adjustment indicator **1706**. In operation, when a user selects a drink type, e.g., a MILKSHAKE, by pressing the corresponding button in drink type selector/indicator panel **1702**, the button’s adjacent indicator illuminates. Also, if the coarse or automatic temperature value associated with a milkshake is about −4.0 degrees Celsius, which corresponds the setting #7 in graph **1800**, then seven indicators (e.g., light bars) will be illuminated in manual adjustment indicator **1706**. The light bars may be dimmed or flash periodically until the target temperature is reached and/or detected by controller **402**. Interface **112** may emit an audible sound, e.g., a beep or beep sequence when a target temperature is reached. A dimmed or flashing illumination may be changed to a brighter and/or steady illumination when a target temperature is reached. In some non-limiting embodiments or aspects, once a target temperature is reached, controller **402** will cycle compressor **214** on and off to keep a temperature of the drink product within a target temperature range above and/or below the target temperature. For example, the range may be greater than or equal to about 0.2, 0.3, 0.5, or 1.0 degrees Celsius above and below the drink product target temperature. As long as the temperature remains within the target temperature range, controller **402** will not initiate an alert (e.g., audible output) or change in status of any indicators of indicator **1706**.

If the user wants to further decrease the target temperature and/or increase the target thickness of the milkshake to setting #10 of FIG. **18**, the user can turn dial **1704** until all 10 light indicators are illuminated. If the user wants to increase the target temperature to setting #3 of FIG. **18** and/or reduce the target thickness of the milkshake, the user can turn dial **1704** until three indicators bars of indicator **1706** are illuminated as illustrated in FIG. **17**. While FIG. **17** shows an interface using a dial **1704** to manually adjust temperature, other types of interfaces may be used such as, without limitation, up/down buttons, a touch screen, or a slider switch.

FIG. **18** also illustrates how each increment of temperature change between each of the temperature settings #1 to #10 may be nonlinear to account for adequate changes in thickness of a cooled or frozen drink product. As temperature decreases, it requires a larger change in temperature to cause a material or proportional change in the amount of frozen drink particles within or the thickness of a drink product. For example, temperature increment **1802** (between settings #4 and #5) is about 0.6 degrees Celsius, while temperature increment **1804** (between setting #8 and #9), in a lower temperature range, is about −1.0 degrees Celsius. In other implementations, the increment of temperature change between settings may be constant, resulting a linear temperature range. While a range including 10 temperature values or settings is illustrated in FIGS. **17** and **18**, any number of settings and/or temperature ranges may be implemented.

FIG. **19** is a graph **1900** of drive motor **208** current and temperature of a drink product vs. time as the drink product is being processed by frozen drink maker **100** of FIG. **1**. Graph **1900** shows changes in drive motor current **1902** and corresponding drink product temperatures **1904** over time as a drink product is being made. Graph **1900** illustrates how the current **1902** applied to drive motor **208** increases as the temperature **1904** decreases, causing the thickness of the drink product to increase, which results in an increased resistance of the drink product to the rotation of dasher **204** which, in turn, requires increased motor power and/or current **1902** to drive dasher **204** against the resistance. When current **1902**, or power, or torque, reaches or exceeds a threshold or motor condition limit **1906**, e.g., about 40 Watts and/or about 0.3 amps current, controller **402** may deactivate the cooling circuit, e.g., stop coolant and/or refrigerant flow to evaporator **202**, to allow temperature **1904** to increase and, thereby reduce the thickness of the drink product to reduce current **1902** of drive motor **208** to below motor condition limit **1906**. Controller **402** may automatically adjust the temperature setting associated with a particular drink type, which may have been fine-tuned by a user selection of a manual temperature adjustment and/or temperature offset, to a new temperature setting corresponding to a second target temperature, where the magnitude of motor current **1902** is lower than motor condition limit **1906**. The second target temperature may be set to, for example, 0.25, 0.5, 0.75, 1, 1.25, 1.5, or 2.0 degrees Celsius above (by a relatively small offset) the initial and/or first target temperature. In this way, controller **402** prevents an overcurrent condition and possible damage to drive motor **208**. This may also enable operation of frozen drink maker **100** and dasher **204** to continue by preventing excessive buildup of ice within mixing vessel **104**, e.g., prevents drive motor **208** from stalling. Otherwise, drive motor **208** would stall and frozen drink maker **100** would be jammed up, blocking slush output from mixing vessel **104** and requiring a user to defrost and/or unblock mixing vessel **104** before normal operations can be resumed. Hence, this stall prevention enables frozen drink maker **100** to provide some slush output. Further, an excessive current or power condition of drive motor **208** caused by an object blocking rotations of dasher **204** can also be prevented. Controller **402** may perform actions in addition to stopping drive motor **208**, such as shutting down compressor **214**. Graph **1900** also shows how controller **402** may continuously and/or periodically monitor temperature associated with a drink product within mixing vessel **104** via temperature sensor(s) **406** to enable continuous control of components such as compressor **214**, and other components, of frozen drink maker **100** to enable automatic control of the temperature of a drink product.

FIG. **20** is a flow diagram of a process **2000** for making a cooled drink product using a recipe for initial or coarse temperature and/or texture control and then using a user input to fine tune the temperature and/or texture of the drink product. In certain implementations, process **2000** includes: receiving, into mixing vessel **104**, a drink product (Step **2002**); mixing, using a mixer and/or dasher **204** driven by drive motor **208**, the drink product within mixing vessel **104** (Step **2004**); cooling, using a cooling circuit such as a refrigeration circuit including evaporator **202**, the drink product within mixing vessel **104** (Step **2006**); detecting, via temperature sensor(s) **406**, a temperature associated with the drink product and outputting a temperature signal (Step **2008**); storing, in memory **404**, a drink object representing a drink type, the drink object specifying a first temperature value and/or setting corresponding to a first target temperature (Step **2010**); receiving, at controller **402**, the temperature signal (Step **2012**); controlling, by controller **402**, the temperature associated with the drink product by controlling the cooling circuit, e.g., by activating or deactivating compressor **214** to initiate or stop refrigerant flow through evaporator **202**, based on the received temperature signal, the first temperature value, and/or a manual temperature adjustment (Step **2014**); and receiving a user input to adjust the manual temperature adjustment (Step **2016**). The user input may be indicative of a desired thickness corresponding to the manual temperature adjustment. In some non-limiting embodiments or aspects, the manual temperature adjustment may be customized per drink type. In certain implementations, the manual adjustment is universal for all drink types. In some non-limiting embodiments or aspects, the manual temperature adjustment is finer and/or for a smaller range specific to a drink type (e.g., corresponding to FIG. **16**) and in other implementations coarser and/or for a larger range not specific to a drink type—e.g., spanning multiple (e.g., all) drink types, thereby enabling a user greater latitude in adjusting thickness and/or temperature.

FIG. **21** is a flow diagram of process **2100** for automatically detecting when drive motor current is too high and/or a drink product is too thick and, in response, adjusting the temperature of the drink product to reduce drive motor current and/or to increase the temperature of the drink product to reduce a thickness of the drink product. In certain implementations, process **2100** includes: receiving, in mixing vessel **104**, the drink product (Step **2102**); mixing, using a mixer and/or dasher **204** driven by drive motor **208**, the drink product within mixing vessel **104** (Step **2104**); cooling, using a cooling circuit such as evaporator **202**, the drink product within mixing vessel **104** (Step **2106**); measuring, via temperature sensor(s) **406**, a temperature associated with the drink product and outputting a temperature signal (Step **2108**); measuring, via motor condition sensor(s) **406**, a motor condition associated with drive motor **208** and outputting a motor condition signal (Step **2110**); storing, in memory **404**, a first temperature value corresponding to a first target temperature and storing a motor condition limit (Step **2112**); receiving, at controller **402**, the temperature signal and the motor condition signal (Step **2114**); and controlling the temperature associated with the drink product by controlling the cooling circuit, e.g., by activating or deactivating compressor **214** to initiate or stop the refrigerant flow through evaporator **202**, based at least on the received temperature signal, the received motor condition signal, the first temperature setting, and the motor condition limit (Step **2116**).

In some non-limiting embodiments or aspects, controller **402** may stop and/or deactivate drive motor **208** to stop rotation of dasher **204** when the motor condition signal exceeds a motor knockdown threshold, e.g., the motor current or power is too high and/or high enough to damage drive motor **208**, which may be caused by an excessive buildup of ice within mixing vessel **104**. Excessive ice build-up may be caused, for example, by filling mixing vessel **104** with only water or a liquid predominantly consisting of water. Shutdown of drive motor **208** may also prevent damage to dasher **204** caused by excessive build-up of hard ice. Controller **402** may perform other actions in addition to deactivating drive motor **208** or alternatively such as issuing an alert, via user interface **112**, to a user to add more ingredients, such as sugar or alcohol, to the drink product or issuing an alert to the user to turn off frozen drink maker **100**. A different motor shutdown threshold for motor **208** may be set higher than the motor knockdown threshold limit. In this way, controller **402** may attempt to increase temperature in mixing vessel **104** when a motor knockdown threshold limit is reached, but only shut down and/or stop drive motor **208** when a motor shutdown threshold is reached to prevent damage to drive motor **208**. Controller **402** may take action based on determining whether the motor knockdown threshold limit or the motor shutdown limit has been reached or exceeded for a period of time, e.g., 0.5, 1.0, 1.5, 2.0, 5 seconds or more. By observing motor current and/or power for a period of time, a false positive and/or reading of current and/or power may be eliminated.

FIG. **22**A shows a dual-use cooling fan **2202** within a housing of a drink maker **2200** including a refrigeration system having a condenser **2208** and compressor **2210**. Drink maker **2200** also includes a drive motor **2204** configured to drive rotation of dasher **2212** during processing of a drink product. Dual-use cooling fan **2202** draws an air flow through condenser **2208** and directs the air flow, via an air channel **2206**, toward drive motor **2204**. The air flow passes over and adjacent to condenser coils as it passes through condenser **2208** to cool the refrigerant passing through condenser **2208** within a closed loop refrigeration system. The air flow also passes along a surface and/or surfaces of drive motor **2204** to effect cooling of drive motor **2204**. While FIG. **22**A shows a configuration where drive motor **2204** and condenser **2208** are positioned at about right angles with respect to dual-use cooling fan **2202**, other configurations, arrangements, or orientations may be implemented such that dual-use cooling fan **2202** provides a cooling air flow to condenser **2208** and drive motor **2204**.

In some non-limiting embodiments or aspects, a drink maker, such as drink maker **2200**, includes a mixing vessel, like mixing vessel **104**, arranged to receive a drink product. Drink maker **2200** includes a mixing component such as dasher **2212** or another type of mixing component, driven by drive motor **2204**, that is arranged to mix the drink product within mixing vessel **104**. A refrigeration system is arranged to cool the drink product within mixing vessel **104** that includes a condenser, such as condenser **2208**. Cooling fan **2202**, e.g., a dual-use cooling fan, is configured to concurrently cool drive motor **2204** and condenser **2208**. Cooling fan **2202** may provide air flow through condenser **2208** to cool refrigerant flowing through condenser **2208**. Cooling fan **2202** may provide air flow along a surface of drive motor **2204** to cool drive motor **2204**. Cooling fan **2202**, drive motor **2204**, and condenser **2208** may be positioned such that air generated by cooling fan **2202** passes serially through condenser **2208** and along a surface of drive motor **2204**.

A first portion of air generated by cooling fan **2202** may cool condenser **2208** and a second portion of air generated by cooling fan **2202** may cool drive motor **2204**. Condenser **2208** may include a plurality of coils that carry coolant and/or refrigerant within a closed loop of the refrigeration circuit. When cooling fan **2202** provides air flow through condenser **2208** to cool refrigerant flowing through condenser **2208**, the air flow may travel adjacent to and/or around the plurality of coils. A cooling channel **2206** may extend between cooling fan **2202** and drive motor **2204** where cooling channel **2206** provides cooling air flow between cooling fan **2202** and drive motor **2204**. Cooling channel **2206** may be at least partially formed by a duct and/or ducting. The ducting may include plastic, metals, composite materials, and the like. A cooling channel **2206** may extend between cooling fan **2202** and condenser **2208**, where the cooling channel provides cooling air flow between cooling fan **2202** and condenser **2208**. The cooling channel may be at least partially formed by a duct. Cooling fan **2202** may include a centrifugal fan, a cross-flow fan, a tangential fan, a volute fan, a backward curved fan, a forward curved fan, a blower fan, a squirrel-cage fan, and/or an axial fan.

In some non-limiting embodiments or aspects, a cooling fan, such as cooling fan **2202**, is configured for cooling a drive motor, such as drive motor **2204**, and a condenser, such as condenser **2208**, within a housing of a drink maker. Cooling fan **2202** may include an air inlet configured to receive an air flow, an impeller configured to generate the air flow; and an air outlet configured to output the air flow through condenser **2208** and along a surface of drive motor **2204**.

FIG. **22**B shows another implementation of a dual-use cooling fan **2222** within the housing of a drink maker **2220** including a drive motor **2224**, a dasher **2226**, a compressor **2230**, and a condenser **2228**. Drive motor **2224** is coupled to and drives rotation of dasher **2226** and also drives rotation of cooling fan **2222** via gears **2236**. Cooling fan **2222** includes an air outlet **2238** that directs air flow from cooling fan **2222** through air channel **2232** which may include ducting **2234** that directs air flow through condenser **2228** to cool refrigerant flowing through condenser **2228**.

FIG. **22**C shows a perspective view **2240** of dual-use cooling fan **2222** within housing **2242** of drink maker **2220**. Cooling fan **2222** may be a centrifugal fan and/or another type of fan, as described herein. Cooling fan **2222** may include an impeller **2244** that draws air flow into cooling fan

2222 via inlet 2236 and then expels air downward at about a right angle via outlet 2238 with respect to inlet 2236. The air flow exiting outlet 2238 flows downward past drive motor 2224, including along a surface of drive motor 2224, and through air channel 2232, which may include ducting 2234 that directs the air flow through condenser 2228 (adjacent to and/or around coils of condenser 2228) to effect cooling of refrigerant passing through the coils.

FIG. 23 is a flow diagram of a process 2300 for operating dual-use cooling fan 2202 or 2222 of FIGS. 22A and 22B respectively. Process 2300 facilitates concurrently cooling condenser 2208 (or condenser 2228) and drive motor 2204 (or drive motor 2224) within a housing of a drink maker using a cooling fan 2202 or 2222 respectively by: activating drive motor 2204 (or drive motor 2224) that is arranged to drive rotation of dasher 2212 (or dasher 2226) within a mixing vessel of a drink maker (Step 2302); activating compressor 2210 (or compressor 2230) of a refrigeration circuit of the drink maker (Step 2304); and activating cooling fan 2202 (or cooling fan 2222) to concurrently generate air flow through condenser 2208 (or condenser 2228) and along a surface of drive motor 2204 (or drive motor 2224) (Step 2306).

As previously mentioned, frozen drink maker 100 may include pour-in opening 106 through which mixing vessel 104 can receive ingredients to be mixed to produce a drink product. An illustrative pour-in-opening 106 for a frozen drink maker 100 is shown in FIGS. 24A-24C. Frozen drink maker 100 includes mixing vessel 104 with a substantially cylindrical chamber and housing 102 with upper housing section 122. FIG. 24A shows a perspective side view of pour-in opening 106. FIG. 24B shows a front view of pour-in opening 106 of FIG. 24A, and FIG. 24C shows a perspective view of pour-in opening 106 of FIG. 24A from the left side of mixing vessel 104 (when viewed from the front view). Pour-in opening 106 may facilitate the addition of fluids, liquids, slush, or other ingredients to mixing vessel 104 while dasher 204 is active, as well as minimizing spillage and preventing finger insertion during use.

In some non-limiting embodiments or aspects, pour-in opening 106 may include cover 160 to seal pour-in opening 106, as shown in FIGS. 24A and 24C. A detailed perspective view of a sample cover 160 for pour-in opening 106 is shown in FIG. 25. If present, cover 160 may be hingedly connected to upper housing section 122 of mixing vessel 104. Cover 160 may be moved between an open position in which pour-in opening 106 is accessible to a user and a closed position in which pour-in opening 106 is not accessible to a user. Although not illustrated in the accompanying figures, pour-in opening 106 may also include a grate to restrict objects from entering aperture 162. If present, a grate may reduce the risk of solids greater than a certain size and/or having one or more certain shapes entering mixing vessel 104, which can cause damage.

FIG. 26 shows a perspective view of a sample pour-in opening 106. Pour-in opening 106 includes surface 164 that inclines radially with respect to a center axis of dasher 204 (shown as axis "A" in FIG. 24A). Surface 164 reduces possible splashing as mixing vessel 104 is filled. Surface 164 also prevents slush contained within mixing vessel 104 from being pushed out of pour-in opening 106. Surface 164 has aperture 162. Although FIG. 26 shows only one aperture 162, additional apertures may also be present. Aperture 162 is in fluid communication with an interior chamber of mixing vessel 104. In some non-limiting embodiments or aspects, aperture 162 extends laterally along surface 164 in a direction parallel to the center axis "A" of dasher 204. Aperture 162 may be shaped as a slot, as shown in FIG. 26, or may have a different shape. If shaped as a slot, aperture 162 may be longer or wider than shown in FIGS. 24A-24C and/or may have a different ratio of length to width than shown. Further, aperture 162, as a slot or another oblong shape, may have its major axis aligned parallel or perpendicular to the axis of mixing vessel 104, or at any other angle relative to the axis of mixing vessel 104. Aperture 162, for example, in the form of a slot, may be sized small enough (at least in width) to not allow passage of a human finger, at least not the entire length of a human finger, to thereby prevent a user from sticking one or more fingers into mixing vessel 104.

Pour-in opening 106 may optionally include one or more lips 166*a*, 166*b* extending up from a perimeter of surface 164 to form a well that feeds into aperture 162, as shown in FIG. 26. One or more lips 166*a*, 166*b* may reduce overflow spill when a liquid is poured into mixing vessel 104. If desired, pour-in opening 106 may also include a grate (not illustrated) covering at least a portion of aperture 162. For safety concerns, users should not contact dasher 204 while it is rotating. The geometry of pour-in opening 106 (including aperture 162 as described above) may inhibit or prevent a user from reaching into mixing vessel 104 even when cover 160 is in an open position and/or dasher 204 is rotating.

Pour-in opening 106 may be positioned on a top of mixing vessel 104, near its rear end, as shown in FIGS. 24A-24C, opposite dispenser assembly 108. Positioning pour-in opening 106 near the rear of mixing vessel 104 avoids interference with slush circulation in the front of frozen drink maker 100, which can lead to waste and non-homogeneous texture. With pour-in opening 106 positioned at the rear of mixing vessel 104, the front ⅔ of mixing vessel 104 has a continuous and smooth internal shape to provide good slush flow and minimize migration of the slush out of the top. By positioning pour-in opening 106 near the rear of mixing vessel 104, opening 106 is located in a position where there is less possible buildup of frozen and/or slush materials, enabling less obstructed pouring and reducing possible buildup of ice and/or slush material at opening 106 during processing.

Surface 164 of pour-in opening 106 is sloped to direct incoming ingredients to enter mixing vessel 104 in an entry direction, which is the same as the direction of dasher 204 rotation. This prevents the rotating frozen mixture from exiting mixing vessel 104 through pour-in opening 106. In some non-limiting embodiments or aspects, when dasher 204 is rotating in a clockwise direction when viewed from the front of frozen drink maker 100, opening 106 is positioned on the right side of mixing vessel 104. Aperture 162 may be positioned to extend laterally along surface 164 in a direction parallel to the center axis (A) of dasher 204, whereas in other implementations, when dasher 204 is rotating in a counter-clockwise direction when viewed from the front of frozen drink maker 100, opening 106 is positioned on the left side of mixing vessel 104.

FIGS. 27A-27D illustrate a sample pour-in opening 106 in which surface 164 of pour-in opening 106 is shaped to slope downwardly toward a rear of mixing vessel 104. In some such implementations, one or more apertures 162 may be positioned at a bottom portion of surface 164. Shaping surface 164 to include a rearward slope can increase the volume capacity of pour-in opening 106 and reduce spillage. In implementations in which surface 164 of pour-in opening 106 is sloped relative to the center axis (A) of dasher 204, surface 164 may be shaped such that a section of surface 164 closest to a front of mixing vessel **104** is positioned farther away from the center axis (A) of dasher **204** than a section of surface **164** closest to a rear of mixing vessel **104**.

FIG. **28** illustrates a sample method **2800** of using a pour-in opening **106** for a frozen drink maker. As shown in FIG. **28**, method **2800** includes optionally opening a cover of the frozen drink maker to provide access to the pour-in opening (block **2802**). Method **2800** also includes introducing one or more ingredients to a mixing vessel of the frozen drink maker via the pour-in opening (block **2804**). The one or more ingredients may be added to the mixing vessel while the mixing vessel is actively mixing (e.g., while the dasher is rotating). Method **2800** further includes dispensing a drink product from the frozen drink maker (block **2806**). The drink product may be dispensed while the dasher is rotating, if desired.

FIGS. **29**A-**29**D show a dispenser assembly **2900** for dispensing a drink product from frozen drink maker **100**, according to a first illustrative implementation of the disclosure. As shown in FIG. **29**A, dispenser assembly **2900** may include dispenser housing **2904** for housing the component parts of dispenser assembly **2900**. Housing **2904** may have a first portion **2904***a* attached to an outer surface of frozen drink maker **100** adjacent to a spout **2902** and a second portion **2904***b* spaced apart from spout **2902** and extending outward from the outer surface. In some non-limiting embodiments or aspects, housing **2904** may have an inverted L-shape. However, the disclosure contemplates other suitable shapes of housing **2904**. Handle **120** of frozen drink maker **100** may have an upper portion **120***a* in the form of a user-actuatable lever **2906** and a lower portion **120***b* attached to second portion **2904***b* of housing **2904**.

As shown in FIGS. **29**B and **29**C, lever **2906** may be rotatable relative to second portion **2904***b* of housing **2904** about a first pivot member **2908**. In some non-limiting embodiments or aspects, first pivot member **2908** may be a rod or pin **2910** extending through second portion **2904***b* of housing **2904** and lower portion **120***b* of handle **120**. However, the disclosure contemplates other suitable types of pivot members **2908**. A link member **2912** may operatively couple to lower portion **120***b* of handle **120**. In some non-limiting embodiments or aspects, link member **2912** may be insertable into lower portion **120***b* of handle **120**. Link member **2912** may be rotatable relative to lever **2906** about a second pivot member **2914**. In some non-limiting embodiments or aspects, second pivot member **2914** may be a rod or pin **2916** extending through link member **2912** and through lower portion **120***b* of handle **120**. However, the disclosure contemplates other suitable types of pivot members **2914**. Bracket member **2918** may operatively couple to link member **2912** and may be attached to first portion **2904***a* of housing **2904**. In some non-limiting embodiments or aspects, link member **2912** may be insertable into a portion of bracket member **2918**. Bracket member **2918** may be rotatable relative to link member **2912** about a third pivot member **2920**. In some non-limiting embodiments or aspects, third pivot member **2920** may be a rod or pin **2922** extending through bracket member **2918** and link member **2912**. However, the disclosure contemplates other suitable types of pivot members **2920**. Bracket member **2918** may also be rotatable relative to first portion **2904***a* of housing **2904** about a fourth pivot member **2924**. In some non-limiting embodiments or aspects, fourth pivot member **2924** may be a rod or pin **2926** extending through first portion **2904***a* of housing **2904** and bracket member **2918**. However, the disclosure contemplates other suitable types of pivot members **2924**. A seal **2928** may attach to bracket member **2918**. Seal **2928** may be configured to seal spout **2902** to prevent inadvertent dispensing of the drink product. In some non-limiting embodiments or aspects, seal **2928** may be a lip seal that covers spout **2902**. However, other suitable types of seals **2928** are contemplated by this disclosure. For example, in some non-limiting embodiments or aspects, seal **2928** may be, or may include, a plug that is made out of one or more relatively dense materials having a relatively high durometer and that extends into spout **2902** to seal spout **2902**. Spout **2902** may include a safety grate **2930** or other mechanism to prevent the user from inadvertently inserting his or her fingers into spout **2902** (FIG. **29**D).

In some non-limiting embodiments or aspects, the dispenser assembly (e.g., dispenser assembly **108**, dispenser assembly **3200**, described herein) may include one or more slots **2909** (FIG. **29**D) configured to engage with connectors (e.g., connectors **3432***a*, **3432***b*, described herein) on a cover (e.g., shroud **116**, spout cover **3410***a*, **3410***b*, described herein). In this manner, covers can be easily attached and unattached from at least a portion of the dispenser assembly, for the purpose of attaching connectors, adaptors, and other parts, and for the purpose of case of cleaning.

To dispense the drink product, in some non-limiting embodiments or aspects, actuation of lever **2906** by the user may cause link member **2912** to move upward relative to housing **2904**. Because bracket member **2918** is attached to both link member **2912** and to housing **2904**, a portion of bracket member **2918** may move upward with link member **2912** while the remainder of bracket member **2918** is forced to pivot about fourth pivot member **2924**. This in turn may cause seal **2928** to move into an open position. When seal **2928** moves into the open position, seal **2928** may uncover spout **2902** to dispense the drink product. Advantageously, in the open position, seal **2928** may be angled at about 45-60 degrees with respect to spout **2902** to direct the drink product downward toward the beverage cup. Release of lever **2906** by the user may allow the components to return to their unactuated position, allowing seal **2928** to again close spout **2902**.

FIGS. **30**A and **30**B show a dispenser assembly **3000** for dispensing a drink product from frozen drink maker **100**, according to a second illustrative implementation of the disclosure. Dispenser assembly **3000** may be substantially similar to dispenser assembly **2900**. For example, as shown in FIG. **30**A, dispenser assembly **3000** may include a dispenser housing **3004** for housing the component parts of dispenser assembly **3000**. Housing **3004** may have a first portion **3004***a* attached to an outer surface of frozen drink maker **100** adjacent to spout **3002** and a second portion **3004***b* spaced apart from spout **3002** and extending outward from the outer surface. Handle **120** may have an upper portion **120***a* in the form of a user-actuatable lever **3006** and a lower portion **120***b* attached to second portion **3004***b* of housing **3004**. Lever **3006** may be rotatable relative to second portion **3004***b* of housing **3004** about a first pivot member **3008**. Link member **3012** may operatively couple to lower portion **120***b* of handle **120**. Link member **3012** may be rotatable relative to lever **3006** about a second pivot member **3014**.

As shown in FIG. **30**B, bracket member **3018** may operatively couple to link member **3012** and may be attached to first portion **3004***a* of housing **3004**. In some non-limiting embodiments or aspects, bracket member **3018** may have an inverted L-shape, as shown. However, the disclosure contemplates other suitable shapes of bracket member **3018**. Bracket member **3018** may be rotatable relative to link member **3012** about a third pivot member

3020. Bracket member **3018** may also be rotatable relative to first portion **3004***a* of housing **3004** about a fourth pivot member **3024**. Seal **3028** may attach to bracket member **3018**. Seal **3028** may be configured to seal spout **3002** in a closed position. In some non-limiting embodiments or aspects, seal **3028** may be a lip seal that covers spout **3002**. However, in some non-limiting embodiments or aspects, seal **3028** may be, or may include, a plug that is made out of one or more relatively dense materials having a relatively high durometer and that extends into spout **3002** to seal spout **3002**.

To dispense the drink product, in some non-limiting embodiments or aspects, actuation of lever **3006** by the user may cause link member **3012** to move upward relative to housing **3004**. Because bracket member **3018** is attached to both link member **3012** and to housing **3004**, a portion of bracket member **3018** may move upward with link member **3012** while the remainder of bracket member **3018** is forced to pivot about fourth pivot member **3024**. This in turn may cause seal **3028** to move into an open position. When seal **3028** moves into the open position, seal **3028** may uncover spout **3002** to dispense the drink product. Advantageously, in the open position, seal **3028** may be angled at about 45-60 degrees with respect to spout **3002** to direct the drink product downward toward the beverage cup. Release of lever **3006** by the user may allow the components to return to their unactuated position, allowing seal **3028** to again close spout **3002**.

Advantageously, unlike other dispenser mechanisms, dispensing assemblies **2900**, **3000** of this disclosure do not rely on leverage against the outer surface of frozen drink maker **100** to open seal **2928**, **3028**. This may reduce wear and tear of the component parts of dispenser assembly **2900**, **3000** and on the outer surface of frozen drink maker **100**. Furthermore, because seal **2928**, **3028** moves both horizontally and vertically with respect to spout **2902**, **3002** to unseal spout **2902**, **3002**, the open position of seal **2928**, **3028** may provide less obstruction to the flow of the drink product from spout **2902**, **3002**.

FIGS. **31**A and **31**B illustrate in greater detail spout cover or shroud **116** for covering a portion of dispenser assembly **2900**, **3000**, according to an illustrative implementation of the disclosure. As shown in FIG. **31**A, shroud **116** may include a first panel section **3102***a* and a second panel section **3102***b* extending substantially parallel to one another. Front section **3104** may extend between panel sections **3102***a*, **3102***b*. In some non-limiting embodiments or aspects, panel sections **3102***a*, **3102***b* may be substantially flat, while front section **3104** may be curved, as shown. In some non-limiting embodiments or aspects, front section **3104** may include an arcuate upper edge **3106** configured such that actuation of handle **120** is not impeded. However, the disclosure contemplates other suitable shapes of upper edge **3106**, such as the rectilinear shape shown in FIG. **1**. As shown in FIG. **31**B, panel sections **3102***a*, **3102***b* may be configured to form a removable snap fit with dispenser housing **2904**, **3004**. A length of shroud **116** may be selected to cover the component parts of dispensing assemblies **2900**, **3000** other than handle **120** to improve the aesthetic appearance of frozen drink maker **100**. The shroud **116** may also aid in directing the drink product downward toward the beverage cup. Shroud **116** may be made of a dishwasher safe material for easy cleaning.

In some non-limiting embodiments or aspects, at least front section **3104** of shroud **116** may be vertically moveable relative to dispenser assembly **2900**, **3000**. For example, in some non-limiting embodiments or aspects, front section **3104** may be moveable relative to first panel section **3102***a* and second panel section **3102***b*. In some non-limiting embodiments or aspects, front section **3104** may be hingedly connected to first and second panel sections **3102***a*, **3102***b* or may be vertically slidable relative first and second panel sections **3102***a*, **3102***b*. Such movement may be useful when dispensing a non-frozen, water-based beverage to prevent the beverage from dispensing at too lateral of a trajectory from spout **2902**, **3002**. Such a lateral trajectory may result in at least a portion of the beverage not dispensing into a receiving vessel located below spout **2902**, **3002**.

It should be appreciated that the various implementations described herein are not limited to making frozen or semi-frozen drinks, but may be applied to produce a cold and/or cooled drink product that is cooler than a received drink product, but not frozen or semi-frozen. For example, in some non-limiting embodiments or aspects, the same or similar mechanisms and/or techniques may be used as part of a cold drink machine and/or cooled drink maker to produce, maintain and dispense cold drinks.

As discussed with respect to FIG. **4**, actions associated with configuring or controlling a frozen drink maker such as frozen drink maker **100** and processes described herein can be performed by one or more programmable processors executing one or more computer programs to control or to perform all or some of the operations described herein. All or part of frozen drink maker **100** systems and processes can be configured or controlled by special purpose logic circuitry, such as, an FPGA and/or an ASIC or embedded microprocessor(s) localized to the instrument hardware.

Non-transitory machine-readable storage media suitable for embodying computer program instructions and data include all forms of non-volatile storage area, including by way of example, semiconductor storage area devices, such as EPROM (erasable programmable read-only memory), EEPROM (electrically erasable programmable read-only memory), and flash storage area devices; magnetic disks, such as internal hard disks or removable disks; magneto-optical disks; and CD-ROM (compact disc read-only memory) and DVD-ROM (digital versatile disc read-only memory).

Elements of different implementations described may be combined to form other implementations not specifically set forth previously. Elements may be left out of the systems described previously without adversely affecting their operation or the operation of the system in general. Furthermore, various separate elements may be combined into one or more individual elements to perform the functions described in this specification.

The disclosure describes a removeable collection tray that may be disposed within the unit underneath the vessel/evaporator. The tray may be configured to collect condensation dripping off the vessel or spills resulting from filling the vessel. The tray may also be used to collect water from cleaning between uses. The tray of this disclosure may be sized to collect up to 16 ounces of liquid. The tray may also be made of a dishwasher safe material for easy cleaning.

This application describes illustrative systems, methods, and devices that provide a removeable condensation tray below vessel to reduce cleaning concerns and increase case of use.

In some non-limiting embodiments or aspects, a removeable collection tray for a frozen drink maker includes a collection chamber for receiving liquid and a handle. The collection chamber is configured to be removably inserted into a slot in a housing of the frozen drink maker adjacent an evaporator. In some non-limiting embodiments or aspects, the collection chamber is vertically spaced from a bottom side of the housing when inserted into the slot. In some non-limiting embodiments or aspects, the collection chamber includes an evaporator-facing surface. The evaporator-facing surface has a shape corresponding to an outer surface of the evaporator. In some non-limiting embodiments or aspects, the shape is semi-cylindrical. In some non-limiting embodiments or aspects, the handle, the evaporator-facing surface, and three other side walls define the collection chamber. In some non-limiting embodiments or aspects, the collection chamber has a liquid volume capacity of 16 ounces. In some non-limiting embodiments or aspects, the removeable collection tray is made from a dishwasher-safe material. In some non-limiting embodiments or aspects, a user-facing surface of the handle is flush with a user interface of the housing when the collection chamber is fully inserted into the slot. In some non-limiting embodiments or aspects, an underside of the handle has one or more ribs for adding structural integrity between the handle and the collection chamber. In some non-limiting embodiments or aspects, the slot is defined between at least one rail and a top surface of the housing.

In some non-limiting embodiments or aspects, a method of removing a collection tray from a frozen drink maker includes removing a mixing vessel from a housing of the frozen drink maker and, after the removing of the mixing vessel from the housing, removing the collection tray from the housing. In some non-limiting embodiments or aspects, removing the mixing vessel from the housing includes removing the mixing vessel and an attached dispenser from the housing. In some non-limiting embodiments or aspects, removing the collection tray from the housing includes pulling the handle toward the user. In some non-limiting embodiments or aspects, pulling the handle toward the user includes sliding the collection tray along a slot in the housing toward the user. In some non-limiting embodiments or aspects, the method also includes fully disengaging the collection tray from the slot in the housing.

The application, in various implementations, addresses deficiencies associated with controlling slush flow within a mixing vessel of a frozen drink maker. This application describes illustrative systems, methods, and devices that use one or more internal baffles positioned within the mixing vessel to direct slush flow for thorough mixing and to prevent blockage within the mixing vessel. The one or more internal baffles, which control the flow of contents within the mixing vessel, can also reduce waste (e.g., waste caused by slush sticking to the vessel instead of dispensing through the spout).

In a first aspect, a mixing vessel for a frozen drink maker is described and the mixing vessel has at least one internal baffle. The mixing vessel includes a curved sidewall defining a substantially cylindrical vessel chamber therein. The vessel chamber includes a front, a rear, a right side, a left side, and a top. The mixing vessel also includes a corner baffle configured to control slush flow within the vessel chamber. The corner baffle is positioned at the front top of the vessel chamber on either the right side or the left side.

The mixing vessel may be configured to accommodate a dasher that rotates within the vessel chamber about a center axis and the corner baffle may be positioned such that the dasher is directed toward the corner baffle while moving upwardly within the vessel chamber. In these and other implementations, the corner baffle is positioned on the left side of the vessel chamber and the dasher is arranged to rotate in a clockwise direction. In select implementations, a distance from the center axis of the dasher to the top of the vessel chamber is less than 16 inches.

The corner baffle may extend out from the front into the vessel chamber at a relatively constant distance. In some non-limiting embodiments or aspects, the mixing vessel also includes a side baffle extending laterally along the vessel chamber from the front to the rear. The side baffle may include a curved surface that protrudes inwardly relative to a cross-section of the vessel chamber when viewed along a center axis of the vessel chamber. In these and other implementations, the side baffle is positioned on the left side or the right side of the vessel chamber. The side baffle and the corner baffle may both be positioned on either the left side or the right side of the vessel chamber. In some non-limiting embodiments or aspects, the mixing vessel also includes a front baffle positioned at the front of the vessel chamber extending across the top. In these and other implementations, the front baffle forms an angle of between 100°-150° relative the front of the vessel chamber. In various implementations in which the front baffle is present, the mixing vessel also includes a side baffle extending laterally along the vessel chamber from the front to the rear, and the corner baffle has a curved surface that extends from the side baffle to the front baffle. The substantially cylindrical vessel chamber may have an oval-shaped cross-section.

In another aspect, a mixing vessel for a frozen drink maker is described and the mixing vessel has at least three internal baffles. The mixing vessel includes a curved sidewall defining a substantially cylindrical vessel chamber therein. The vessel chamber includes a front, a rear, a right side, a left side, and a top. The mixing vessel includes a corner baffle positioned at the front top of the vessel chamber on either the right side or the left side. The mixing vessel also includes a side baffle extending laterally along the vessel chamber from the front to the rear. The mixing vessel further includes a front baffle positioned at the front of the vessel chamber extending across the top.

In some non-limiting embodiments or aspects, the side baffle and the corner baffle are both positioned on either the left side or the right side of the vessel chamber. In these and other implementations, the mixing vessel is configured to accommodate a dasher that rotates within the vessel chamber about a center axis. The corner baffle and the side baffle are positioned such that the dasher is directed toward the corner baffle and the side baffle while moving upwardly within the vessel chamber. In these and other implementations, the corner baffle and the side baffle are positioned on the left side of the vessel chamber and the dasher is arranged to rotate in a clockwise direction. In select implementations, a distance from the center axis of the dasher to the top of the vessel chamber is less than 16 inches. The corner baffle may extend out from the front into the vessel chamber at a relatively constant distance.

In yet another aspect, a frozen drink maker is described. The frozen drink maker includes a mixing vessel, a housing, a dasher, and a dispenser assembly. The mixing vessel has a front, a rear, and a curved sidewall defining a vessel chamber therein. The housing has an upper housing section abutting the rear of the mixing vessel. The dasher is arranged to rotate within the mixing vessel about a center axis. The dispenser assembly is at the front of the mixing vessel. The mixing vessel includes at least two internal baffles configured to control slush flow within the vessel chamber.

In some non-limiting embodiments or aspects, the mixing vessel includes at least three internal baffles configured to control slush flow within the vessel chamber. In some such implementations, the at least three internal baffles include:

(1) a corner baffle positioned at a front top of the vessel chamber on either a right side or a left side, (2) a side baffle extending laterally along the vessel chamber from the front to the rear, and (3) a front baffle positioned at the front of the vessel chamber extending across the top. In these and other implementations, the dasher rotates in a clockwise direction, and the corner baffle and the side baffle are positioned on a left side of the vessel chamber when viewed from a front of the frozen drink maker.

The application, in various implementations, addresses deficiencies associated with controlling temperatures of drink products using recipes in a more adaptive and user-specific manner.

This application describes illustrative systems, methods, and devices that enable a drink maker to automatically control a temperature of a drink product based on a preset recipe target temperature stored in memory, while further allowing a user to adjust the preset temperature via a user input to enable the frozen drink maker to more flexibly achieve desired temperatures and/or textures tailored to the preferences of different users. The application also describes illustrative systems, methods, and devices that enable a drink maker to automatically control a temperature of a drink product based on a preset recipe target temperature stored in memory, while further monitoring a condition of a drive and/or dasher motor, such as current or power, and, if the current or power is too high, increasing the temperature of the drink product to reduce the thickness of the drink product and, thereby, reduce the current and/or power used by the drive and/or dasher motor to prevent damage to the drive motor.

In one aspect, a drink maker includes a mixing vessel arranged to receive a drink product and a dasher, driven by a drive motor, that is arranged to mix the drink product within the mixing vessel. The drink maker also includes a cooling circuit and/or device arranged to cool the drink product within the mixing vessel, a temperature sensor arranged to measure a temperature associated with the drink product and output a temperature signal, and a memory arranged to store a drink object representing a drink type, the drink object specifying a first temperature value corresponding to a first target temperature. A controller, in communication with the memory, is arranged to: i) receive the temperature signal, and ii) control the temperature associated with the drink product by controlling the cooling circuit based on the received temperature signal, the first temperature value, and/or a manual temperature adjustment and/or temperature offset. The frozen drink maker also includes a user interface arranged to receive a user input to adjust the manual temperature adjustment.

The temperature associated with the drink product may include a temperature of the drink product, a temperature of a cooling element used to cool the drink product, and/or a temperature of a refrigerant used to cool the drink product. The controller may adjust the first target temperature by adding the manual temperature adjustment to the first target temperature. The manual temperature adjustment may include positive or negative temperature value. The manual temperature adjustment may include a range of temperatures at, above, and below the first target temperature. The manual temperature adjustment may be adjustable in increments of greater than or equal to 0.1, 0.2, 0.3, 0.4, 0.5, 1, and/or 2 degrees Celsius.

In some non-limiting embodiments or aspects, the memory includes a plurality of recipes, each of the recipes including a temperature value corresponding to a target temperature. The cooling circuit and/or device may include a refrigeration circuit including an evaporator. The evaporator may be part of the closed loop refrigeration circuit and/or system including a condenser and a compressor. The controller may be configured to control the temperature associated with the drink product by activating the compressor to circulate refrigerant through the evaporator to cool the drink product and deactivating the compressor to stop a flow of refrigerant through the evaporator to stop cooling of the drink product. The controller may control the temperature associated with the drink product by comparing the received temperature signal to the first temperature value, adjusted based on the manual temperature adjustment, and, in response, activating or deactivating the cooling circuit to match the received temperature signal to the first temperature value, adjusted by the manual temperature adjustment, and, thereby, adjust the temperature associated with the drink product to about the target temperature adjusted by the manual temperature adjustment. In some non-limiting embodiments or aspects, the cooling circuit includes a thermal energy cooling (TEC) system implementing, for example, the Peltier effect.

In another aspect, a method for making a drink product includes: receiving, into a mixing vessel, the drink product; mixing, using a dasher driven by a drive motor, the drink product within the mixing vessel; cooling, using a cooling circuit, the drink product within the mixing vessel; measuring, via a temperature sensor, a temperature associated with the drink product and outputting a temperature signal; storing, in a memory, a drink object representing a drink type, the drink object specifying a first temperature value corresponding to a first target temperature; receiving, at a controller, the temperature signal; controlling, by the controller, the temperature associated with the drink product by controlling the cooling circuit based on the received temperature signal, the first temperature value, and a manual temperature adjustment; and receiving a user input to adjust the manual temperature adjustment.

In a further aspect, a drink maker includes a mixing vessel arranged to receive a drink product and a dasher, driven by a drive motor, arranged to mix the drink product within the mixing vessel. The drink maker also includes a cooling circuit arranged to cool the drink product within the mixing vessel, a temperature sensor arranged to measure a temperature associated with the drink product and output a temperature signal, a motor condition sensor arranged to measure a motor condition associated with the drive motor and output a motor condition signal, and a memory arranged to store a first temperature value corresponding to a first target temperature and store a motor condition limit. A controller, in communication with the memory, is arranged to: i) receive the temperature signal, ii) receive the motor condition signal, and iii) control the temperature associated with the drink product by controlling the cooling circuit based at least on the received temperature signal, the received motor condition signal, the first temperature value, and the motor condition limit.

In some non-limiting embodiments or aspects, the controller deactivates the cooling circuit when a magnitude (e.g., a current or power level) of the received motor condition signal is equal to or greater than the motor condition limit. The controller may determine a second temperature value corresponding to a second target temperature, where the magnitude of the received motor condition signal is lower than the motor condition limit. The controller may control the temperature associated with the drink product by controlling the cooling circuit based on the second temperature value. In some non-limiting embodiments or aspects, the controller deactivates the cooling circuit until the temperature associated with the drink product is about equal to the second target temperature.

The motor condition may include current, power, torque, speed of rotation, acceleration of rotation, noise, and/or thermal output. The motor condition sensor may include a motor current sensor, motor voltage sensor, motor torque sensor, motor rotation sensor, acoustic sensor, and/or temperature sensor. A user interface may be arranged to receive a user input to adjust a manual temperature adjustment. The controller may control the temperature associated with the drink product by controlling the cooling circuit based on the received temperature signal, the received motor condition signal, the first temperature value, the motor condition limit, and/or the manual temperature adjustment. The controller may adjust the first target temperature by adding the manual temperature adjustment to the first target temperature.

In yet a further aspect, a method for making a drink product includes: receiving, in a mixing vessel, the drink product; mixing, using a dasher driven by a drive motor, the drink product within the mixing vessel; cooling, using a cooling circuit, the drink product within the mixing vessel; measuring, via a temperature sensor, a temperature associated with the drink product and output a temperature signal; measuring, via a motor condition sensor, a motor condition associated with the drive motor and outputting a motor condition signal; storing, in a memory, a first temperature value corresponding to a first target temperature and storing a motor condition limit; receiving, at a controller, the temperature signal and the motor condition signal; and controlling the temperature associated with the drink product by controlling the cooling circuit based on the received temperature signal, the received motor condition signal, the first temperature value, and/or the motor condition limit.

The application, in various implementations, addresses deficiencies associated with cooling components of a drink maker.

This application describes illustrative systems, methods, and devices whereby a dual-use cooling fan concurrently provides cooling air flow to both a drive motor used to drive rotation of a dasher, and a condenser used to cool refrigerant of a refrigeration circuit and/or system of the drink maker.

In one aspect, a drink maker includes a mixing vessel arranged to receive a drink product and a dasher, driven by a drive motor, arranged to mix the drink product within the mixing vessel. A refrigeration circuit is arranged to cool the drink product within the mixing vessel including a condenser. A cooling fan is configured to concurrently cool the drive motor and the condenser. In some non-limiting embodiments or aspects, the cooling fan is driven by the drive motor either directly or via a gear assembly, and therefore is activated when the drive motor is activated.

The cooling fan may provide air flow through the condenser to cool refrigerant flowing through the condenser. The cooling fan may provide air flow along a surface of the drive motor to cool the drive motor. The cooling fan, drive motor, and condenser may be positioned such that air flow generated by the cooling fan passes serially through the condenser and along a surface of the drive motor. A first portion of air flow generated by the cooling fan may cool the condenser and a second portion of air flow generated by the cooling fan may cool the drive motor. In another implementation, air flow generated by the cooling fan passes in parallel through the condenser and along a surface of the drive motor such that a first portion of the air flow passes through the condenser, while a second portion of the air flow passes along a surface of the drive motor. The condenser may include one or more coils wound in a serpentine arrangement. Each of the one or more coils may include a plurality of thermal transfer fins. When the cooling fan provides air flow through the condenser to cool refrigerant flowing through the condenser, the air flow may travel adjacent to and/or around the plurality of coils.

A cooling channel may extend between the cooling fan and the drive motor, where the cooling channel provides cooling air flow between the cooling fan and the drive motor. The cooling channel may be at least partially formed by a duct. A cooling channel may extend between the cooling fan and the condenser, where the cooling channel provides cooling air flow between the cooling fan and the condenser. The cooling channel may be at least partially formed by a duct. The cooling fan may include a centrifugal fan, a cross-flow fan, a tangential fan, a volute fan, a backward curved fan, a forward curved fan, a blower fan, a squirrel-cage fan, and/or an axial fan.

In another aspect, a cooling fan is configured for cooling a drive motor and a condenser within a housing of a drink maker, where the drive motor is configured to drive rotation of a dasher within a mixing vessel of the drink maker and the condenser is configured to cool a refrigerant circulating within a refrigeration system of the drink maker. The cooling fan includes an air inlet configured to receive an air flow from the ambient environment, an impeller configured to generate the air flow, and an air outlet configured to output the air flow through the condenser and along a surface of the drive motor. The cooling fan may include an air channel arranged to direct the air flow through the condenser and along the surface of the drive motor. The air channel may be at least partially formed by an air duct. The cooling fan may include a centrifugal fan, a cross-flow fan, a tangential fan, a volute fan, a backward curved fan, a forward curved fan, a blower fan, a squirrel-cage fan, and/or an axial fan.

In a further aspect, a method for concurrently cooling a condenser and a drive motor within a housing of a drink maker using a cooling fan includes: activating the drive motor that is arranged to drive rotation of a dasher within a mixing vessel of the drink maker; activating a compressor of a refrigeration system of the drink maker; and activating the cooling fan to concurrently generate air flow through the condenser and along a surface of the drive motor. In some non-limiting embodiments or aspects, the cooling fan is coupled to and/or driven to rotate by the drive motor. The method may include receiving a user input to activate the drive motor, the compressor, and the cooling fan. The user input may initiate a recipe and/or computer program, controlled by a controller, that automatically activates the drive motor, the compressor, and the cooling fan.

One of ordinary skill will recognize that the systems, methods, and devices described herein may apply to other types of food products, such as to the making and/or processing of, without limitation, ice cream, frozen yogurt, other creams, and the like. While the present disclosure describes examples of a drink maker processing various frozen and/or semi-frozen drink products, the systems, devices, and methods described herein are not limited to such drink products and are capable of processing and/or making other types of drink products, such as cooled drink products and/or chilled drink products. The terms “mix,” “mixed” or “mixing” as used herein are not limited to combining multiple ingredients together, but also include mixing a drink product or liquid having a single or no added ingredients. For example, a drink product may consist of only water that is mixed by a dasher during processing, e.g., portions of the water are churned and/or intermingled as the dasher rotates. This may, for example, advantageously enable a more uniform temperature of the water and/or liquid as a whole within the mixing vessel by intermingling portions of the water and/or liquid having different temperatures.

The application, in various implementations, addresses deficiencies associated with fluid inlets for frozen drink makers. Previous frozen drink makers were typically sized for commercial applications. Commercial frozen drink makers have significant headspace above the slush in the vessel. In a commercial frozen drink maker, ingredients can be roughly poured into an open top of the vessel without concern of losing liquids due to splashing or ingredient expansion generated by impact force.

This application describes illustrative systems, methods, and devices that address shortcomings of how liquids are added to a vessel for a frozen drink maker. In particular, a pour-in opening for a frozen drink maker is described that allows ingredients to be added to the vessel in a controlled manner, thereby minimizing or preventing slush overflow. The disclosed pour-in opening can be used with both commercial frozen drink makers or residential frozen drink makers having a smaller vessel capacity and less available headspace than commercial units. The pour-in opening advantageously avoids external splatter and spillage of ingredients as they are added to the vessel and prevents finger insertion (to protect users from moving componentry within the vessel). The pour-in opening also prevents slush contained within the vessel from being pushed out of the vessel.

In some aspects, a pour-in opening for a frozen drink maker is described. The frozen drink maker has a dasher configured to rotate within a mixing vessel about a center axis. The pour-in opening includes a surface that inclines radially with respect to the center axis of the dasher. The pour-in opening also includes an aperture positioned on the surface in fluid communication with an interior of the mixing vessel. The surface may be sloped to direct fluids entering the mixing vessel to enter in the direction of dasher rotation. In some non-limiting embodiments or aspects, the aperture extends laterally along the surface in a direction parallel to the center axis of the dasher. The aperture may be shaped as a slot. In some non-limiting embodiments or aspects, the surface incline directs ingredients to enter the mixing vessel in an entry direction and the entry direction is the same as a rotation direction of the dasher. In some such implementations, the rotation direction of the dasher is clockwise when viewed from a front of the frozen drink maker. In these and other implementations, the aperture is positioned on a right side of the mixing vessel when viewed from the front of the frozen drink maker. A grate may cover at least a portion of the aperture, if desired. In these and other implementations, there may also be a cover moveable between an open position in which the pour-in opening is accessible to a user and a closed position in which the pour-in opening is not accessible to the user. In select implementations, the pour-in opening may also include a lip extending up from a perimeter of the surface to form a well that feeds into the aperture. The pour-in opening may be located approximate to a rear of the mixing vessel when viewed from a front of the frozen drink maker. In these and other implementations, a rotation of the dasher moves contents of the mixing vessel from the rear of the mixing vessel to a front of the mixing vessel.

In another aspect, a vessel for a frozen drink maker is described. The vessel includes a chamber and a pour-in opening. The chamber is a substantially cylindrical chamber sized to accommodate a dasher configured to rotate within the vessel about a center axis. The pour-in opening is positioned on a top section of the vessel. The pour-in opening includes a surface and an aperture. The surface inclines radially with respect to the center axis of the dasher. The aperture is positioned on the surface in fluid communication with the chamber. In some non-limiting embodiments or aspects, the pour-in opening is positioned at a rear of the vessel. In these and other implementations, the surface of the pour-in opening inclines radially to direct incoming ingredients to enter the vessel in an entry direction, and the entry direction is the same as a rotation direction of the dasher. In select implementations, the rotation direction of the dasher is clockwise and the aperture is positioned on a right side of the vessel when viewed from a front of the vessel. In some non-limiting embodiments or aspects, the vessel also includes a cover positioned over the pour-in opening and the cover is moveable between an open position in which the pour-in opening is accessible to a user and a closed position in which the pour-in opening is not accessible to the user.

In further aspects, a frozen drink maker is described. The frozen drink maker includes a mixing vessel having a substantially cylindrical chamber, a dasher configured to rotate within the mixing vessel about a center axis, and a pour-in opening positioned on a top of the mixing vessel. The pour-in opening has a surface that inclines radially with respect to the center axis of the dasher and an aperture positioned on the surface in fluid communication with the chamber. In some non-limiting embodiments or aspects, the center axis of the dasher extends in a horizontal direction. In these and other implementations, the aperture extends laterally along the surface in a direction parallel to the center axis of the dasher. The surface inclines radially to direct incoming ingredients to enter the mixing vessel in an entry direction, and the entry direction is the same as a rotation direction of the dasher. In select implementations, the rotation direction of the dasher is clockwise when viewed from a front of the frozen drink maker and the aperture is positioned on a right side of the mixing vessel when viewed from the front of the frozen drink maker.

The disclosure describes a low-maintenance dispensing system for a frozen drink maker that uses a lip seal rather than a plunger seal. The dispensing mechanism includes several pivoting linkages that operate to swing the seal up when a user actuates a dispensing lever. In the open position, the seal is angled about 45-60 degrees with respect to the spout, which helps to direct the dispensed drink product downward. The spout opening also includes a safety grate to prevent the user from inadvertently inserting his or her fingers into the spout.

This application describes illustrative systems, methods, and devices that provide a dispenser assembly for dispensing a drink product through a spout of a frozen drink maker.

In some exemplary implementations, a dispenser assembly for a frozen drink maker of this disclosure includes a housing having a first portion attached to an outer surface of the frozen drink maker adjacent to a spout, and a second portion spaced apart from the spout and extending outward from the outer surface. A lever attaches to the second portion of the housing. The lever is rotatable relative to the second portion of the housing about a first pivot member. A seal operatively couples to the lever. The seal is configured to seal the spout in a closed position. Rotation of the lever causes the seal to move into an open position to allow dispensing of a drink product through the spout.

In some non-limiting embodiments or aspects, a link member operatively couples to the lever. The link member is rotatable relative to the lever about a second pivot member. A bracket member operatively couples to the link member and attaches to the first portion of the housing. The bracket member is rotatable relative to the link member about a third pivot member and rotatable relative to the first portion of the housing about a fourth pivot member. The seal is attached to the bracket member.

In some non-limiting embodiments or aspects, the second pivot member is a pin extending through the lever and through the link member. In some non-limiting embodiments or aspects, the third pivot member is a pin extending through the bracket member and through the link member. In some non-limiting embodiments or aspects, the fourth pivot member is a pin extending through the first portion of the housing and through the bracket member. In some non-limiting embodiments or aspects, the seal is angled at about 45-60 degrees with respect to the spout when the seal is in the open position. In some non-limiting embodiments or aspects, the spout includes a grate configured to prevent a user from inserting fingers into the spout. In some non-limiting embodiments or aspects, the first pivot member includes a pin extending through the second portion of the housing and through the lever. In some non-limiting embodiments or aspects, the seal is a lip seal. In some non-limiting embodiments or aspects, the bracket member is L-shaped. In some non-limiting embodiments or aspects, the housing is L-shaped.

In some non-limiting embodiments or aspects, a method of dispensing a drink product through a spout of a frozen drink maker of this disclosure includes rotating a lever about a first pivot member relative to a second portion of a housing of a dispenser assembly. The housing further includes a first portion attached to an outer surface of the frozen drink maker adjacent to the spout and the second portion spaced apart from the spout and extending outward from the outer surface. Rotating the lever causes a seal operatively coupled to the lever to move into an open position to allow dispensing of the drink product through the spout. The seal is configured to seal the spout in a closed position.

In some non-limiting embodiments or aspects, the dispenser assembly further includes a link member operatively coupled to the lever, and rotating the lever causes the link member to rotate relative to the lever about a second pivot member. In some non-limiting embodiments or aspects, the dispenser assembly further includes a bracket member operatively coupled to the link member and attached to the first portion of the housing, and rotating the lever causes the bracket member to rotate relative to the link member about a third pivot member and to rotate relative to the first portion of the housing about a fourth pivot member. In some non-limiting embodiments or aspects, the seal is attached to the bracket member.

The disclosure describes a shroud for attaching to a dispenser assembly of a frozen drink maker. The shroud is configured to direct the drink product downward toward a beverage cup without interfering with the movement of the dispenser lever. The shroud also hides components of the dispenser assembly for a more pleasing aesthetic appearance and is removable for easy cleaning.

This application describes illustrative systems, methods, and devices that provide a shroud for attaching to a dispenser assembly for directing a drink product downward toward a beverage cup.

In some non-limiting embodiments or aspects, a shroud for a dispenser assembly of a frozen drink maker of this disclosure includes a first panel section and a second panel section extending substantially parallel to the first panel section. A front section extends between the first and second panel sections. The first and second panel sections are configured to form a removable snap fit with a dispenser housing of the dispenser assembly.

In some non-limiting embodiments or aspects, the front section is curved. In some non-limiting embodiments or aspects, a vertical position of at least the front section of the shroud is adjustable relative to the dispenser assembly. In some non-limiting embodiments or aspects, the front section defines an upper edge. A shape of the upper edge is configured to allow actuation of a handle of the dispenser assembly. In some non-limiting embodiments or aspects, a shape of the upper edge is arcuate. In some non-limiting embodiments or aspects, a shape of the upper edge is rectilinear. In some non-limiting embodiments or aspects, a length of the shroud is selected to cover component parts of the dispenser assembly. In some non-limiting embodiments or aspects, a length and shape of the shroud is selected to direct a drink product dispensed from a spout of the frozen drink maker downward. In some non-limiting embodiments or aspects, the shroud is made from a dishwasher safe material. In some non-limiting embodiments or aspects, the front section is moveable relative to the first panel section and the second panel section. In some non-limiting embodiments or aspects, the front section is hingedly connected to the first panel section and the second panel section. In some non-limiting embodiments or aspects, the front section is vertically slidable relative to the first panel section and the second panel section. In some non-limiting embodiments or aspects, the first panel section is flat. In some non-limiting embodiments or aspects, the second panel section is flat.

Referring now to FIGS. **32**A-**32**C, shown is a cross-sectional profile view of dispenser assembly **3200**, according to some non-limiting embodiments or aspects. For example, FIG. **32**A shows a cross-sectional profile view of dispenser assembly **3200** in a rest position, FIG. **32**B shows a cross-sectional profile view of dispenser assembly **3200** in a first position, and FIG. **32**C shows a cross-sectional profile view of dispenser assembly **3200** in a second position. Each of FIGS. **32**A-**32**C may represent a stage of a multi-stage dispensing operation of dispenser assembly **3200**.

With further reference to FIGS. **32**A-**32**C, dispenser assembly **3200** may include an upper member **3222** configured to connect to a lever (e.g., lever **2906**) via lever platform **3226**. In some-non limiting embodiments or aspects, lever platform **3226** may be formed coextensively with a lower portion of the lever that is pulled by the user. Additionally, or alternatively, lever platform **3226** may be connected to the lever that is pulled by the user. Upper member **3222** may include a bracket, wherein an upper portion of the bracket is lever platform **3226**. Upper member **3222** may include a first opening **3224**. In some non-limiting embodiments or aspects, first opening **3224** may be configured to engage with (e.g., anchor) a handle, a spring (not shown), and/or the like. Additionally or alternatively, a spring (not shown) may engage with dispenser assembly **3200** to provide a biasing force to return a connected lever to a rest position. In some non-limiting embodiments or aspects, the spring may be a butterfly spring (e.g., centered on pivot member **3210**).

In some non-limiting embodiments or aspects, upper member **3222** may include a first pivot member **3220**, through which may be received a pin or rod to affix upper member **3222** rotatably about first pivot member **3220**. When the user pulls the lever in a first rotational direction (e.g., away from mixing vessel **104**), upper member **3222** may rotate about first pivot member **3220**. A lower portion of upper member **3222** may be connected to and/or extend into a first link member **3218**. First link member **3218** may be connected at a first end to the lever (e.g., via lever platform **3226**) and may include a cam at a second end opposite the first end, the cam being defined by a cambered slot **3216** within a wall of first link member **3218**.

In some non-limiting embodiments or aspects, dispenser assembly **3200** may include second link member **3212**. Second link member **3212** may be connected at a first end of second link member **3212** to bracket member **3206**. Bracket member **3206** may be configured to support seal **3202** (e.g., where seal **3202** is mounted on bracket member **3206**). A second end of second link member **3212** may be slidably connected to the cam of first link member **3218**. For example, a second end of second link member **3212** may include pin **3214**, which may be received in cambered slot **3216** of the cam. In some non-limiting embodiments or aspects, cambered slot **3216** may be a V-shaped (e.g., 50-130 degree interior angle, such as in FIG. **39**), and/or L-shaped slot (e.g., 80-100 degree interior angle, such as in FIG. **38**) including a first slot length (e.g., depicted horizontally in FIG. **32**A) and a second slot length (e.g., depicted vertically in FIG. **32**A). In some non-limiting embodiments or aspects, the second slot length may be angled between 80 and 100 degrees (e.g., of an interior angle) relative to the orientation of the first slot length. In some non-limiting embodiments or aspects, the second slot length may be angled greater than 80 degrees relative to the orientation of the first slot length, including greater than 120 degrees relative to the orientation of the first slot length, including 130 degrees (see, e.g., FIG. **39**). As second link member **3212** is pulled into motion by the action of pin **3214** in cambered slot **3216**, second link member **3212** may cause bracket member **3206**, and thereby seal **3202**, to rotate about second pivot member **3210**. Bracket member **3206** may be rotatably fixed at second pivot member **3210** with a rod or pin **3214** secured therein.

In this manner, as the lever is pulled forward, lever platform **3226** is pulled forward, which pulls first link member **3218** downward, which causes pin **3214** to slide along the first slot length of cambered slot **3216** to arrive at the vertex of cambered slot **3216** (e.g., the first position shown in FIG. **32**B). This motion from FIG. **32**A to FIG. **32**B may present a first (e.g., lesser) resistance to the user's pulling, as pin **3214** slides horizontally across the first slot length of cambered slot **3216**. As the lever is pulled farther forward, lever platform **3226** is pulled further forward in the same direction, which pulls first link member **3218** farther forward, which causes pin **3214** to slide along the second slot length of cambered slot **3216**, arriving at a second position as depicted in FIG. **32**C. The motion from FIG. **32**B to FIG. **32**C may present a second (e.g., greater) resistance to the user's pulling, as pin **3214** slides along a steeper angled surface of the cambered slot **3216**. The shape of the cambered slot **3216** combined with a spring (e.g., a butterfly spring) may allow the user to pull at a first strength to move dispenser assembly **3200** from a rest position in FIG. **32**A to a first position in FIG. **32**B, and then to pull at a second strength to move dispenser assembly **3200** from the first position in FIG. **32**B to the second position in FIG. **32**C. The user may control the overall opening of a spout through selective control and pulling of the lever, including pulling through each point at and between the rest position, first position, and second position. For example, the user may allow only minimal drink product flow by pulling the lever a distance less than the first distance required to reach the first position. Similarly, the user may pull the lever to a distance past the first position but not to the second position, to allow a more moderate flow of drink product.

In some non-limiting embodiments or aspects, seal **3202** may entirely close spout **3230** of mixing vessel **104** of the drink maker when the lever is in the rest position (shown in FIG. **32**A). As shown in FIG. **32**A, an entire perimeter of seal **3202** may provide a fluid seal against fluid flow of drink product, preventing leakage from spout **3230**. The biasing force of a spring (e.g., connected to first opening **3224**, centered on pivot member **3210**, centered on pivot member **3220**, etc.) may hold seal **3202** firmly against spout **3230**.

In some non-limiting embodiments or aspects, the user may apply a force to (e.g., pull) the lever (e.g., attached to lever platform **3226**) in a first rotational direction (e.g., away from mixing vessel **104**). As the user applies a first force, the user will meet a first resistance, caused by the configuration of the biasing force of the spring and the interconnection of pin **3214** in cambered slot **3216** (e.g., the first slot length thereof). When the user's applied force overcomes the biasing force of the spring, the lever may cause lever platform **3226** to move with the lever, which may cause upper member **3222** and first link member **3218** to tilt, which may cause pin **3214** to move along a first slot length of cambered slot **3216**, until pin **3214** comes to rest in a vertex of cambered slot **3216**, as shown in FIG. **32**B. In some non-limiting embodiments or aspects, the cam (e.g., cambered slot **3216**) and/or pin **3214** may be formed of a low-friction, durable material (e.g., polyoxymethylene (POM), polytetrafluoroethylene (PTFE), ultra-high molecular weight polyethylene (UHMWPE), polyetheretherketone (PEEK), polycarbonate (PC), nylon (polyamide), and/or the like).

In some non-limiting embodiments or aspects, when dispenser assembly **3200** has arrived at the configuration of FIG. **32**B, the lever and dispenser assembly **3200** may be in a first position, and seal **3202** may be at a first angle. In the first position, a majority of an outer perimeter of seal **3202** may continue to contact and block a majority of liquid flow from spout **3230** of mixing vessel **104**, however, a recessed channel **3204** in seal **3202** may permit some liquid flow through recessed channel **3204** of the seal **3202**. The first position, therefore, may be particularly well configured for dispensing low-viscosity liquids (e.g., chilled, non-frozen or partially frozen drink product) and/or releasing drink product slowly. Instead of opening wide, allowing drink product to spill uncontrollably from spout **3230**, the configuration of the first position promotes clean, controlled dispensing by urging the released drink product from spout **3230** and along recessed channel **3204** of seal **3202**. Controlled release from spout **3230** prevents loss of drink product as it travels to a container, such as a beverage container.

In some non-limiting embodiments or aspects, the user may apply a second force to (e.g., pull) the lever (e.g., attached to lever platform **3226**) in the first rotational direction (e.g., away from mixing vessel **104**). As the user applies a second force (e.g., greater than the first force), the user will meet a second resistance (e.g., greater than the first resistance), caused by the configuration of the biasing force of the spring and the interconnection of pin **3214** in cambered slot **3216** (e.g., the second slot length thereof). When the user's applied force overcomes the biasing force of the spring, the lever may cause lever platform **3226** to move with the lever, which may cause upper member **3222** and first link member **3218** to tilt further, which may cause pin

3214 to move along a second slot length of cambered slot 3216, until pin 3214 comes to rest at an end of cambered slot 3216, as shown in FIG. 32C.

In some non-limiting embodiments or aspects, when dispenser assembly 3200 has arrived at the configuration of FIG. 32C, the lever and dispenser assembly 3200 may be in a second position, and seal 3202 may be at a second angle (e.g., greater than the first angle). In the second position, seal 3202 may be fully retracted from spout 3230 and may release a majority of possible liquid flow from spout 3230 of mixing vessel 104. Drink product may continue to come into contact with seal 3202 as it is released in the second position, however, so the angled retracted and recessed channel 3204 may continue to serve the purpose of guiding the fluid flow downward. Furthermore, a portion of bracket member 3206 may extend below seal 3202, so as to act as a further guiding surface to control fluid flow within and from dispenser assembly 3200. To that end, drink product (e.g., low-viscosity, liquid drink product) may impact against the lower portion of bracket member 3206 and be further redirected downward, such as toward a beverage container.

It will be appreciated that, due to the rendering of a cross-sectional area, only one first link member 3218 or second link member 3212 may be shown, however, in some non-limiting embodiments or aspects, duplicate first link members 3218 or second link members 3212 may be provided, to promote additional strength and stability for the motion between the various positions. In particular, see FIGS. 34A and 34B for renderings of a lower portion of dispenser assembly 3200 that includes two second link members 3212.

Referring now to FIG. 33, shown is a cross-sectional profile view of dispenser assembly 3200 with shroud 3300, according to some non-limiting embodiments or aspects. As shown in FIG. 33, dispenser assembly 3200 may include an upper member 3222 having a first opening 3224 (e.g., to which a spring or lever may be connected on one end). Upper member 3222 may further include a first pivot member 3220, about which upper member 3222 and first link member 3218 may rotate. An upper surface of upper member 3222 may include lever platform 3226, on which the lever may be formed and/or connected. First link member 3218 may include, on an end distal from first pivot member 3220, a cam, which may be defined by the area surrounding and including cambered slot 3216. In some non-limiting embodiments or aspects, cambered slot 3216 may include a V-shaped and/or L-shaped slot that receives pin 3214, which is connected to a second end of second link member 3212. A first end of second link member 3212 may connect to bracket member 3206. Bracket member 3206 may rotate about second pivot member 3210.

In some non-limiting embodiments or aspects, bracket member 3206 may support seal 3202 (e.g., seal 3202 may be mounted to bracket member 3206, affixed to bracket member 3206, formed coextensively with bracket member 3206, and/or the like). Seal 3202 may include recessed channel 3204, which may control a liquid flow (F) of drink product released from spout 3230, particularly when dispenser assembly 3200 and the lever are in a first position, as shown in FIG. 33. The liquid flow (F) is shown in FIG. 33 as a dashed arrowed line beginning in mixing vessel 104, continuing through spout 3230, along an outer surface of seal 3202 (e.g., through recessed channel 3204), and down into and onto shroud 3300. Also, as shown in FIG. 33, a portion of bracket member 3206 extends below seal 3202, to provide additional control of liquid flow (F) as it leaves spout 3230. As is further shown in FIGS. 36 and 37, shroud 3300 is formed to promote gentle release of drink product by helping to redirect a horizontal liquid flow (F) into a downward liquid flow (F).

Referring now to FIGS. 34-37, shown are various views of a shared output connector assembly 3400, according to non-limiting embodiments or aspects. In particular, FIG. 34 shows a front view of a shared output connector assembly 3400, and FIG. 35 shows a front view of a shared output connector assembly 3400 attached to a first dispenser assembly 108*a* (e.g., dispenser assembly 108, 3200) and a second dispenser assembly 108*b* (e.g., dispenser assembly 108, 3200). Moreover, FIG. 36 shows a perspective view of a shared output connector assembly 3400, and FIG. 37 shows a perspective view of a shared output connector assembly 3400 attached to first dispenser assembly 108*a* and second dispenser assembly 108*b*. It will be appreciated that FIG. 37 shows only a front section of mixing vessels 104*a*, 104*b*, and mixing vessels 104*a*, 104*b* may continue further back than as shown in FIG. 37 (see, e.g., FIG. 1).

With further reference to FIGS. 34-37, shared output connector assembly 3400 may be configured to removably attach (e.g., as an adaptor, removable connector, etc.) to first dispenser assembly 108*a* and second dispenser assembly 108*b*. Each dispenser assembly that shared output connector assembly 3400 is adapted to be attached to may include, but is not limited to, a lever (e.g., including a handle, such as handle 120*a*, 120*b*, shown in FIG. 35), a pivot, a seal (e.g., seal 3202, shown in FIG. 33), and one or more additional components shown and described in connection with FIGS. 32A-33.

In some non-limiting embodiments or aspects, shared output connector assembly 3400 may include a primary lever 3401. Primary lever 3401 may include a handle 3402, for a user to manually grasp and actuate (e.g., by pulling in a rotational direction toward the user). Primary lever 3401 may be connected to a pivot 3426, which may allow primary lever 3401 to be rotated during actuation by a user. Pivot 3426 may be connected to, and at least partly concealed by, a pivot cover 3424. Pivot cover 3424 may include a curved surface configured to envelop at least part of pivot 3426, to prevent incursion by food/liquid and provide further stability. Pivot cover 3424 may further include indicia 3428 (e.g., a surface ornamentation or formation to convey information), such as an indication to user what will result from actuating primary lever 3401. As shown, indicia 3428 may include the text "1|2", indicating that both first dispenser assembly 108*a* and second dispenser assembly 108*b* will dispense if the user actuates primary lever 3401. Pivot 3426 and/or primary lever 3401 may be connected to a biasing element (e.g., spring, elastic, etc.) configured to provide a biasing force to primary lever 3401 such that, when primary lever 3401 is released by the user, the biasing element causes primary lever 3401 to return to a neutral position.

In some non-limiting embodiments or aspects, one or more fixed components of shared output connector assembly 3400 may be co-formed in one or more pieces of a same structure, to reduce seams (which may catch food/liquid) and provide stability. For example, one or more components of shared output connector assembly 3400 may be formed of a durable, food-safe, and non-brittle material, such as, but not limited to, polypropylene, high-density polyethylene, low-density polyethylene, nylon, and/or the like. In some non-limiting embodiments or aspects, the following components may be formed of a single structure: first spout cover 3410*a*, first face panel 3414*a*, pivot cover 3424, second face panel 3414*b*, and second spout cover 3410*b*. In some non-limiting embodiments or aspects, the following components may be formed of a single structure: first channel **3416***a*, shared output **3420**, and second channel **3416***b*. In some non-limiting embodiments or aspects, the following components may be formed of a single structure: primary lever **3401**, primary handle **3402**, first projection segment **3408***a*, and second projection segment **3408***b*.

In some non-limiting embodiments or aspects, shared output connector assembly **3400** may include at least one projection (e.g., first projection segment **3408***a*, second projection segment **3408***b*) configured to mechanically connect (e.g., interact to transfer mechanical force, such as through contact, but not limited to a fixed connection) primary lever **3401** to levers (e.g., handles **120***a*, **120***b*) of at least two dispenser assemblies (e.g., dispenser assemblies **108***a*, **108***b*). For example, the at least one projection may include first projection segment **3408***a* configured to mechanically connect primary lever **3401** to a first lever (e.g., handle **120***a*) of first dispenser assembly **108***a*. By way of further example, the at least one projection may include second projection segment **3408***b* configured to mechanically connect primary lever **3401** to a second lever (e.g., handle **120***b*) of second dispenser assembly **108***b*. The at least one projection may be linear, curvilinear, and/or the like, and may be sufficiently rigid to transfer a force from primary lever **3401** to each lever (e.g., handles **120***a*, **120***b*) of each dispenser assembly **108***a*, **108***b*. In some non-limiting embodiments or aspects, the at least one projection may include a horizontal projection extending from a first lever (e.g., handle **120***a*) to a second lever (e.g., handle **120***b*) and may include both first projection segment **3408***a* and second projection segment **3408***b*. The horizontal projection may be connected to and/or co-formed with primary handle **3402**. The first lever and the second lever may be configured to be actuatable without movement of (e.g., actuating) primary lever **3401**.

In some non-limiting embodiments or aspects, the at least one projection may include a first recess **3404***a* configured to receive first handle **120***a* of the first lever of first dispenser assembly **108***a*. The at least one projection may further include a second recess **3404***b* configured to receive second handle **120***b* of the second lever of second dispenser assembly **108***b*. A shape of recesses **3404***a*, **3404***b* may be configured to correspond to a shape of a rear surface area of handles **120***a*, **120***b*, such that recesses **3404***a*, **3404***b* cradle handles **120***a*, **120***b*, preventing slippage or side-to-side movement of projection segments **3408***a*, **3408***b* relative to handles **120***a*, **120***b*. In some non-limiting embodiments or aspects, recesses **3404***a*, **3404***b* may be form-fitted to snap around at least part of a corresponding handle **120***a*, **120***b*. Additionally, or alternatively, recesses **3404***a*, **3404***b* may include a means of securing to handles **120***a*, **120***b*, such as a magnet, snap-fit connector, and/or the like. Handles **120***a*, **120***b* may have pivots and/or levers connected to a biasing element (e.g., spring, clastic, etc.) that urges handles **120***a*, **120***b* back to an upright, neutral position when projection segments **3408***a*, **3408***b* are not applying an active displacing force to handles **120***a*, **120***b*. In this manner, when primary handle **3402** is released and returns to an upright, neutral position, handles **120***a*, **120***b* also are released and return to an upright, neutral position.

In some non-limiting embodiments or aspects, when shared output connector assembly is attached to first dispenser assembly **108***a* and second dispenser assembly **108***b*, the first lever (e.g., handle **120***a*) and the second lever (e.g., handle **120***b*) may be independently actuatable (e.g., able to be pulled by a user). When primary lever **3401** (e.g., handle **3402**) is actuated, the at least one projection (e.g., projection segments **3408***a*, **3408***b*) may cause the first lever (e.g., handle **120***a*) and the second lever (e.g., handle **120***b*) to actuate. For example, when a user manually pulls primary handle **3402** in a first direction (e.g., away from a mixing vessel **104**, toward the user, and/or the like), first projection segment **3408***a* and second projection segment **3408***b* may cause the first handle **120***a* and the second handle **120***b*, respectively, to simultaneously or substantially simultaneously move a distance in the first direction that is proportional to a distance (e.g., measured linearly, radially, etc.) moved by the primary handle **3402**. By way of further example, if the user pulls primary handle **3402** forward by 10 degrees, projection segments **3408***a*, **3408***b* may cause first handle **120***a* and second handle **120***b* to tilt forward 10 degrees, or approximately an equal distance thereto. In this manner, and for embodiments where handles **120***a*, **120***b* may control the rate of flow of drink product, a user can simultaneously control the flow from both dispenser assemblies **108***a*, **108***b* through controlled use of primary handle **3402**.

In some non-limiting embodiments or aspects, shared output connector assembly **3400** may further include first spout cover **3410***a* and second spout cover **3410***b*. First spout cover **3410***a* may be configured to be mounted over (e.g., attached in a manner that partly encloses) at least part of first dispenser assembly **108***a* (e.g., a seal **2928**, **3028**, **3202**, a bracket member **2918**, **3018**, **3206**, an outlet of mixing vessel **104***a*, etc.). Second spout cover **3410***b* may be configured to be mounted over at least part of second dispenser assembly **108***b* (e.g., a seal **2928**, **3028**, **3202**, a bracket member **2918**, **3018**, **3206**, an outlet of mixing vessel **104***a*, etc.). First spout cover **3410***a* may define a first cavity **3406***a*, through which first handle **120***a* may be inserted when attaching shared output connector assembly **3400** to dispenser assemblies **108***a*, **108***b*. Second spout cover **3410***b* may define a second cavity **3406***b*, through which second handle **120***b* may be inserted when attaching shared output connector assembly **3400** to dispenser assemblies **108***a*, **108***b*. Moreover, each cavity **3406***a*, **3406***b* may provide space for portions of dispenser assemblies **108***a*, **108***b* to be covered by spout covers **3410***a*, **3410***b* and operate therein. Spout covers **3410***a*, **3410***b* may also partly form a channel to direct drink product from dispenser assemblies **108***a*, **108***b* downward to channels **3416***a*, **3416***b*. In some non-limiting embodiments or aspects, an interior form of spout covers **3410***a*, **3410***b* may be similar to or the same as shroud **116** or shroud **3300**, to act as a cover and a spout.

In some non-limiting embodiments or aspects, each spout cover **3410***a*, **3410***b* may include indicia **3412***a*, **3412***b* to indicate to user what will happen if user actuates the corresponding handle **120***a*, **120***b* associated with the respective spout cover **3410***a*, **3410***b*. For example, first spout cover **3410***a* may have first indicia **3412***a* in the form of the text "1", to indicate that only first dispenser assembly **108***a* will release drink product if the user actuates first handle **120***a*. By way of another example, second spout cover **3410***b* may have second indicia **3412***b* in the form of the text "2", to indicate that only second dispenser assembly **108***b* will release drink product if the user actuates second handle **120***b*. It will be appreciated that many configurations and forms of user feedback are possible.

In some non-limiting embodiments or aspects, first spout cover **3412***a* and second spout cover **3412***b* may include connectors (e.g., first connectors **3432***a*, second connectors **3432***b*) configured to engage with first dispenser assembly **108***a* and second dispenser assembly **108***b*. For example, first spout cover **3412***a* may include first connectors **3432***a* (e.g., snap-fit connectors, magnetic connectors, mechanically interlocking connectors) configured to engage with first dispenser assembly **108***a* (e.g., in slots thereof, with magnetic connectors thereof, with corresponding mechanical interlocking points thereof) to cause first spout cover **3412***a* to become removably attached to mixing vessel **104***a* and/or first dispenser assembly **108***a*. By way of further example, second spout cover **3412***b* may include second connectors **3432***b* (e.g., snap-fit connectors) configured to engage with second dispenser assembly **108***b* (e.g., in slots thereof) to cause second spout cover **3412***b* to become removably attached to mixing vessel **104***b* and/or second dispenser assembly **108***b*. First connectors **3432***a* and second connectors **3432***b* may include one or more connectors, which may be on either or both sides of an interior of a spout cover **3412***a*, **3412***b*. In some non-limiting embodiments or aspects, connectors **3432***a*, **3432***b* may include snap-fit connectors (e.g., cantilever snap-fit connectors, annular snap-fit connectors, hinge snap-fit connectors, etc.) that are configured to engage with (e.g., mate with) slots on dispenser assemblies **108***a*, **108***b*. Additionally, or alternatively, connectors **3432***a*, **3432***b* may include magnetic connectors (e.g., embedded magnets with a first pole orientation) that are configured to engage with (e.g., magnetically attract) corresponding magnetic connectors (e.g., embedded magnets with a second pole orientation) of dispenser assemblies **108***a*, **108***b*). Additionally, or alternatively, connectors **3432***a*, **3432***b* may include connectors that mechanically interlock with corresponding interlocking points on dispenser assemblies **108***a*, **108***b*, such as, but not limited to, friction/interference fit connectors, spring-loaded latch connectors, lever-lock clamp connectors, and/or the like.

In some non-limiting embodiments or aspects, shared output connector assembly **3400** may further include face panels **3414***a*, **3414***b* to connect pivot cover **3424** to spout covers **3410***a*, **3410***b*. For example, shared output connector assembly **3400** may include a first face panel **3414***a* extending between pivot cover **3424** and first spout cover **3410***a*. Shared output connector assembly **3400** may further include a second face panel **3414***b* extending between pivot cover **3424** and second spout cover **3410***b*. Face panels **3414***a*, **3414***b* may provide structural support and strength to shared output connector assembly **3400**, as well as may at least partly cover a gap and/or space between first dispenser assembly **108***a* and second dispenser assembly **108***b*. In some non-limiting embodiments or aspects, face panels **3414***a*, **3414***b* may be co-formed (e.g., molded in a single piece) with spout covers **3410***a*, **3410***b*.

In some non-limiting embodiments or aspects, shared output connector assembly **3400** may further include a shared outlet **3420** configured to output drink product released from first dispenser assembly **108***a*, the second dispenser assembly **108***b*, or any combination thereof. Shared outlet **3420** may include one or more attachment protrusions **3421** that are configured to connect a nozzle attachment (not shown) to shared outlet **3420**. Additionally, or alternatively, attachment protrusions **3421** may be configured to allow attachment of a receiving container (e.g., a bottle) to shared output connector assembly **3400**, to receive dispensed drink product. If a user pulls first handle **120***a* or primary handle **3402** and releases drink product from first dispenser assembly **108***a*, the drink product may flow from first dispenser assembly **108***a* and ultimately output from shared outlet **3420**. If a user pulls second handle **120***b* or primary handle **3402** and releases drink product from second dispenser assembly **108***b*, the drink product may flow from second dispenser assembly **108***b* and ultimately output from shared outlet **3420**. Shared outlet **3420** allows for single placement and/or attachment of a receiving container such that drink product can be received from first mixing vessel **104***a* and/or vessel **104***b* at a single point, below intersection **3422**.

In some non-limiting embodiments or aspects, shared output connector assembly **3400** may further include channels **3416***a*, **3416***b* to guide drink product from spout covers **3410***a*, **3410***b* to shared outlet **3420**. For example, shared output connector assembly **3400** may include first channel **3416***a* that is configured to guide the drink product released from first dispenser assembly **108***a* to shared outlet **3420**. Additionally, shared output connector assembly **3400** may include second channel **3416***b* that is configured to guide the drink product released from second dispenser assembly **108***b* to shared outlet **3420**. Channels **3416***a*, **3416***b* may include sloped chutes (e.g., curved partial pipes) and may extend from the bottom of spout covers **3410***a*, **3410***b* and connect to shared outlet **3420**. An interior surface of channels **3416***a*, **3416***b* may be low-friction (e.g., smoothed) to promote a consistent flow of drink product from dispenser assemblies **108***a*, **108***b* to shared outlet **3420**. Channels **3416***a*, **3416***b* may meet at intersection **3422**, which is located at or proximal to shared outlet **3420**. Channels **3416***a*, **3416***b* may be joined and/or continuous along intersection **3422**.

In some non-limiting embodiments or aspects, shared output connector assembly **3400** may further include channel covers **3418***a*, **3418***b* configured to at least partly enclose channels **3416***a*, **3416***b*. For example, shared output connector assembly **3400** may include first channel cover **3418***a* configured to extend at least partially along first channel **3416***a* and at least partly enclose a first flow volume (e.g., a space through which drink product flows) of first channel **3416***a*. Additionally, shared output connector assembly **3400** may include second channel cover **3418***b* configured to extend at least partially along second channel **3416***b* and at least partly enclose a second flow volume (e.g., a space through which drink product flows) of second channel **3416***b*. A width of channel cover **3418***a*, **3418***b* may be equal to a width of a corresponding channel **3416***a*, **3418***b*. Moreover, channel covers **3418***a*, **3418***b* may be curved (e.g., arched, vaulted, etc.) to increase a flow volume through and along channels **3416***a*, **3416***b*. In some non-limiting embodiments or aspects, channel covers **3418***a*, **3418***b* may be formed of an at least partially transparent material (e.g., polycarbonate, Tritan copolyester, and/or the like), to provide a viewing window for a user to monitor the flow of drink product through channels **3416***a*, **3416***b*. Additionally, or alternatively, channel covers **3418***a*, **3416***b* may be formed of an at least partially opaque material (e.g., polypropylene, acrylonitrile butadiene styrene (ABS), high-density polyethylene (HDPE), polycarbonate, and/or the like). Transparent channel covers **3418***a*, **3418***b* may promote cleanability of channels **3416***a*, **3416***b* by allowing visible inspection inside the drink product pathways. Opaque channel covers **3418***a*, **3418***b* may reduce visual clutter and provide a more uniform aesthetic presentation, and may further reduce incursions by light that may prematurely warm the drink product being dispensed.

In some non-limiting embodiments or aspects, first dispenser assembly **108***a* and second dispenser assembly **108***b* may belong to a same drink maker, such as a dual-tank drink maker that includes two dispenser assemblies **108** and separate mixing vessels (e.g., a first mixing vessel **104***a* and a second mixing vessel **104***b*). In some non-limiting embodiments or aspects, first dispenser assembly **108***a* may belong to a first drink maker (e.g., drink maker **100**) and second dispenser assembly **108***b* may belong to a second drink maker (e.g., drink maker **100**), which may each include a mixing vessel **104** (e.g., first mixing vessel **104***a*, second mixing vessel **104***b*) and an independent dispenser assembly **108**. Shared output connector assembly **3400** allows a container to be filed from a single shared outlet **3420** whether user pulls a first handle **120***a*, a second handle **120***b*, or primary handle **3402**. If user pulls only first handle **120***a*, a first drink product may be dispensed from first mixing vessel **104***a*. If user pulls only second handle **120***b*, a second drink product may be dispensed from second mixing vessel **104***b*. If user pulls primary handle **3420**, both the first drink product and the second drink product may be dispensed from first mixing vessel **104***a* and second mixing vessel **104***b*, respectively, meeting at intersection **3422** and being output at shared outlet **3420**.

When attaching shared output connector assembly **3400** to one or more drink makers (e.g., one or more dispenser assemblies **108***a*, **108***b* thereof), one or more components of shared output connector assembly **3400** may detach, such as pivot cover **3424**, first face panel **3414***a*, second face panel **3414***b*, first spout cover **3410***a*, and/or second spout cover **3410***b*, allowing shared output connector assembly **3400** to be slipped over one or more levers (e.g., including handles **120***a*, **120***b*) of first dispenser assembly **108***a* and second dispenser assembly **108***b*, via first cavity **3406***a* and second cavity **3406***b*. Additionally, or alternatively, shared output connector assembly **3400** may be configured to attach to first dispenser assembly **108***a* and second dispenser assembly **108***b* without removing components thereof. For example, shared output connector assembly **3400** may be rotated about a horizontal axis such that a top of shared output connector assembly **3400** is inclined away from the user and a bottom of shared output connector assembly **3400** is inclined toward the user. Through the axial rotation, handles **120***a*, **120***b* of dispenser assemblies **108***a*, **108***b* may be threaded up and through cavities **3406***a*, **3406***b* (e.g., by lowering shared output connector assembly **3400** onto and around dispenser assemblies **108***a*, **108***b*). Once handles **120***a*, **120***b* of dispenser assemblies **108***a*, **108***b* are inserted through cavities **3406***a*, **3406***b*, the axial rotation may be reversed, such that shared output connector assembly **3400** is returned to a vertical orientation. Furthermore, connectors **3432***a*, **3432***b* may engage with dispenser assemblies **108***a*, **108***b* to removably connect shared output connector assembly **3400** to dispenser assemblies **108***a*, **108***b* (see, e.g., slots **2909** in FIG. **29**D, which may receive connectors **3432***a*, **3432***b* and allow shared output connector assembly **3400** to be held in place during use). To remove shared output connector assembly **3400**, the above steps may be reversed, beginning with disengaging any connectors **3432***a*, **3432***b* and proceeding with tilting shared output connector assembly **3400**.

Although embodiments have been described in detail for the purpose of illustration, it is to be understood that such detail is solely for that purpose and that the disclosure is not limited to the disclosed embodiments or aspects, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present disclosure contemplates that, to the extent possible, one or more features of any embodiment or aspect can be combined with one or more features of any other embodiment or aspect.

What is claimed is:

1. A shared output connector assembly for connecting a first dispenser assembly and a second dispenser assembly, the shared output connector assembly comprising:
- a primary lever; and
- at least one projection configured to mechanically connect the primary lever to (i) a first lever of the first dispenser assembly and (ii) a second lever of the second dispenser assembly,
- wherein, when the shared output connector assembly is attached to the first dispenser assembly and the second dispenser assembly, the first lever and the second lever are independently actuatable, and when the primary lever is actuated, the at least one projection causes the first lever and the second lever to actuate.

2. The shared output connector assembly of claim **1**, further comprising:
- a first spout cover configured to be mounted over at least part of the first dispenser assembly;
- a second spout cover configured to be mounted over at least part of the second dispenser assembly.

3. The shared output connector assembly of claim **2**, wherein the first spout cover comprises a first connector configured to engage with the first dispenser assembly and the second spout cover comprises a second connector configured to engage with the second dispenser assembly.

4. The shared output connector assembly of claim **3**, wherein the first connector and the second connector each comprise a snap-fit connector configured to engage with a corresponding slot on the first dispenser assembly and the second dispenser assembly, respectively.

5. The shared output connector assembly of claim **3**, wherein the first connector and the second connector each comprise a magnetic connector configured to engage with a corresponding magnetic connector on the first dispenser assembly and the second dispenser assembly, respectively.

6. The shared output connector assembly of claim **3**, wherein the first connector and the second connector are configured to mechanically interlock with the first dispenser assembly and the second dispenser assembly, respectively.

7. The shared output connector assembly of claim **2**, further comprising:
- a pivot connected to the primary lever; and
- a pivot cover configured to at least partly conceal the pivot.

8. The shared output connector assembly of claim **7**, further comprising:
- a first face panel extending between the pivot cover and the first spout cover; and
- a second face panel extending between the pivot cover and the second spout cover.

9. The shared output connector assembly of claim **1**, further comprising a shared outlet configured to output drink product released from at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof.

10. The shared output connector assembly of claim **9**, further comprising:
- a first channel configured to guide the drink product released from the first dispenser assembly to the shared outlet; and
- a second channel configured to guide the drink product released from the second dispenser assembly to the shared outlet.

11. The shared output connector assembly of claim **10**, wherein the first channel comprises a first sloped chute configured to guide the drink product released from the first dispenser assembly to the shared outlet, and wherein the second channel comprises a second sloped chute configured to guide the drink product released from the second dispenser assembly to the shared outlet.

12. The shared output connector assembly of claim **10**, further comprising:

a first channel cover configured to extend at least partially along the first channel and at least partly enclose a first flow volume of the first channel; and

a second channel cover configured to extend at least partially along the second channel and at least partly enclose a second flow volume of the second channel.

13. The shared output connector assembly of claim **12**, wherein the first channel cover and the second channel cover are formed of an at least partially transparent material.

14. The shared output connector assembly of claim **12**, wherein the first channel cover and the second channel cover are formed of an at least partially opaque material.

15. The shared output connector assembly of claim **1**, wherein the at least one projection comprises:

a first projection segment configured to mechanically connect the primary lever to the first lever of the first dispenser assembly; and

a second projection segment configured to mechanically connect the primary lever to the second lever of the second dispenser assembly.

16. The shared output connector assembly of claim **15**, wherein the at least one projection comprises a horizontal projection extending from the first lever to the second lever, the horizontal projection comprising the first projection segment and the second projection segment.

17. The shared output connector assembly of claim **15**, wherein the first projection segment comprises a first recess configured to receive a first handle of the first lever, and wherein the second projection segment comprises a second recess configured to receive a second handle of the second lever.

18. The shared output connector assembly of claim **1**, wherein the primary lever comprises a primary handle, the first lever comprises a first handle, and the second lever comprises a second handle, and wherein the primary handle, the first handle, and the second handle are configured to be manually pulled by a user when the shared output connector assembly is attached to the first dispenser assembly and the second dispenser assembly.

19. The shared output connector assembly of claim **18**, wherein, when the user manually pulls the primary handle in a first direction, the at least one projection is configured to cause the first handle and the second handle to simultaneously move in the first direction.

20. The shared output connector assembly of claim **19**, wherein, when the user manually pulls the primary handle in the first direction, the at least one projection is configured to cause the first handle and the second handle to simultaneously move a distance in the first direction that is proportional to a distance moved by the primary handle.

21. The shared output connector assembly of claim **1**, further comprising:

a pivot connected to the primary lever; and

a biasing element connected to the primary lever, the biasing element configured to provide a biasing force to the primary lever wherein, when the primary lever is released by the user, the biasing element causes the primary lever to return to a neutral position.

22. The shared output connector assembly of claim **21**, wherein the biasing element comprises a spring.

23. The shared output connector assembly of claim **22**, wherein a first drink maker comprises the first dispenser assembly, and wherein a second drink maker comprises the second dispenser assembly.

24. The shared output connector assembly of claim **1**, wherein a first mixing vessel comprises the first dispenser assembly, and wherein a second mixing vessel comprises the second dispenser assembly.

25. The shared output connector assembly of claim **1**, wherein the first lever and the second lever are each configured to be actuatable without movement of the primary lever.

26. A shared output connector assembly for connecting a first dispenser assembly and a second dispenser assembly, the shared output connector assembly comprising:

a shared outlet configured to output drink product released from at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof;

a first channel configured to guide the drink product released from the first dispenser assembly to the shared outlet; and

a second channel configured to guide the drink product released from the second dispenser assembly to the shared outlet.

27. The shared output connector assembly of claim **26**, wherein the shared output connector assembly is configured to be removably attachable to at least one of the first dispenser assembly, the second dispenser assembly, or any combination thereof.

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